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Appendix R.7.5–1 Testing strategy for specific system/organ toxicity: example of neurotoxicity assessment

Content of Appendix R.7.5-1

1. General aspects

For some specific system/organ effects the testing methods of the Annex to the EU Test Methods (TM) Regulation (Council Regulation (EC) No 440/2008) or of the OECD may not provide for adequate characterisation of the toxicity. There may be indications of such effects in the standard studies for systemic toxicity, or from SAR. For adequate characterisation of the toxicity and, hence, the risk to human health, it may be necessary to conduct studies using other published test methods, in-house methods or specially designed tests. Some references are given in Table R.7.5–3. Before initiating a study to investigate specific organ/system toxicity, it is important that the study design is presented to the Agency, in order that the need for (and scope/size of) studies using live animals can be particularly carefully considered.

Specific investigation of organ/systemic toxicity is to some extent undertaken as part of the repeated dose toxicity tests conducted according to test guidelines of the OECD and the Annex to the EU TM Regulation. Specific investigation (or further investigation) of any organ/system toxicity (e.g. immune, endocrine or nervous system) may sometimes be necessary and should be addressed on a case-by-case basis. As an example of a testing strategy the approach for neurotoxicity is given below.

2. Definition of neurotoxicity and indication of neurotoxicity potential from REACH information requirements for repeated dose toxicity

Neurotoxicity is the induction by a substance of adverse effects in the central or peripheral nervous system, or in sense organs. It is useful for the purpose of hazard and risk assessment to differentiate sense organ-specific effects from other effects which lie within the nervous system. A substance is considered *neurotoxic* if it induces a reproducible lesion in the nervous system or a reproducible pattern of neural dysfunction.

The starting point for the testing strategy are the REACH requirements specified in Annexes VIII, IX and X and detailed in Section R.7.5.6.3.

Depending on the tonnage level, these requirements may trigger a 28-day and/or a 90-day test (e.g. OECD TGs 407, 408 / EU B.7, B.26). These protocols include a number of nervous system endpoints (e.g. clinical observations of motor and autonomous nervous system activity, histopathology of nerve tissue), which should be regarded as the starting point for evaluation of a substance potential to cause neurotoxicity. It should be recognised that the standard 28-/90-day tests only measure some aspects of nervous system structure and function, e.g. Functional Observational Battery, while other aspects, e.g. learning and memory and sensory function is not or only superficially tested. SAR considerations may prompt the introduction of additional parameters to be tested in standard toxicity tests or the immediate request of studies such as delayed neurotoxicity (OECD TG 418 or 419 / EU B.37 or B.38; see below).

If there are no indications of neurotoxicity from available information i.e. adequately performed repeated dose toxicity tests, other testing systems (e.g. *in vitro*), non-testing systems ((Q)SAR and read-across) or human data, it will not be necessary to conduct any special tests for neurotoxicity.

The approach presented below is a hierarchical, stepwise strategy to investigate the potential neurotoxicity of a substance. It should be pointed out that the requirements outlined in steps 1 and 2 are met by the tonnage-based information requirements in Annexes VIII, IX and X to the REACH Regulation.

3. Structure-activity considerations

Structural alerts are only used as a positive indication of neurotoxic potential. Substance classes with an alert for neurotoxicity may include organic solvents (for chronic toxic encephalopathy), organophosphorus substances (for delayed neurotoxicity) and carbamates (for cholinergic effects). Several estimation techniques are available, one of which is the rule-based DEREK (Deductive Estimation of Risk from Existing Knowledge) system. The rulebase comprises the following hazards and structural alerts: Organophosphate (for direct and indirect anticholinesterase activity), N-methyl or N,N-dimethyl carbamate (for direct anticholinesterase activity), gamma-diketones (for neurotoxicity).

3. Assessment of available information or results from initial testing

Signs of neurotoxicity in standard acute or repeated dose toxicity tests may be secondary to other systemic toxicity or to discomfort from physical effects such as a distended or blocked gastrointestinal tract. Nervous system effects seen at dose levels near or above those causing lethality should not be considered, in isolation, to be evidence of neurotoxicity. In acute toxicity studies where high doses are administered, clinical signs are often observed which are suggestive of effects on the nervous system (e.g. observations of lethargy, postural or behavioural changes), and a distinction should be made between specific and non-specific signs of neurotoxicity.

Neurotoxicity may be indicated by the following signs: morphological (structural) changes in the central or peripheral nervous system or in special sense organs; neurophysiological changes (e.g. electroencephalographic changes); behavioural (functional) changes; neurochemical changes (e.g. neurotransmitter levels).

A *Weight-of-Evidence* approach should be taken into account for the assessment of the neurotoxicity and the type, severity, number and reversibility of the effect should be considered. A consistent pattern of neurotoxic findings rather than a single or a few unrelated effects should be taken as persuasive evidence of neurotoxicity.

It is important to ascertain whether the nervous system is the primary target organ. The reversibility of neurotoxic effects should also be considered. The potential for such effects to occur in exposed humans (i.e. the exposure pattern and estimated level of exposure are *acute*) should be considered in the risk characterisation. Reversible effects may be of high concern depending on the severity and nature of effect. In this context it should be kept in mind that effects observed in experimental animals that appear harmless might be of high concern in humans depending on the setting in which they occur (e.g. sleepiness in itself may not be harmful, but in relation to operation of machinery it is an effect of high concern). Furthermore the possibility that a permanent lesion has occurred cannot be excluded, even if the overt effect is transient. The nervous system possesses reserve capacity, which may compensate for the damage, but the resulting reduction in the reserve capacity should be regarded as an adverse effect. Irreversible neurotoxic effects are of high concern and usually involve structural changes, though, at least in humans, lasting functional effects (e.g. depression, involuntary motor tremor) are suspected to occur as a result of neurotoxicant exposure, apparently without morphological abnormalities.

For the evaluation of organophosphate pesticides, the WHO/FAO Joint Meeting of Experts on Pesticide Residues (JMPR) has published recommendations on "Interpretation of Cholinesterase Inhibition" (FAO, 1998; 1999). The applicability of these recommendations, outlined below, could also be extended to other substances that inhibit cholinesterase. It should be pointed out that for substances that may have a structural alert for cholinesterase inhibition, the measurement of acetylcholinesterase activity as recommended by JMPR can be included in the list of parameters for the standard 28- or 90-day testing protocols required by REACH, irrespective of the route of exposure.

5. Recommendations from the WHO/FAO Joint Meeting of Experts on Pesticide Residues (JMPR)

The inhibition of brain acetylcholinesterase activity and clinical signs are considered to be the primary endpoints of concern in toxicological studies on substances that inhibit acetylcholinesterases. Inhibition of erythrocyte acetylcholinesterase is also considered to be an adverse effect, insofar as it is used as a surrogate for brain and peripheral nerve acetylcholinesterase inhibition, when data on the brain enzyme are not available. The use of erythrocyte acetylcholinesterase inhibition as a surrogate for peripheral effects is justified for acute exposures resulting in greater acetylcholinesterase inhibition in erythrocytes than in the brain. However, reliance on inhibition of erythrocytic enzyme in studies of repeated doses might result in an overestimate of inhibition on peripheral tissues, because of the lower rate of resynthesis of the enzyme in erythrocytes than in the nervous system. Plasma acetylcholinesterase inhibition is considered not relevant. Regarding brain and erythrocyte acetylcholinesterase inhibition, the experts defined that statistically significant inhibition by 20% or more represents a clear toxicological effect and any decision to dismiss such findings should be justified. JMPR also agreed on the convention that statistically significant inhibition of less than 20% or statistically insignificant inhibition above 20% indicate that a more detailed analysis of the data should be undertaken. The toxicological significance of these findings should be determined on a case-by-case basis. One of the aspects to consider is the dose-response characteristic.

5. Further neurotoxicity testing

If the data acquired from the standard systemic toxicity tests required by REACH provide indications of neurotoxicity which are not adequate for a hazard assessment, risk characterisation or C&L, the nature of further investigation will need to be considered. If a 90-day study is triggered to meet the requirements of Annex IX to the REACH Regulation following a standard 28-day study, a number of endpoints assessing the nervous system endpoints should be included, irrespective of the administration route. In some cases, it may be necessary to conduct a specific study such as a neurotoxicity test using the OECD TG 424 with possible inclusion of a satellite group for assessment of reversibility of effects. The OECD TG 424 is intended for confirmation or further characterisation of potential neurotoxicity identified in previous studies. The OECD guideline allows for a flexible approach, in which the number of simple endpoints which duplicate those already examined during standard testing may be minimised, and where more effort is put into in-depth investigation of more specific endpoints by inclusion of more specialised tests. Adjustment of dose levels to avoid confounding by general toxicity should be considered.

If data from standard toxicity studies are clearly indicative of specific neurotoxicity, e.g. neurotoxicity occurring at lower dose levels than systemic toxicity, further specific neurotoxicity testing is required to confirm and extend the findings from the general toxicity studies and to establish an NOAEL for neurotoxicity. Again, the neurotoxicity test according to OECD TG 424 is considered appropriate for this situation.

Certain substances and/or certain effects are best investigated in particular species. Pyridine derivatives are neurotoxic to humans and primates but not to rats. Among other neurotoxic substances, organophosphorus substances are a group with known delayed neurotoxic properties, which need to be assessed in a specified test for delayed neurotoxicity, to be performed preferentially in the adult laying hen according to EU B.37 or OECD TG 418 (Delayed neurotoxicity of organophosphorus substances following acute exposure) and B.38 or OECD TG 419 (Delayed neurotoxicity of organophosphorus substances: 28-day repeated dose study). Such studies are specifically required for biocidal substances of similar or related structures to those capable of inducing delayed neurotoxicity. If anticholineesterase activity is detected, a test for response to reactivating agent may be required.

Standard exposure conditions may not always be adequate for neurotoxicity studies. The duration of exposure needed to induce specific neurotoxic effects in an animal experiment will depend on the underlying mechanism of action. Short-term peak exposures can be important for certain types of substance/effect. When the test substance is administered as a bolus *via* the intravenous, subcutaneous or oral route it is essential to determine the time-effect course, and to perform measurements of neurotoxicity parameters preferentially at the time of peak effect.

For example, the neurotoxicity associated with short-term exposure to some volatile organic solvents has largely been identified following human exposure, particularly occupational exposure. Acute inhalation studies, using protocols designed to detect the expected effects, are ideal for such substances/effects. For some neurotoxic substances a long exposure period is necessary to elicit neurotoxicity.

The most appropriate methods for further investigation of neurotoxicity should be determined on a case-by-case basis, guided by the effects seen in the standard systemic toxicity tests and/or from SAR-based predictions. Extensive coverage of methods that may be used can be found in the documents issued by the OECD (2004), WHO (1986) and ECETOC (1992), and some methods are summarised in Table R.7.5–3.

Table R.7.5–3 Methods for investigation of neurotoxicity

Effect	Methods available	References*
Morphological changes	Neuropathology. Gross anatomical techniques. Immunocytochemistry. Special Stains	Krinke, 1989; Odonoghue, 1989; Mattson *et al.*, 1990
Physiological changes	Electrophysiology (e.g. nerve conduction velocity (NCV), Electroencephalogram (EEG), evoked potentials	Fox *et al.*, 1982; Rebert, 1983; Mattson and Albee, 1988
Behavioural changes	Functional observations. Sensory function tests. Motor function tests (e.g. locomotor activity). Cognitive function tests	Robbins, 1997; Tilson *et al.*, 1980; Cabe and Eckerman, 1982; Pryor *et al.*, 1983 Moser and MacPhail, 1990; Moser 1995
Biochemical changes	Neurotoransmitter analysis. Enzyme/protein activity. Measures of cell integrity.	Dewar and Moffet, 1979; Damstra and Bondy, 1982; Cooper *et al.*, 1986; Costa, 1998.

*Given in full in ECETOC (1982), IPCS (1986) or Mitchell (1982)

7. References

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Appendix R.7.5–2 (Q)SARs for the prediction of repeated dose toxicity

A number of *in silico* tools are available for the prediction of repeated dose toxicity.

As already stated in the main text of this Section, the use of these tools should be mainly for obtaining screening and mechanistic information. Some of them are presented in Table R.7.5–4 below. A more exhaustive review of the available databases, literature and *in silico* models is given in a JRC report from Lapenna *et al.*, 2010.

Table R.7.5–4 *in silico* tools for the prediction of repeated dose toxicity

Tool	Model/module	Description
QSAR Toolbox (Free) http://www.qsartoolbox.org/	Profilers and databases	Co-developed by ECHA and OECD, the QSAR Toolbox includes specific profilers (e.g. Repeated dose HESS) and databases (e.g. Fraunhofer ITEM) for repeated dose toxicity. These modules facilitate the selection of analogues with repeated dose toxicity experimental data for filling data gaps via read-across or trend-analysis.
ADMET Predictor (Simulation Plus) (Commercial) http://www.simulations-plus.com/Products.aspx?pID=13&mID=27	Toxicity	The toxicity module in ADMET Predictor includes a series of models for various organ toxicities (e.g. cardiac, liver).
Derek Nexus (Lhasa) (Commercial) https://www.lhasalimited.org/products/derek-nexus.htm	Models for organ toxicity	Derek Nexus includes several specific organ toxicity models related to repeated dose toxicity (e.g. liver).
Discovery Studio (BIOVIA) (Commercial) http://accelrys.com/products/collaborative-science/biovia-discovery-studio/qsar-admet-and-predictive-toxicology.html	TOPKAT	TOPKAT (TOxicity Prediction by Komputer Assisted Technology) includes a model for Rat chronic LOAEL.
Leadscope (Commercial) http://www.leadscope.com/index.php	Various organs adverse effects statistical models	Leadscope includes several specific organ toxicity models related to repeated dose toxicity (e.g. hepatobiliary tract).

Reference

Lapenna S, Fuart-Gatnik M and Worth A (2010) Review of QSAR models and software tools for predicting acute and chronic systemic toxicity. JRC report EUR 24639 EN. Publications Office of the European Union. Available at: http://publications.jrc.ec.europa.eu/repository/handle/JRC61930

R.7.6 Reproductive toxicity

R.7.6.1 Introduction

Reproductive hazards of chemicals are of obvious concern for the general population. Similarly, to the individual, an impairment of the ability to reproduce and the occurrence of developmental disorders are self-evidently serious health constraints. Therefore it is important that the potential hazardous properties and risks with respect to reproduction are established for substances. The REACH information requirements have two core objectives:

- to have adequate information in order to decide whether classification and labelling, including categorisation, as a reproductive toxicant is warranted;
- to have sufficient information for the purpose of risk assessment.

REACH information requirements for reproductive toxicity were amended in 2015[113] and the recitals of that amendment describe the motivation of the legislator. Recitals are considered a complementary part of the guidance aiming to allow a comprehensive understanding of the objectives of the legislation. Some of them are referred to in this guidance as necessary.

The terminology used in various legislation and in context related to reproductive toxicity differs. In this guidance document the term "reproductive toxicity*"* is used to cover both the effects on fertility and development. Fertility is seen as a broad concept covering all the effects on the reproductive cycle except for developmental toxicity. Development, referred to as "developmental toxicity" is defined in the text below.

In REACH, the Chemical Safety Report (CSR) format includes the terms "effects on fertility" and "developmental toxicity" under the main heading of "toxicity to reproduction". Also in other texts in REACH, such as in the REACH Annexes, reproductive toxicity is divided into fertility and developmental toxicity[114]. It is worth noting that in IUCLID the main heading for reproductive toxicity (7.8) is "Toxicity to reproduction", the subheading for fertility (7.8.1) is "Toxicity to reproduction" and the subheading for developmental toxicity (7.8.2) is "Developmental toxicity / teratogenicity".

In Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures (CLP Regulation), the term "*reproductive toxicity"* as defined in CLP Annex I, is used to describe the adverse effects induced (by a substance) on sexual function and fertility in adult males and females, the development of the offspring and adverse effects on or mediated *via* lactation. Thus, in the CLP Regulation, the differentiation within reproductive toxicity differs from the one stipulated in REACH, namely that lactation effects are considered separately. Hence, for the purpose of classification, reproductive toxicity is divided into three main differentiations, which relate to (i) impairment of male and female reproductive functions or capacity (fertility), (ii) the induction of non-heritable harmful effects on the progeny (developmental toxicity), and (iii) effects on or *via* lactation.

It is necessary to distinguish as far as possible effects on fertility and developmental toxicity for a substance and information on both types of effects is required by REACH above certain tonnage levels. The term "*fertility*" is used in the present guidance document instead of "*sexual function and fertility*" as explained above in order to follow the terminology used in

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113 Commission Regulation (EU) 2015/282 of 20 February 2015 amending Annexes VIII, IX and X to Regulation (EC) No 1907/2006 of the European Parliament and of the Council on the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) as regards to Extended one-generation reproductive toxicity study

114 in Column 2 (see REACH Annexes VIII, IX and X, 8.7.1, Column 2).

REACH. The term "*sexual function and fertility"* is not used in REACH, however, in specific places, where classification and labelling is discussed, "*sexual function and fertility*" is used as a hazard class in the same context as "*fertility*" used alone. It is to be noted that fertility (as a REACH endpoint) covers functional fertility, morphological and histological changes related to reproductive organs in males and females as well as the ability to produce offspring and to nurse them.

In the following text, endpoints for fertility and developmental toxicity are explained based on the description provided in the CLP Regulation. In practical terms, reproductive toxicity is characterised by multiple diverse endpoints, which relate to impairment of male and female reproductive functions or capacity (fertility), the induction of non-heritable harmful effects on the progeny (developmental toxicity), and effects on or *via* lactation.

Adverse effects on sexual function and fertility include any effect of a substance that has the potential to interfere with sexual function and fertility. This includes, but is not limited to, alterations to the female and male reproductive system, adverse effects on onset of puberty, gamete production and transport, reproductive (oestrus) cycle normality, sexual behaviour, fertility, gestation length, parturition, pregnancy outcomes, premature reproductive senescence, or modifications in other functions that are dependent on the integrity of the reproductive system.

Developmental toxicity includes, in its widest sense, any effect interfering with normal development of the organism, before or after birth and resulting from exposure of either parent prior to conception, or exposure of the developing organism during prenatal development, or postnatal development, to the time of sexual maturation – thus generally speaking, these effects can be manifested at any point in the life span of the organism. However, it is considered that classification under the heading of developmental toxicity is primarily intended to provide a hazard warning for pregnant women, and for men and women of reproductive capacity. The major manifestations of developmental toxicity include (1) death of the developing organism, (2) structural abnormality, (3) altered growth, and (4) functional deficiency.[115]

This guidance provides advice on how the registrant can address the reproductive toxicity of the substance and how the information requirements of REACH can be met, thereby providing data on the hazardous properties that can be used for classification purposes and in the risk assessment.

R.7.6.2 Information requirements and testing approaches for reproductive toxicity

Article 10 of REACH specifies the information that is to be submitted for general registration purposes. This information includes minimum information requirements on physicochemical, toxicological and ecotoxicological properties, which are dependent on the tonnage of the registration (Article 10(a) (vi) and (vii) read with Article 12(1) of REACH).

The standard information requirements for the lowest tonnage level are given in Annex VII of REACH. Whenever a higher tonnage level is reached, the minimum requirements of the corresponding REACH Annex (i.e. the REACH Annex for the higher tonnage level) have to be fulfilled in addition to those in all preceding REACH Annexes (see Annex VI of REACH).

For reproductive toxicity, as for any endpoint, all available information must be collected, including data from literature searches. This should then be evaluated with regard to its reliability and relevance, and whether it fulfils the information requirements and their

[115] As written in 3.7.1.3 and 3.7.1.4 in Annex I to CLP (the definition for developmental toxicity is shortened here).

adaptations (triggers and waivers), as well as its use for the purpose of classification, risk assessment and risk management measures.

R.7.6.2.1 REACH information requirements

To examine effects on reproduction, REACH requires information on fertility and developmental toxicity via the "standard information requirements" which are specified in Column 1 of the respective REACH Annexes.

These standard information requirements are minimum information requirements. If there are concerns ("triggers" or "conditions") further testing might be needed to assure availability of appropriate information for chemical safety assessment (including risk characterisation, classification and labelling and other risk management measures).

The term "triggers" is used here as a general term instead of various other possible terms (such as alert, condition, indication, indication of concern, serious concern, or a particular concern) which are used in the REACH Regulation and some of which are used in this Guidance document as described below. A discussion on the evaluation of triggers is given in Appendix R.7.6–5 of this Guidance. For clarification purposes when reading this Guidance document, the terms are used as follows:

- triggers: general term covering all other terms describing findings/conditions which raise concerns;
- alerts: previous term used in this guidance; means the same as triggers but may also include aspects relating to waiving;
- conditions: a specific term used e.g. in REACH Annex IX/X for triggering the extension of Cohort 1B, and which includes aspects which are not findings.

Certain specific adaptation rules described in Column 2 for reproductive toxicity specify when further testing is needed or may be needed at that tonnage level.

REACH information requirements can also be fulfilled by adaptations that reduce the requirement for testing. Adaptation possibilities are either specified in Column 2 of the information requirement or in REACH Annex XI.

An approach on how to fulfil the information requirements is presented in Section R.7.6.2.3 "*Adaptation and testing approaches*" of this Guidance.

The information requirements specified in Column 1 (standard information requirements) are generally cumulative with increasing tonnage levels. Column 2 adaptations are linked with the corresponding Column 1 requirement in the respective REACH Annex and should be considered together with the Column 1 requirement. For reproductive toxicity the standard information requirements (Column 1) combined with specific Column 2 adaptations that require different or further testing are as follows:

REACH Annex VIII (applicable for any registration of 10 tonnes or more per year)

- *Screening for reproductive/developmental toxicity*[116]*, one species (OECD TGs 421 or 422*[117]*) if there is no evidence from available information on structurally related substances, from (Q)SAR estimates or from in vitro methods that the substance may be a developmental toxicant;*

If there are serious concerns about the potential for adverse effects on fertility or development, the registrant may propose:

an extended one-generation reproductive toxicity study (B.56 of the Commission Regulation on test methods as specified in Article 13(3) or OECD TG 443) if there are serious concerns about the potential for adverse effects on fertility or peri-postnatal development;

or

a prenatal developmental toxicity study (B.31 of the Commission Regulation on test methods as specified in Article 13(3) or OECD TG 414) if there are serious concerns about the potential for adverse effects on prenatal development[118];

REACH Annex IX (applicable for any registration of 100 tonnes or more per year)

- *Prenatal developmental toxicity study, one species, most appropriate route of administration, having regard to the likely route of human exposure*[118] *(B.31 of the Commission Regulation on test methods as specified in Article 13(3) or OECD TG 414);*

and if Column 2 of REACH Annex IX, Section 8.7.2 applies for a second species:

- *Prenatal developmental toxicity study, second species (B.31 of the Commission Regulation on test methods as specified in Article 13(3) or OECD TG 414);*
- *Extended one-generation reproductive toxicity study (B.56 of the Commission Regulation on test methods as specified in Article 13(3) or OECD 443), basic test design (cohorts 1A and 1B without extension to include a F2 generation), one species, most appropriate route of administration, having regard to the likely route of human exposure*[12]*, if the available repeated dose toxicity studies (e.g. 28-day or 90-day studies, OECD TGs 421 or 422 screening studies) indicate adverse effects on reproductive organs or tissues or reveal other concerns in relation with reproductive toxicity.*

see REACH Annex IX, Section 8.7.3, Column 2 for the triggers (conditions) when to extend the Cohort 1B to mate the F1 animals and produce the F2 generation, and the triggers (conditions) when to include the Cohorts 2A/2B and/or Cohort 3. For further information on the study design see Appendix R.7.6–2 of this Guidance.

and if Column 2 of REACH Annex IX, Section 8.7.3 applies for a second species/strain:

- *Extended one-generation reproductive toxicity study on a second strain or a second species (exceptional cases only).*

It should be noted that regarding the requirement of a second species, the EU B.56, OECD TG 443 prefers the rat and notes that if another species is to be used, justification should

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116 Later referred also as a screening study.

117 To date there are no corresponding EU test methods available.

118 It is strongly recommended that the registrant considers conducting a screening study in addition to the prenatal developmental toxicity study to cover the fertility and early peri/post natal development if an extended one-generation reproductive toxicity study is not conducted.

be given and appropriate modifications to the protocol will be necessary. There is currently (at the time of publication July 2015), still very limited experience of the protocol and only in rats. This will of course change in the future and registrants should check for new protocols and updates. It is stated in the OECD TG 443 paragraph 9 that "*When a sufficient number of studies are available to ascertain the impact of this new study design, the Test Guideline will be reviewed and if necessary revised in light of experience gained*."

REACH Annex X (applicable for any registration of 1000 tonnes or more per year)

- *Developmental toxicity study, one* [additional] *species, most appropriate route of administration, having regard to the likely route of human exposure (OECD TG 414);*
- *Extended one-generation reproductive toxicity study (B.56 of the Commission Regulation on test methods as specified in Article 13(3) or OECD 443), basic test design (cohorts 1A and 1B without extension to include a F2 generation), one species, most appropriate route of administration, having regard to the likely route of human exposure, unless already provided as part of REACH Annex IX requirements.*

see REACH Annex X, Section 8.7.3, Column 2 for the triggers (conditions) when to extend the Cohort 1B to mate the F1 animals and produce the F2 generation, and the conditions when to include the Cohorts 2A and 2B and/or Cohort 3. For further information on the study design see Appendix R.7.6–2 of this Guidance.

and if Column 2 of REACH Annex IX, Section 8.7.3. applies for a second species/strain:

- *Extended one-generation reproductive toxicity study on a second strain or a second species, in exceptional cases if not already provided as part of REACH Annex IX requirements. (for further explanation see REACH Annex IX above).*

A simplified summary of the information requirements for reproductive toxicity is presented in the following Table R.7.6–1. The standard information requirements of REACH Annexes VIII to X, Section 8.7 Column 1 are indicated, combined with specific Column 2 adaptations that require different or further testing.

Table R.7.6–1 Summary of information requirements for reproductive toxicity in REACH (Annexes VII to X).

Study	Annex VII (<10 t/yr)	Annex VIII (≥10 t/yr)	Annex IX (≥100 t/yr)	Annex X (≥1000 t/yr)
Screening test for reproductive /developmental toxicity (OECD TGs 421 or 422)		Required. If a prenatal developmental toxicity study is available or proposed, it is strongly recommended to consider conducting a screening study in addition to the prenatal developmental toxicity[1] study. If an extended one-generation reproductive toxicity study is available or is proposed, a screening study may not need to be conducted.	Strongly recommended if no higher tier study (such as OECD TG 443) is/will be available to address fertility and peri/post natal development	(a higher tier study is required)
Prenatal developmental toxicity study (EU B.31, OECD TG 414)		May be proposed in cases of serious concern[2] for prenatal developmental toxicity instead of the screening study.	Required in <u>one</u> species; second species may be triggered[3]	Required in <u>two</u> species
Extended one-generation reproductive toxicity study (EU B.56, OECD TG 443)[4]		May be proposed in cases of serious concern for fertility instead of the screening study[2]	Required in one species if triggered[5]; second species/strain may be triggered in exceptional cases	Required in one species unless already conducted at previous Annex level; second species/strain may be triggered in exceptional cases

NOTES for Table R.7.6-1

[1] See discussion at Stage 4.3 (i) Reproduction/developmental toxicity screening test under Section R.7.6.2.3.2 of this Guidance.

[2] Column 1 and Column 2 provisions at REACH Annex VIII, 8.7.1 need to be considered together. Serious concern reflects a high likelihood for adverse effects on reproductive health.

[3] For discussion on triggers see Stage 4.4 (ii), Prenatal developmental toxicity study under Section R.7.6.2.3.2 of this Guidance.

[4] Basic study design addressing fertility, and developmental toxicity effects manifested after birth, with Cohort 1A and Cohort 1B without extension of Cohort 1B, see Stage 4.4 (iii) and Stage 4.5 (ii) Extended one-generation reproductive toxicity study of this Guidance under Section R.7.6.2.3.2 for an overview and Appendix R.7.6–2 and Appendix R.7.6–3 for details and when the study needs to be expanded.

[5] For description of triggers see Stage 4.4 (iii), Extended one-generation reproductive toxicity study under Section R.7.6.2.3.2 of this Guidance.

R.7.6.2.2 Key objectives and information produced by the test methods referred to in REACH

Key objectives and information produced by the test methods referred to in the REACH Regulation for reproductive toxicity are explained in short below in the text and in Table R.7.6–2. More information on how these studies are to be used in a REACH context and important aspects to consider during planning and evaluation are described in Section R.7.6.4.2 of this Guidance.

REACH Annex IX and REACH Annex X level studies and other studies considered not to be screening level studies, require a testing proposal.

R.7.6.2.2.1 Reproduction/Developmental Toxicity Screening Test

The purpose of the reproduction/developmental toxicity screening tests (OECD TGs 421 and 422) is to provide initial information of the effects on male and female reproductive performance such as gonadal function, mating behaviour, conception and parturition and histopathological information on reproductive organs. Initial information on the offspring is limited to mortality, abnormal behaviour and body weight of pups after birth, a macroscopic examination and additional parameters for endocrine disrupting modes of action as given in the revised TGs (2015)[119]. These screening tests are not meant to provide complete information on all aspects of reproduction and development.

R.7.6.2.2.2 Prenatal developmental toxicity study

The prenatal developmental toxicity study (EU B.31, OECD TG 414) provides a focused evaluation of potential effects following prenatal exposure, although only effects that are manifested before birth can be detected. More specifically, this study is designed to provide information on substance-induced effects on growth and survival of the foetuses, and increased incidences in external, skeletal and soft tissue malformations and variations in foetuses.

R.7.6.2.2.3 Extended one-generation reproductive toxicity study

The extended one-generation reproductive toxicity study (EOGRTS, EU B.56, OECD TG 443) allows evaluation of effects of the test substance on the integrity and performance of the adult male and female reproductive system, prenatal effects manifested postnatally and postnatal effects of substances on development as well as a thorough evaluation of systemic toxicity in pregnant and lactating females and young and adult offspring. The study also includes certain parameters for endocrine disrupting modes of action. The extended one-generation reproductive toxicity study is a modular study design with various investigational options.

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[119] OECD TGs 421 and 422 are in the process of being revised: adoption and publication is expected by the end of 2015.

The basic study design, which is the standard information requirement at REACH Annexes IX and X[120], focuses on evaluation of the fertility of parental animals (F0 animals) and of defined parameters on postnatal development of F1 animals until adulthood (see the test method, EU B.56, OECD TG 443). The basic study design does not include mating of F1 animals (extension of Cohort 1B) or cohorts for developmental neurotoxicity (Cohorts 2A and 2B) or developmental immunotoxicity (Cohort 3). Conditions for triggering extension of Cohort 1B and Cohorts 2 and 3 are adaptations to the standard information requirement, and must be proposed by the registrant if the triggers (conditions) described in Column 2 are met. A check list for information that should be presented in the dossier in order to establish the existence or the nonexistence of the conditions and triggers specifying the study design for an extended one-generation reproductive toxicity study regarding the extension of Cohort 1B, inclusion of Cohort 2 and/or Cohort 3 is provided in Appendix R.7.6–1 of this Guidance. More detailed information and examples of triggers and conditions for extension of Cohort 1B and the need to include Cohort 2 and/or Cohort 3, are presented in Appendix R.7.6–2 of this Guidance.

The focus of the study in the REACH Annexes is on fertility[121], which should be considered in the study design of the extended one-generation reproductive toxicity study. Thus, as a starting point, a ten-week premating exposure duration and a highest dose level with the aim to induce some toxicity for all variant study designs of an extended one-generation reproductive toxicity study should be proposed. However based on substance specific justifications the premating exposure duration may be shorter than ten weeks but should not be shorter than two weeks (see Appendix R.7.6–3 of this Guidance). Regarding the highest dose level, it is important to ensure that toxicity in both female and male animals is considered to ensure that reproductive toxicity in either gender is not overlooked.

The extension of the Cohort 1B (mating of the Cohort 1B animals to produce the F2 generation) provides information on the fertility of the offspring, (i.e. the F1 generation), which has been exposed already during primordial germ cell and germ line formation, pre-implantation, *in utero* and postnatal periods. The fertility of Cohort 1B animals, if mated, is evaluated after exposure of full spermatogenesis.

Cohorts 2A and 2B provide information on developmental neurotoxicity and Cohort 3 on developmental immunotoxicity; this information is not covered by any other study within REACH requirements, but might be useful for further hazard and risk assessment.

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120 Recital (6) of Commission Regulation (EU) 2015/282 of 20 February 2015 amending Annexes VIII, IX and X to Regulation (EC) No 1907/2006 of the European Parliament and of the Council on the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) as regards the Extended one-generation reproductive toxicity study: "*The standard information requirement in Annexes IX and X to Regulation (EC) No 1907/2006 should be limited to the basic configuration of EOGRTS. Nevertheless, in certain specific cases, where justified, the registrant should be able to propose and the European Chemicals Agency (ECHA) should be able to request the performance of the F2 generation, as well as the DNT and DIT cohorts.".*

121 Recital (7) of Commission Regulation (EU) 2015/282 of 20 February 2015 amending Annexes VIII, IX and X to Regulation (EC) No 1907/2006 of the European Parliament and of the Council on the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) as regards the Extended one-generation reproductive toxicity study: "*It should be ensured that the reproductive toxicity study carried-out under point 8.7.3 of Annexes IX and X to Regulation (EC) No 1907/2006 will allow adequate assessment of possible effects on fertility. The premating exposure duration and dose selection should be appropriate to meet risk assessment and classification and labelling purposes as required by Regulation (EC) No 1907/2006 and Regulation (EC) No 1272/2008 of the European Parliament and of the Council.*"

Table R.7.6–2 Overview of *in vivo* EU test methods and OECD test guidelines for reproductive toxicity referred to in REACH

Test	Design	Focus of examination
Reproduction/ developmental toxicity screening test (OECD TGs 421 and 422)	Exposure from 2 weeks prior to mating (P) until a specified post-natal day (F1) 3 dose levels plus control Preferred species rat Preferred route oral[1] N = 10 mating pairs per dose group	*Parental (P) generation:* Growth, survival, fertility (limited) Pregnancy length and litter size Histopathology and weight of reproductive organs Histopathology and weight of major non-reproductive organs (OECD TG 422 only) *Offspring (F1):* Growth and survival until a specified post-natal day Certain parameters for endocrine modes of action.[122]
Prenatal developmental toxicity study (EU B.31, OECD TG 414)	Maternal exposure at least from implantation to one or two days before expected delivery 3 dose levels plus control Preferred species rat and rabbit Preferred route oral[1] N = 20 pregnant females per dose group	*Maternal animals:* Growth, survival, (effects on implantation only if dosing is started before implantation), maintenance of pregnancy *Offspring:* Resorptions, foetal deaths foetal growth Morphological variations and malformations (external, skeletal and visceral)
Extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) REACH requires a "basic study design" with a focus on fertility and defines specific conditions for the extension of Cohort 1B and/or inclusion of Cohorts 2A and 2B and/or Cohort 3 (see Section R.7.6.4.2.3 and Appendix R.7.6–2 of this Guidance)	Exposure of 10 weeks prior to mating[2] (P) until post-natal day 90-120 (Cohorts 1A and 1B). If the extension of Cohort 1B is triggered, then until post-natal day 4 or 21 (F2)[3]. 3 dose levels plus control; highest dose level must be chosen with the aim to induce some toxicity. Preferred species rat Preferred route oral[1] N = sufficient mating pairs to produce 20 pregnant animals per dose group (P generation)	*Parental (P) generation:* Growth, survival, fertility Oestrus cyclicity and sperm quality Pregnancy length and litter size Histopathology and weight of reproductive and non-reproductive organs Haematology and clinical chemistry *Offspring (F1):* Growth, survival and sexual maturation Histopathology and weight of reproductive and non-reproductive organs (Cohort 1A) Weight of reproductive organs and optional histopathology (Cohort 1B) Haematology and clinical chemistry

[122] OECD TGs 421 and 422 are in the process of being revised: adoption and publication is expected by the end of 2015.

	N = 20 mating pairs (extension of Cohort 1B, if triggered) N = 10 males and 10 females per dose group (Cohorts 2A, 2B and 3, if triggered)	Fertility of F1 animals to produce F2 generation (extension of Cohort 1B) under certain conditions Developmental neurotoxicity (Cohorts 2A and 2B or a separate study) in cases of a particular concern Developmental immunotoxicity (Cohort 3 or a separate study) in cases of a particular concern Certain parameters for endocrine modes of action

NOTES for Table R.7.6-2

[1] See Stage 4.1 (iv) for discussion on route of administration (Section R.7.6.2.3.2 of this Guidance).

[2] Unless data to support a shorter pre-mating period (see discussion in Appendix R.7.6–3 of this Guidance).

[3] According to the test method EU B.56 (OECD TG 443) the F2 generation may be terminated on postnatal day 4 or 21. For further details see Section R.7.6.4.2.3 of this Guidance, under "Further aspects".

R.7.6.2.3 Adaptation and testing approaches

R.7.6.2.3.1 Overview

This section describes how to use testing approaches and adaptations to achieve the core objectives of REACH (to fulfil information requirements for adequate risk assessment and classification and labelling purposes) with effective use of the gathered information and for designing potential actions needed to fulfil information requirements and to ensure the safe use of substances.

While Column 1 describes the standard information requirements, Column 2 sets certain rules if further or different information is triggered or if information may be omitted thus, Column 2 specific adaptation rules should be considered together with Column 1 standard information requirements. Adaptation may mean further or less information needs than specified in Column 1. If, where the specific adaptation rules in Column 2 or general adaptation rules in REACH Annex XI are not met, the standard information requirements must be fulfilled.

The Registrant is guided in a step-by-step tiered manner on how to meet the information requirements within the production tonnage and influenced by triggers (or conditions). These may increase the need for information or conditions which may allow adaptation of standard information requirements by means of replacing, omitting or adapting in another way. Adaptations of information requirements always need to be clearly stated and supported by adequate justification demonstrating the fulfilment of applicable conditions established by REACH.

As an initial step, all available information relevant to reproductive toxicity must be collected for substances manufactured or imported at tonnage levels ≥1 t/y (REACH Annexes VII-X)(see REACH Annex VI, Step 1). Information from literature may assist identifying the presence or absence of hazardous properties of the substance. In addition, information on exposure, uses and risk management measures should be collected. This information needs to be evaluated with regard to relevance and reliability and to decide if it is adequate for the purposes of risk assessment and classification for reproductive toxicity, including a comparison with the criteria for classification (Annex I of CLP); (see also the *Guidance on the Application of the CLP criteria* and *Guidance on IR&CSA* Chapter R.3 on Information gathering and Chapter R.4 on Evaluation of available information). Considering all the information together, the registrant will be able to determine the need to generate further information in order to fulfil the information requirements.

Consistent with the information requirements defined within REACH Annexes VII to X, testing for reproductive toxicity is not required as a standard approach for registrations of chemicals for the manufacture or import at tonnage levels below 10 tonnes per year (REACH Annex VII). At higher production volumes (i.e. ≥10 t/y, ≥100 t/y or ≥1000 t/y), standard information requirements are staggered according to tonnage levels of the registrations. Flexibility to adopt the most appropriate testing regime for any single substance is maintained by using adaptation rules provided by Column 2 and REACH Annex XI. The adaptation rules are the key components of the testing approaches.

However, regardless of tonnage level, before any testing is carried out careful consideration by the registrants of the following is required: all the available toxicological data, the classification for reproductive toxicity, carcinogenicity and germ cell mutagenicity (EU harmonised or self-classification), human exposure characteristics and current risk management procedures; these are necessary to ascertain whether the information requirements can already be met (see the *Guidance on IR&CSA* Chapter R.5 on *Adaptation of information requirements*). If it is concluded that testing is required in order to fulfil the information requirements, for reasons such as triggers, data gaps which cannot be adapted (for the purpose of classification and/or risk assessment), or increases in production volumes resulting in an REACH Annex upgrade. A series of decision points are defined and described below to help shape the scope of an appropriate testing programme. The REACH approach provides a four-stage process for clear decision-making, relevant for all tonnage levels.

Stage 1: Consider hazardous CMR properties meeting the classification criteria to Category 1A or 1B to decide on the need for further reproductive toxicity testing. Based on Column 2 adaptation of Section 8.7 in REACH Annexes further information on reproductive toxicity may be omitted in certain conditions described in Column 2. Therefore, dependent on the outcome of this analysis, it is possible that some chemicals may not progress beyond Stage 1.

Stage 2: Clarify the standard information requirements relevant for manufactured/imported tonnage level of a single registrant or a SIEF[123].

Stage 3: Evaluate the available toxicology database and consider reproductive toxicity findings and conditions that may serve as triggers or allow omitting further studies. This evaluation should also consider information from substances with a similar structure or causing toxicity via similar mechanisms/modes of action. The aim of this stage is to ensure that the applicable REACH information requirements are identified and to determine the scope of the reproductive toxicity testing necessary to adequately clarify the reproductive toxicity properties. Following this review in conjunction with the analysis in Stage 1 or if sufficient data for risk assessment/risk management and classification purposes are available allowing adaption based on Column 2 or REACH Annex XI adaptation rules, it is possible that no further testing may be necessary.

If the specific adaptation rules in Column 2 or general adaptation rules in REACH Annex XI are not met, the standard information requirements must be fulfilled. Thus, any scientific or other substance-specific justifications for adaptation must follow Column 2 or REACH Annex XI adaptation rules.

Stage 4: Plan and conduct a screening study or plan and propose a prenatal developmental toxicity study or an extended one-generation reproductive toxicity study or specific other studies in exceptional cases. In accordance with Article 12.1d-e/Article 22.1h of REACH, a testing proposal must be submitted to ECHA.

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[123] SIEF is a substance information exchange forum.

R.7.6.2.3.2 Procedure for adaptations and testing approaches

Collection of data

At all REACH Annex levels, the available information from human, animal and non-animal studies and testing approaches need to be collected, including data from literature searches which needs to be evaluated and documented (see REACH Annex I, Step 1 of REACH).

Stage 1: Genotoxic carcinogenicity, germ cell mutagenicity and reproductive toxicity (CMR- properties) to be considered before deciding whether any testing for reproductive toxicity potential is required (relevant for <u>all</u> tonnage levels)

If the answer at the Stage 1.1 and/or Stage 1.2 is yes, i.e. the substance has been already classified to Category 1 for any of the CMR property (as described below), no further testing for reproductive toxicity may be needed if the conditions are fulfilled and appropriate risk management measures are in place.

Stage 1.1

Has the substance already been classified[124] for effects on sexual function and fertility *and* developmental toxicity (Reproductive toxicity Category 1A or 1B (H360FD))?

If the answer is no, proceed to Stage 1.2: if the answer is yes and the available data are adequate to support a robust risk assessment, then no further testing may be necessary. However, if the substance is classified for fertility only, further testing for developmental toxicity must be considered and if the substance is classified for developmental toxicity only, further testing for fertility must be considered; then proceed to Stage 2 via Stage 1.2. If the available data are not adequate to support a robust risk assessment then proceed to Stage 2.

Stage 1.2

Is the substance known to be[125] a genotoxic carcinogen (Carcinogenicity Category 1A and at least Germ cell mutagenicity Category 2; or Carcinogenicity Category 1B and at least Germ cell mutagenicity Category 2) or as a germ cell mutagen (Germ cell mutagenicity Category 1A or 1B) and appropriate risk management measures are implemented?

If the answer is no, proceed to Stage 2. If the answer is yes, it is important to establish that appropriate risk management measures addressing potential carcinogenicity, genotoxicity and reproductive toxicity have been implemented and therefore further specific testing for reproductive and/or developmental toxicity will not be necessary.

Stage 2: Clarify the standard information requirements

At this stage it is necessary to understand what the standard information requirements are at the tonnage level relevant to the registrant. The registrant must fulfil the standard information requirements unless the Column 2 or REACH Annex XI adaptions rules are met to omit the study. In addition to standard information requirements presented in Column 1, Column 2 adaptation rules may indicate triggers (or conditions) for further studies or if certain study design must be proposed.

Stage 3: Conduct a detailed review of the available relevant toxicological data to identify conditions to adapt standard information requirements for reproductive toxicity

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[124] Harmonised classification or self-classification meeting the classification criteria.

[125] Harmonised classification or self-classification meeting the classification criteria.

At Stage 3, the available relevant data is examined to verify if any of the adaptations rules beyond "CMR classification adaptations" explained at Stage 1 are met. Adaptation rules may allow omitting the study or indicate when further information may be needed or must be proposed.

Before any testing is conducted, a thorough data review should be conducted.

Following the adaptation based on CMR classification considered in Stage 1, further general adaptation possibilities of REACH Annex XI and specific adaptation possibilities for omitting the testing provided in Column 2 of the REACH Annexes should be explored. These adaptation rules are described in Stage 3.1 in Appendix R.7.6–4 of this Guidance. These adaptation rules apply to substances for which standard information requirements apply because they passed the Stage 1.

It is important to consider both Column 2 and REACH Annex XI adaptation possibilities because new tests on vertebrates must only be conducted or proposed as a last resort when all other data sources have been exhausted (REACH Annex VI, Step 4).

If sufficient data are available to permit an adaptation according to Column 2 and/or REACH Annex XI rules, then no further testing is required. If the rules for adaptation according to Column 2 or REACH Annex XI are not met and there is a data gap, then the testing strategy for reproductive and/or developmental toxicity in Stage 4 should be followed.

Standard information requirements are described in Column 1 at each REACH Annex. At REACH Annex IX, if there are triggers for reproductive toxicity (fertility and postnatal development) an extended one-generation reproductive toxicity study must be proposed. For definition of triggers and how to evaluate them, see Appendix R.7.6–5 of this Guidance. The examples for triggers for an extended one-generation reproductive toxicity study at REACH Annex IX are described in this Section, under Stage 4.4 (iv), extended one-generation reproductive toxicity study.

If the data are insufficient, which study (or studies) is most appropriate? This decision must take account of both the tonnage-related standard information requirements, the nature of the trigger(s) and total assessment of data.

REACH standard information requirements are minimum information requirements and triggers for reproductive toxicity may indicate a need for further information. Where there is an information gap that needs to be filled, new data must be generated (REACH Annexes VII and VIII) or a testing approach must be proposed (REACH Annexes IX and X). Note that other data sources need to be explored and new tests on vertebrates must only be conducted or proposed as a last resort when all other data sources have been exhausted (REACH Annex VI, Step 4). Whether the registrant must or should or may propose/conduct further information beyond the standard information requirements depends on the REACH Annex level and the provisions in Column 2 and any further concerns. These are further explained at Stage 3.2 and Appendix R.7.6–5 of this Guidance.

Stage 3.1 Substances for which the standard information requirements apply after Stage 1 – options for adaptation rules which may apply instead of conducting new studies

These are substances which are not classified as Category 1 for CMR properties as described in Stage 1 (i.e. are not genotoxic Category 1 carcinogens, germ cell Category 1 mutagens or Category 1 reproductive toxicants (fertility and development)). See Appendix R.7.6–4 of this Guidance for details of adaptation possibilities for these substances. In Appendix R.7.6–4, Stages 3.1.1-3.1.7 describe REACH Annex XI adaptations based on:

1) existing information from non-GLP or test methods not referred in the test method regulation;

2) existing historical human data;

3) existing information in a *Weight-of-Evidence* approach;

4) non-animal approaches such as QSAR approaches and *in vitro* methods;

5) grouping and read across;

6) technical reasons, and substance-tailored exposure driven testing.

Stage 3.1.8 describes adaptations based on Column 2 rules others than based on CMR classification described at Stage 1.

Stage 3.2 Substances for which there are triggers for further information needs beyond the standard information requirements (Column 1)

Whereas Column 1 describes the standard information requirements (and triggers for those), Column 2 includes triggers for further information needs (in addition to provision to omit studies which are described at Stage 3.1.8 in Appendix R.7.6–4 of this Guidance).

Column 2 triggers may have various levels of requirements/consequences:

1) the registrant must act;

2) the registrant should act;

3) the registrant may act.

The consequence level depends on the wording in Column 2. If there is further concern on reproductive toxicity beyond the information requirements (Column 1 and 2 provisions), it is the responsibility of the registrant to consider how to address the concern to ensure the safe use of that substance. The various triggers related to reproductive toxicity and how to evaluate them are described in Appendix R.7.6–5 of this Guidance, Evaluation of Triggers, giving further information needs beyond the standard information requirements.

Stage 4. Reproductive toxicity tests triggered by tonnage level or by findings/conditions which raise concerns for further studies identified in Stages 1-3

Stage 4.1 Preliminary considerations

(i) Introduction

It has to be noted that if studies listed in REACH Annexes IX and X like the prenatal developmental toxicity study or the extended one-generation reproductive toxicity study are intended to be performed, a testing proposal must be submitted to ECHA. Furthermore, before the result from a study for which a testing proposal is submitted to ECHA will be available, risk management measures have to be put in place, recorded in the chemical safety report and recommended to downstream users according to REACH Annex I, 0.5.

A brief description of the protocols for the studies listed in REACH Annexes is presented at Stages 4.2, 4.3 and 4.4 according to registration tonnage levels. When planning any reproductive toxicity studies, considerations such as the properties of the substance, dose levels, vehicle, adequate study design, route and animal species, are needed. Some of these considerations which are especially relevant for reproductive toxicity testing are presented below.

(ii) Range-finding studies

It is recommended that the dose range-finding studies are reported together with the main studies (in IUCLID) to provide sufficient information and justification for the doses selected for testing. The findings from a range-finding study may also support the interpretation of the results from the main study.

(iii) Selection of vehicle

Most of the test methods guide on selection of vehicle if that is needed. For use of all other vehicles except for water a justification is needed and has to be documented. The vehicle should not cause any adverse effects itself as that may interfere with the interpretation of the results and may invalidate the study. Also, the vehicle must not react with the substance or interfere with toxicokinetics of the substance or affect significantly the nutritional status of the animals. The control group should receive the same vehicle and at the same dosing volume as the treated groups.

(iv) Route of administration for reproductive toxicity studies

REACH specifies that the reproductive toxicity studies should be conducted via the "*most appropriate route of administration, having regard to the likely route of human exposure*". "*Likely routes of human exposure*" within REACH are oral, inhalation and dermal. The selection of the "most appropriate route of administration" focuses on identification of hazards (see the Introduction to this Guidance, R7a and sub-section "Selection of the appropriate route of administration for toxicity testing", under R.7.2 Human health properties or hazards) and depends on the most appropriate route for identification of the intrinsic properties of the substance for reproductive hazard.

According to the test methods for reproductive toxicity which focus on the detection of reproductive hazards, the oral route (gavage, in diet, or in drinking water) is the "default" route, except for gases. For the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) dietary administration may be an appropriate route to model human exposure. If another route of administration other than oral is used, the registrant should provide justification and reasoning for its selection. In practice, testing via the oral route is usually performed with liquids and dusts and testing via inhalation route is usually performed with gases and liquids with very high vapour pressure. Testing via dermal route might be necessary under specific circumstances, for example for substances with high dermal penetration and indications for a specific toxicity following dermal absorption. Dermal application or inhalation route using nose-only administration may need specific considerations to ensure that the administration can be adequately conducted without causing confounding factors, for example, cause additional stress to the pregnant animals. Case-specific deviations from the default approach must be justified, such as in the case of available information on route-specific toxicity or toxicokinetics indicating that the use of oral administration of substance would not be relevant for assessing the human health hazards via inhalation, which would be the main route of exposure.

It is to be noted that corrosive or highly irritating substances should be tested preferentially via the oral route, however it must be noted that in vivo testing with corrosive substances at concentration/dose levels causing corrosivity must be avoided (see REACH Annex VII-X preamble). The vehicle should be chosen to minimise gastrointestinal irritation. For some substances dietary administration may allow adequate dosing without irritation compared with oral gavage dosing. In certain cases, testing of neutral salts of alkaline or acidic substances may be appropriate and allows investigation of intrinsic properties at adequate dose levels. If immediate hydrolysis of a substance occurs, it may be possible to provide information on all the cleavage products. For this read-across approach adequate justification and documentation is needed according to REACH Annex XI, 1.5. For corrosive or irritating vapours or gases for which oral testing is not possible, the highest concentration for inhalation should be chosen carefully to induce some toxicity (or mild irritation).

(v) Selection of species

The most common species used for reproductive toxicity testing is the rat. There is good historical background information for various rat strains which may be used to support the interpretation of the results. The strain selected should have an adequate fecundity and not too high an incidence of spontaneous malformations or any other specific feature that may reduce the adequacy of the strain to study reproductive toxicity of a substance in question. In order to make integrated data interpretation including information from other studies, it is recommended to use the same strain both in reproductive toxicity testing as well as repeated dose toxicity studies.

For prenatal developmental toxicity studies, testing in two species is a standard information requirement for registrations at 1000 or more tonnes per year (and might be triggered at lower tonnage levels). According to the test methods (EU B.31, OECD TG 414), the rat is the preferred rodent species and the rabbit the preferred non-rodent species. The extended one-generation reproductive toxicity study may need to be conducted using a second strain or species in certain exceptional cases. (For details see Stage 4.5 (ii) under Section R.7.6.2.3.2 of this Guidance). The most sensitive species and/or strain should be used as a first species taking into account the human relevancy, if known.

However, in choosing the appropriate species or strain of animal, consideration must be given to the suitability of the species and strain for the test protocol, and the availability of background information on the species and strain for the test protocol. The species/strain selection should be justified if the default species referred to in a test method is not used.

(vi) Dose level selection

Like in repeated dose toxicity studies the highest dose level should be chosen with the aim to induce some toxicity unless limited by physical or chemical properties of the substance (e.g. flammability and explosivity limits). Regarding the highest dose level, it is important to ensure that toxicity in both female and male animals is considered to ensure that reproductive toxicity in either gender is not overlooked. Generally at least three dose levels and a concurrent control must be used, except where a limit test (1000 mg/kg bw/day which is generally referred to as the oral limit dose level) is conducted. Expected human exposure may indicate the need for a higher dose level to be used than a 1000 mg/kg bw/day[126]. The conditions for applicability of a limit test are provided in the individual test methods for reproductive toxicity. For inhalation exposure, OECD Guidance document 39 may be used.

Dose level selection is assisted by the information from existing studies as well as from specific dose range-finding studies that may need to be conducted. Toxicokinetic information may provide reasons to adjust for example, the dosing route and regime. In addition, it should be considered that toxicity and toxicokinetics in pregnant animals may differ to that in non-pregnant animals. This may cause challenges in selecting the highest dose level for the study as at various phases of the study the sensitivity of the animals may differ.

For fertility as well as developmental toxicity it is important to investigate whether these reproductive toxicity effects are considered to be a secondary non-specific consequence of other toxic effects seen, such as, maternal toxicity, which may occur at the same dose level as the reproductive effects. However, in general, all findings on reproductive toxicity should be considered for classification purposes even if they are seen in the presence of parental toxicity. A comparison between the severity of the effects on fertility/development and the severity of other toxicological findings must then be performed[127]. Thus, it is important to get information

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[126] CLP, Annex I, Sections 3.7.2.5.7 – 3.7.2.5.9 state on the limit dose and very high dose levels the following: "There is general agreement about the concept of a limit dose, above which the production of an adverse effect is considered to be outside the criteria which lead to classification, but not regarding the inclusion within the criteria of a specific dose as a limit dose. However, some guidelines for test methods, specify a limit dose, others qualify the limit dose with a statement that higher doses may be necessary if anticipated human exposure is sufficiently high that an adequate margin of exposure is not achieved. Also due to species differences in toxicokinetics, establishing a specific limit dose may not be adequate for situations where humans are more sensitive than the animal model." Section 3.7.2.5.8: "In principle, adverse effects on reproduction seen only at very high dose levels in animal studies (for example doses that induce prostration, severe inappetence, extensive mortality) would not normally lead to classification, unless other information is available, e.g. toxicokinetics information indicating that humans may be more susceptible than animals, to suggest that classification is appropriate. Please also refer to the section on maternal toxicity (3.7.2.4) for further guidance in this area." And section 3.7.2.5.9 continues: "However, specification of an actual 'limit dose' will depend upon test method that has been employed to provide the test results, e.g. in the OECD Test Guideline for repeated dose toxicity studies by oral route, an upper dose of 1000 mg/kg has been recommended as a limit dose, unless expected human response indicates the need for a higher dose level."

[127] See the *Guidance on the Application of the CLP criteria*, i.e. the intro to section 3.7.2.2.1.1 "*Effects to be considered in the presence of marked systemic effects*".

about the reproductive toxicity profile of a substance including the spectrum of reproductive toxicity effects related to different dose levels as well as information to allow evaluation of the potency for reproductive toxicity of a substance. Therefore, the highest dose level should be intended to produce some toxicity to provide adequate information on reproductive toxicity for the purpose of both classification (including categorisation within the Reproductive toxicity hazard class) and risk assessment. For further information and clarification see the CLP criteria for classification (Section 3.7, Annex I of CLP) and Section 3.7 in the *Guidance on the Application of the CLP criteria*.

In reproductive toxicity studies local irritating effects at the site of administration may not allow investigating the reproductive toxicity in relation to systemic toxicity. In addition the irritation may affect the behaviour of the animals confounding the interpretation. Therefore, testing of corrosive or highly irritating substances at dose levels causing corrosivity or irritation must be avoided as far as possible (see REACH Annex VII-X preamble).

Dose level selection (and vehicle used) must be justified and documented to allow independent evaluation of the choice made.

Stage 4.2 Registrations of 1 to 10 tonnes per year (REACH Annex VII)

For substances manufactured or imported at tonnage levels ≥1-<10 t/y (REACH Annex VII) there are no specific standard information requirements for reproductive toxicity. However, the available relevant information needs to be evaluated and the classification for reproductive toxicity should be considered and applied if the classification criteria are met. If no information on reproductive toxicity is available, relevant non-animal approaches like validated *in vitro* tests, (Q)SAR predictions, or other available *in vivo* studies with the substance or with structurally related substances may be used to evaluate if there are triggers for reproductive toxicity. If the available information indicates a concern (trigger) for reproductive toxicity and relevant human exposure occurs, an animal study like the reproduction/developmental toxicity screening test (OECD TGs 421 or 422) should be considered to be performed to address the concern as an option. If an REACH Annex IX or X level study, such as prenatal development toxicity study (EU B.31, OECD TG 414) or extended-one-generation reproductive toxicity study (EU B.56, OECD TG 443) is considered necessary to address the concern, a testing proposal should be submitted to ECHA. A thorough scientific justification on how the concern has been addressed should be adequately documented.

Stage 4.3 Registrations of 10 to 100 tonnes per year (REACH Annexes VII and VIII)

At this tonnage level, progression beyond Stages 1-3, will trigger the reproduction/ developmental toxicity screening test (OECD TG 421) or a combined repeated dose toxicity study with the reproduction/developmental toxicity screening test (OECD TG 422).

(i) Reproduction/developmental toxicity screening test

If a 28-day study (EU B.7, OECD TG 407) is not already available, the conduct of a combined repeated dose toxicity study with the reproduction/developmental toxicity screening test (OECD TG 422) is preferred to the reproduction/developmental toxicity screening test (OECD TG 421). This approach offers the possibility to avoid carrying out a 28-day study, because the OECD TG 422 can at the same time fulfil the information requirement of REACH Annex VIII, 8.7.1 and that of REACH Annex VIII, 8.6.1.

If available information indicates serious concerns[128] (trigger(s)) about the potential of a substance for adverse effects on fertility or development, a screening test (OECD TG 421 or 422; REACH Annex VIII, Section 8.7.1) may not need to be performed. Instead, a testing

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[128] Serious concern reflects a high likelihood for adverse effects on reproductive health.

proposal for either a prenatal developmental toxicity study (EU B.31, OECD TG 414; REACH Annex IX, Section 8.7.2) or an extended one-generation reproductive toxicity study (EU B.56, OECD TG 443; REACH Annex IX, Section 8.7.3) may be submitted to ECHA depending on the type of trigger(s). Trigger(s) indicating serious concerns that the substance may be toxic to reproduction could stem from non-animal approaches[129] or *in vivo* information with the substance under consideration or from structurally related substances. Triggers for fertility (see Appendix R.7.6–5 of this Guidance for discussion on triggers), could also stem for example, from existing repeated dose toxicity studies showing histopathological changes in gonads, and/or effects in sperm parameters. The correct study to be proposed depends on the concern: if there is a concern for hazardous effects on fertility and/or development leading to developmental toxicity effects manifested after birth, an extended one-generation study should be proposed; if there is a concern for hazardous effects on embryonic or foetal development, a prenatal developmental toxicity study should be proposed. However, since the fertility and reproductive performance and developmental toxicity manifested shortly after birth are not assessed in a prenatal developmental toxicity study, it is strongly recommended to also conduct an OECD TGs 421 or 422 screening study as already discussed earlier (a testing proposal is not needed for a screening study). If an extended one-generation reproductive toxicity study is proposed (all various study designs) then it covers all the same parameters, exposure duration and statistical power compared to that of the screening study and thus, an additional screening study is not required.

If a reproduction/developmental toxicity screening test (OECD TGs 421 or 422) for an REACH Annex VIII substance provides no triggers for reproductive and developmental toxicity, then further testing for reproductive toxicity is not required at this tonnage level. Similarly, if a clear and unequivocal reproductive and/or developmental toxicity effect is observed in a screening test which is deemed sufficient to enable a scientifically robust decision on classification and categorisation to 1B for reproductive toxicity and risk assessment, then no further testing beyond the screening test is recommended at this tonnage level.

However, if a screening test (OECD TGs 421 or 422) shows effects which are deemed not sufficient to enable a scientifically robust decision on classification and risk assessment, further studies may be considered. Based on the type of trigger, a testing for either a prenatal developmental toxicity study (REACH Annex IX, Section 8.7.2) or an extended one-generation study (REACH Annex IX, Section 8.7.3) may be proposed. Specifically, if a clear and unequivocal reproductive and/or developmental toxicity effect is observed in a screening test which is deemed sufficient for classification in Category 2 for reproductive toxicity, then this is a serious concern and either a prenatal developmental toxicity study (REACH Annex IX, Section 8.7.2) or an extended one-generation study (REACH Annex IX, Section 8.7.3) may be proposed.

Stage 4.4 Registrations of 100 to 1000 tonnes per year (REACH Annexes VII to IX)

At this tonnage level, progression beyond Stages 1-3 will trigger a prenatal developmental toxicity study in a first species (EU B.31, OECD TG 414) and, if the available repeated dose toxicity studies indicate adverse effects on reproductive organs or tissues or reveal other concerns in relation to reproductive toxicity, will also trigger an extended one-generation reproductive toxicity study (EU B.56, OECD TG 443). For further information on triggers for an extended one-generation toxicity study at REACH Annex IX level, see point (iii) below.

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129 In order to be considered providing "*serious concern*", information from non-animal approaches should be reliable, relevant and from validated studies with appropriate applicability domain (for QSAR models a formal validation process is not required) . Based on case-by-case scientific justification results from non-validated studies and non-guideline tests, it may be acceptable. Generally several information sources may be needed.

If the results from existing studies (prenatal developmental toxicity test or repeated-dose studies) are sufficient to support classification to Category 1B for effects on developmental toxicity and/or sexual function and fertility and the risk assessment, the Column 2 adaptation rules for REACH Annex IX, point 8.7 should be followed. If the classification criteria for sexual function and fertility are met, then further testing for developmental toxicity must be considered and vice versa. For details, see Stage 1.

(i) Reproduction/developmental toxicity screening test

A reproduction/developmental toxicity screening test (OECD TGs 421 or 422) is a standard information requirement at REACH Annex VIII level. Since the Column 1 requirements in the REACH Annexes are cumulative, a screening test should also be available at REACH Annex IX and X level. However, if a prenatal developmental toxicity study, a two-generation reproductive toxicity study or an extended one-generation study is available, the screening study can be omitted based on REACH Annex VIII, Section 8.7.1., Column 2 adaptation rules (at REACH Annex VIII).

Where a screening test is omitted based on a prenatal developmental toxicity study and an extended one-generation reproduction toxicity study is not triggered at REACH Annex IX level, then information on fertility would be limited to evaluation of the reproductive organs after repeated dosing, if those studies are available. Where information from a reproductive toxicity study addressing a fertility endpoint is not available, it is strongly recommended that a screening study is considered to fulfil this endpoint.

(ii) Prenatal developmental toxicity study

A prenatal developmental toxicity study (EU B.31, OECD TG 414), conducted in one species, is a standard data requirement at REACH Annex IX level.

Consideration of existing information and the testing approach is required to select the appropriate species for the prenatal developmental toxicity study (see especially Stage 4.1(v) above). According to the test methods (EU B.31, OECD TG 414), the rat is the preferred rodent species and the rabbit the preferred non-rodent species. Since most of the toxicity studies (e.g. acute, repeated-dose, and toxicokinetic studies) are conducted in the rat, it may be considered that the first prenatal developmental toxicity study should also be conducted in this species. Findings from previous studies may be useful in dose selection, or the identification of additional endpoints for evaluation. In addition, the outcome of the prenatal developmental toxicity study may be helpful in the interpretation of other reproductive toxicity studies, for which the rat is generally the preferred species.

In certain cases the rabbit might be selected as the species for the first prenatal developmental toxicity study. This may be done for example, if the rabbit is considered to be a more sensitive species than the rat for that specific substance. The selection of the species for the prenatal developmental toxicity study should be made taking into account substance-specific aspects. If a species other than the rat and the rabbit is selected as the first or second species, the selection should be justified.

A decision on the need to perform a study on a second species at REACH Annex IX level should be based on the outcome of the first study and all other relevant available data. A study on a second species might be necessary if the available data contain triggers for prenatal developmental toxicity. For example, performance of a prenatal developmental toxicity study in a second species may be justified if developmental effects that are not sufficient to meet classification criteria to Category 1B reproductive toxicant (but maybe sufficient to Category 2 reproductive toxicant) were observed in the prenatal developmental toxicity study with the first species. Further triggers may stem from non-animal approaches, structurally similar substances, mechanisms/modes of action or results from a screening study. However, if there are no triggers and no indication of prenatal developmental toxicity in the first prenatal developmental toxicity study, no study on a second species is necessary at REACH Annex IX level.

If a study on a second species is found to be necessary by the registrant, a testing proposal needs to be submitted. Testing in a second species should be performed in a non-rodent species (rabbit) if the first species was a rodent species (rat) and vice versa. Further considerations on the species selection are provided in Section R.7.6.4.2.2 of this Guidance.

(iii) Extended one-generation reproductive toxicity study

An extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) is required at REACH Annex IX level if the available repeated dose toxicity studies (e.g. 28- or 90-days studies or OECD TGs 421 or 422 screening tests) indicate adverse effects on reproductive organs or tissues or reveal other concerns in relation with reproductive toxicity. Information from non-animal approaches are thus not listed as triggers for this study at REACH Annex IX level in the REACH Annex text. However, if there is a serious concern based on available information from non-animal approaches or structurally analogous substances, the study may be triggered.

Triggers for the study at REACH Annex IX level

A detailed review of the available data is required to identify any reproductive toxicity triggers (see Appendix R.7.6–5 of this Guidance for evaluation and determination of triggers), furthermore, examples of triggers for an extended one-generation reproductive toxicity study at REACH Annex IX level are provided below.

The legal text does not especially specify that the adverse effects should be seen in intact animals, however, it is considered that findings observed in non-intact animals should generally be used as triggers unless there is evidence that the findings would not be relevant for intact animals and/or humans. Experiments with non-intact animals may include animals with removal of an endocrine organ, such as ovary (ovariectomy). Another possibility is hormonal manipulation, for example causing decrease or increase of organ weight. These animal models may be very sensitive to detect a change in hormonal response, however, it should be considered whether the same applies in intact animals.

Examples (not an exhaustive list) of triggers to conduct an extended one-generation reproductive toxicity study at REACH Annex IX level (considered as adverse, and which are in line with other data and not considered secondary to systemic or maternal toxicity) are as follows:

From a screening study or equivalent:

- Changes in reproductive or other endocrine organ weight in intact animals;
- Effects in spermatogenesis or folliculogenesis *in vivo* and/or histopathological findings in reproductive organs and/or accessory sex organs;
- Effects in histopathology of the thyroid;
- Effects on sperm parameters analysis or oestrous cycle;
- Biologically relevant changes in hormone levels *in vivo* (related to reproductive toxicity);
- Reduced mating, fertility or litter size;
- Increased incidence of abortions compared to controls;
- Changes in gestation length;
- Reduced survival of offspring;
- Reduced body weight of offspring independent of litter size;
- Reduced maternal care;
- Changes in anogenital distance unrelated to body weight/size;

- Changes in nipple retention;
- Indication of other endocrine disrupting modes of action related to reproductive toxicity.

From a repeated dose toxicity study:

- Changes in reproductive or other endocrine organ weight in intact animals;
- Effects in spermatogenesis or folliculogenesis *in vivo* and/or histopathological findings in reproductive organs and/or accessory sex organs;
- Effects on sperm parameters analysis or oestrous cycle
- Biologically relevant changes in hormone levels (related to reproductive toxicity)*;*
- Indication of other endocrine disrupting modes of action related to reproductive toxicity.

From in vivo studies from non-intact animals (if the findings are considered relevant for intact animals/humans):

- Changes in reproductive or other endocrine organ weight.
- Indication of other endocrine disrupting modes of action related to reproductive toxicity

Study design for the extended one-generation reproductive toxicity study

If triggers are identified that require performance of an extended one-generation reproductive toxicity study, the appropriate study design as described in Column 1 and 2 and in Recital (7) of Commission Regulation (EU) 2015/282 amending REACH, needs to be defined, justified and documented. Specification is required for 1) length of the premating exposure duration and dose level selection, 2) the need to extend Cohort 1B and termination time for F2 generation, 3) the need to include Cohorts 2A and 2B, and 4) the need to include Cohort 3.

The study design of the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) specified in REACH in Column 1 as a standard information requirement, is the so called "basic" study design and a one-generation study including Cohorts 1A and 1B. Recital (7) of Commission Regulation (EC) 2015/282 amending REACH, states that the extended one-generation reproductive toxicity study should allow adequate assessment of fertility and that premating exposure duration and dose levels should be appropriate to meet the risk assessment and classification and labelling purposes (including categorisation)[130]. The focus of the study in the REACH Annexes is on fertility, which should be considered in the study design of the extended one-generation reproductive toxicity study, thus, as a starting point, a ten-week premating exposure duration and a highest dose level with the aim to induce some toxicity for all variant study designs of an extended one-generation reproductive toxicity study should be proposed. Regarding the highest dose level, it is important to ensure that toxicity in both female and male animals is considered to ensure that reproductive toxicity in either gender is not overlooked. The basic study design, including the premating exposure duration according to Appendix R.7.6–3 of this Guidance, should be proposed by registrants unless the conditions specified in Column 2 are met.

The extension of the Cohort 1B (mating of the Cohort 1B animals to produce the F2 generation) must be proposed by the registrant if the conditions specified in Column 2 are

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[130] Recital (7) of Commission Regulation (EU) 2015/282 of 20 February 2015 amending Annexes VIII, IX and X to Regulation (EC) No 1907/2006 of the European parliament and of the Council on the Registration, evaluation, Authorisation and Restriction of Chemicals (REACH) as regards the Extended one-generation reproductive toxicity study: "*It should be ensured that the reproductive toxicity study carried-out under point 8.7.3 of Annexes IX and X to Regulation (EC) No 1907/2007 will allow adequate assessment of possible effects on fertility. The premating exposure duration and dose selection should be appropriate to meet risk assessment and classification and labelling purposes as required by Regulation (EC) No 1907/2006 and Regulation (EC) No 1272/2008 of the European parliament and of the Council.*"

met. Based on specific triggers for neurotoxicity defined in Column 2, developmental neurotoxicity cohorts (Cohorts 2A and 2B) must be proposed by the registrant. Respectively, based on specific triggers for immunotoxicity defined in Column 2, developmental immunotoxicity cohort (Cohort 3) must be proposed by the registrant.

The registrant may also propose a separate developmental neurotoxicity and/or developmental immunotoxicity study instead of the cohorts for developmental neurotoxicity and/or developmental immunotoxicity.

The conditions specifying the study design are listed in REACH Annex IX, 8.7.3, Column 2 and explained in more detail in Appendix R.7.6–2 of this Guidance and discussed in Section R.7.6.4.2.3 "Extended one-generation reproductive toxicity study" of this Guidance. Appendix R.7.6–1 of this Guidance lists the information that should be considered and, where available, presented in the dossier in order to establish the existence or the nonexistence of the conditions (triggers) specifying the study design. It is the registrant's responsibility to evaluate all the available information and to propose an adaptation of the standard information requirement following conditions described in Column 2 of REACH Annex IX/X, 8.7.3.

The justification of the study design that is most appropriate for evaluation of the reproductive toxicity of a substance must be adequately documented. This documentation must include justifications why the registrant holds the conditions of deviations from the basic study design not to be fulfilled taking into account all the available information.

A study on a second species or strain

REACH Annex IX specific rules for adaptation states that the need to perform an extended one-generation reproductive toxicity study (EU B.56; OECD TG 443) in a second strain or a second species, either at this tonnage level or the next, may be considered, and a decision should be based on the outcome of the first test and any other relevant available data.

It is recognised that the extended one-generation reproductive toxicity study is designed to be conducted in rats and it may be challenging to use other species. Thus, it has been made possible to conduct a second study using another rat strain instead of a second species. The need to conduct the study using a second species or strain will be in exceptional cases only.

A study on a second strain or species might be necessary if the available data contain triggers which have not been addressed in the study on the first species. For example, performance of a study in a second strain or species may be justified if effects were observed in the study with the first species cause further serious concern but are not sufficient to meet classification criteria to Category 1B reproductive toxicant. Further triggers may stem from validated[131] non-animal approaches, reliable and relevant QSAR models with adequate applicability domain, structurally similar substances, modes of action or results from a screening study. However, if there are no triggers and no indication of adverse effects on reproductive toxicity in the first study and other available data, no study on a second species or strain is necessary at REACH Annex IX level.

If a study on a second species or strain is found to be necessary by the registrant, a testing proposal should be submitted.

Stage 4.5 Registrations of 1000 tonnes or more per year (REACH Annexes VII to X)

Progression beyond Stage 1-3 will trigger a prenatal developmental toxicity study (EU B.31, OECD TG 414) on a second species, if not conducted at the previous tonnage level, and an extended one-generation reproductive toxicity study (EU B.56, OECD TG 443), if not already conducted at the previous tonnage level.

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[131] Case-by-case scientific justification must be provided when non-validated or non-guideline methods are used.

(i) Prenatal developmental toxicity study

At REACH Annex X level, a prenatal developmental toxicity study (EU B.31, OECD TG 414) conducted on a second species is a standard information requirement in addition to a prenatal developmental toxicity study in a first species that is required at REACH Annex IX level. Availability of information on two species allows a more comprehensive evaluation of prenatal developmental toxicity. The prenatal developmental toxicity study in a second species can be omitted, if, taking into account the outcome of the first test and all other relevant available data, an adaptation pursuant to REACH Annex X, Section 8.7, Column 2 or pursuant to REACH Annex XI can be justified.

According to the test methods (EU B.31, OECD TG 414), the rat is the preferred rodent species and the rabbit the preferred non-rodent species. Depending on whether the rat or the rabbit is selected as a first species, and/or is already available, the other should be the preferred second species. In certain cases the rabbit might be selected as the species for the first prenatal developmental toxicity study. This may be done for example if the rabbit is considered to be the more sensitive species than the rat for that specific substance. The selection of the species for the prenatal developmental toxicity study should be made taking into account substance-specific aspects. If a species other than the rat and the rabbit is selected as the first or second species, the selection must be justified.

(ii) Extended one-generation reproductive toxicity study

The extended one-generation reproductive toxicity study (EU B.56; OECD TG 443) is a standard information requirement at REACH Annex X level.

<u>Study design for the extended one-generation reproductive toxicity study</u>

The criteria for the study design for the extended one-generation reproductive toxicity study are the same at REACH Annex IX and X levels. Thus, the description of the study design here is identical to that at REACH Annex IX level (Stage 4.4 (iii)).

The appropriate study design as described in Column 1 and 2 and in Recital (7) of Commission Regulation (EU) 2015/282 amending REACH, needs to be defined, justified and documented. Specification is required for 1) length of the premating exposure duration and dose level selection, 2) the need to extend Cohort 1B and termination time for F2 generation, 3) the need to include Cohorts 2A and 2B, and 4) the need to include Cohort 3.

The study design of the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) specified in REACH in Column 1 as a standard information requirement is the so called "basic" study design and a one-generation study including Cohorts 1A and 1B. Recital (7) of Commission Regulation (EC) 2015/282 amending REACH, states that the extended one-generation reproductive toxicity study should allow adequate assessment of fertility and that premating exposure duration and dose levels should be appropriate to meet the risk assessment and classification and labelling purposes[132]. The focus of the study in the REACH

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[132] Recital (7) of Commission Regulation (EU) 2015/282 of 20 February 2015 amending Annexes VIII, IX and X to Regulation (EC) No 1907/2006 of the European parliament and of the Council on the Registration, evaluation, Authorisation and Restriction of Chemicals (REACH) as regards the Extended one-generation reproductive toxicity study: "*It should be ensured that the reproductive toxicity study carried-out under point 8.7.3 of Annexes IX and X to Regulation (EC) No 1907/2007 will allow adequate assessment of possible effects on fertility. The premating exposure duration and dose selection should be appropriate to meet risk assessment and classification and labelling purposes as required by Regulation (EC) No 1907/2006 and Regulation (EC) No 1272/2008 of the European parliament and of the Council.*"

Annexes is on fertility, which should be considered in the study design of the extended one-generation reproductive toxicity study. Thus, as a starting point, a ten -week premating exposure duration and a highest dose level with the aim to induce some toxicity for all variant study designs of an extended one-generation reproductive toxicity study should be proposed. Regarding the highest dose level, it is important to ensure that toxicity in both female and male animals is considered to ensure that reproductive toxicity in either gender is not overlooked. The basic study design, including the premating exposure duration according to Appendix R.7.6–3 of this Guidance, should be proposed by registrants unless the conditions specified in Column 2 are met.

The extension of the Cohort 1B (mating of the Cohort 1B animals to produce the F2 generation) must be proposed by the registrant if the conditions specified in Column 2 are met. Based on specific triggers for neurotoxicity defined in Column 2, developmental neurotoxicity cohorts (Cohorts 2A and 2B) must be proposed by the registrant. Respectively, based on specific triggers for immunotoxicity defined in Column 2, developmental immunotoxicity cohort (Cohort 3) must be proposed by the registrant.

The registrant may also propose a separate developmental neurotoxicity and/or developmental immunotoxicity study instead of the cohorts for developmental neurotoxicity and/or developmental immunotoxicity.

The conditions specifying the study design are listed in REACH Annex X, 8.7.3, Column 2 and are explained in more detail in Appendix R.7.6–2 of this Guidance and discussed in Section R.7.6.4.2.3 "Extended one-generation reproductive toxicity study" of this Guidance. Appendix R.7.6–1 of this Guidance lists the information that should be considered and, where available, presented in the dossier in order to establish the existence or the non-existence of the conditions (triggers) specifying the study design. It is the registrant's responsibility to evaluate all the available information and to propose an adaptation of the standard information requirement following conditions described in Column 2 of REACH Annex IX/X, 8.7.3.

The justification of the study design that is most appropriate for evaluation of the reproductive toxicity of a substance must be adequately documented. This documentation must include justifications why the registrant holds the conditions of deviations from the basic study design not to be fulfilled taking into account all the existing information.

A study on a second species or strain

REACH Annex IX specific rules for adaptation states that the need to perform an extended one-generation reproductive toxicity study (EU B.56; OECD TG 443) in a second strain or a second species, either at REACH Annex IX tonnage level or at REACH Annex X tonnage level, may be considered and a decision should be based on the outcome of the first test and any other relevant available data. It is recognised that the extended one-generation reproductive toxicity study is designed to be conducted in rats and it may be challenging to use other species. Thus, it has been made possible to conduct a second study using another rat strain instead of a second species. The study on a second species or strain is needed in exceptional cases only.

A study on a second strain or species might be necessary if the available data contain triggers which have not been addressed in the study on first species. For example, performance of a study in a second strain or species may be justified if effects were observed in the study with the first species cause further serious concern but are not sufficient to meet classification criteria to Category 1B reproductive toxicant. Further triggers may stem from validated[133] non-animal approaches, reliable and relevant QSAR methods with adequate applicability domain, structurally similar substances, modes of action or results from a screening study. However, if

133 Case-by-case scientific justification must be provided when non-validated or non-guideline methods are used.

there are no triggers and no indication of adverse effects on reproductive toxicity in the first study and other available data, no study on a second species or strain is necessary at REACH Annex X level.

If a study on a second species or strain is found to be necessary by the registrant, a testing proposal must be submitted.

R.7.6.3 Information sources on reproductive toxicity

Information on reproductive toxicity can be obtained from various source categories, which are indicated below. Examples from each source category are provided. Evaluation of this information is described in Section R.7.6.4 of this Guidance. Where *in vivo* testing is required, registrants must follow the EU Directive 2010/63 in selecting the test(s) requiring fewest animals and the least suffering.

R.7.6.3.1 Information on reproductive toxicity from non-animal approaches

Limited information of supportive nature may be inferred from numerous non-animal approaches (tests not using whole animals including embryos and foetuses after a certain developmental stage). For evaluation of the quality of the information, see Section R.7.6.4 of this Guidance where reference to ECHA guidance on evaluation of available information is given (*Guidance on IR&CSA, Chapter R.4 "Evaluation of available information"*):

- physico-chemical characteristics of a substance (distribution, accumulation);
- information on structurally analogue substances and (Q)SAR models;
- *in silico* and *in chemico* models (with adequate applicability domain);
- *in vitro* tests (with relevant concentrations) in reproductive toxicity or relevant modes on action; e.g.:
  - Performance-based test guideline for stably transfected transactivation *in vitro* assays to detect oestrogen receptor agonists (OECD TG 455, updated 2012);
  - BG1Luc Estrogen receptor transactivation test method for identifying oestrogen receptor agonists and antagonists (OECD TG 457);
  - H295R steroidogenesis assay (EU B.57, OECD TG 456);
  - *in vitro* embryotoxicity tests;
  - *in vitro* organ and cell cultures.
- Where possible, well developed and justified reverse toxicokinetic models may be used to support results from *in vitro* tests to estimate exposures needed to achieve bioactive blood concentrations.

Approaches combining various methodologies, e.g. from adverse outcome pathway (AOP) concept (OECD GD 184).

R.7.6.3.2 Information on reproductive toxicity in humans

If human information is available, it must be presented and if possible in the form of a table as stated in REACH Annex I, 1.2.

Information may stem from epidemiological and/or occupational studies, medical records, case studies and accidents. For evaluation of the quality of the information, see Section R.7.6.4 of this Guidance where reference to ECHA guidance on evaluation of available information is given (*Guidance on IR&CSA, Chapter R.4 "Evaluation of available information"*).

R.7.6.3.3 Information on reproductive toxicity from *in vivo* animal studies

Data may be available from a wide variety of animal studies, with standard or non-standard study design, which give different amounts of direct or indirect information on the potential reproductive toxicity of a substance. For evaluation of the quality of the information, see Section R.7.6.4 of this Guidance where reference to ECHA guidance on evaluation of available information is given (*Guidance on IR&CSA*, *Chapter R.4* "*Evaluation of available information*").

In vivo studies referred to in REACH and providing information on reproductive toxicity:

- Extended one-generation reproductive toxicity study (EU B.56, OECD TG 443);
- Two-generation reproductive toxicity study (EU B.35, OECD TG 416);[134]
- Prenatal developmental toxicity study (EU B.31, OECD TG 414).

In vivo studies referred to in REACH and providing preliminary information on reproductive toxicity:

- A reproduction/developmental toxicity screening test (OECD TG 421);[135]
- Combined repeated dose toxicity study with the reproductive/developmental toxicity screening test (OECD TG 422)[136].

Other *in vivo* study on reproductive toxicity with EU and OECD test guidelines:

- One-generation reproductive toxicity study (EU B.34, OECD TG 415).

Repeated dose toxicity studies which may include parameters relevant for reproductive toxicity:

- 28- and 90-day repeated-dose toxicity studies (EU B.7; EU B.10), where relevant parameters are included, for example semen analysis, oestrous cyclicity, organ weights of reproductive organs and accessory sex organs, and/or reproductive organ histopathology.

Short-term *in vivo* tests on endocrine disrupting modes of action in intact or non-intact animals, e.g.:

- Uterotrophic bioassay in rodents: a short-term screening test for oestrogenic properties (EU B.54, OECD TG 440; OECD GD 71 for anti-oestronicity);
- Hershberger bioassay in rats: a short-term screening assay for (anti)androgenic properties (EU B.55, OECD TG 441 and GD 115);
- Studies on juvenile/peripubertal animals.

Other studies which may provide relevant information, e.g.:

- Chernoff/Kavlock tests (see Hardin *et al.,* 1987);
- a modified one-generation study by NTP (National Toxicology Program, U.S. Department of Health and Human Services; http://ntp.niehs.nih.gov/testing/types/mog/index.html)

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[134] Existing two-generation reproductive toxicity studies (EU B.35, OECD TG 416) started before 15 March 2015 fulfil the standard information requirement for Annex IX/X, 8.7.3 but new studies for REACH must be proposed according to an extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) as described in Annex IX/X, 8.7.3.

[135] To date there are no corresponding EU testing methods available.

[136] At the time of publication (July 2015), there are no corresponding EU testing methods available.

- Reproductive Assessment by Continuous Breeding (RACB) protocol (e.g. Chapin and Sloane 1997);
- peri-postnatal studies;
- male or female fertility studies of non-standard design;
- dominant lethal assay (EU B.22, OECD TG 478);
- mechanistic studies;
- toxicokinetic studies (EU B.36, OECD TG 417);
- studies in fish (e.g. Fish Sexual Development Test (OECD TG 234));
- studies in amphibians (e.g. Amphibian Metamorphoses Assay (OECD TG 231) or Larval Amphibian Growth and Development Assay (under development));
- studies in other non-mammalian species.

Studies with focus on developmental neurotoxicity and developmental immunotoxicity:

- developmental neurotoxicity studies (such as EU B.53, OECD TG 426);
- developmental immunotoxicity studies (see Section R.7.6.4.2.7 of this Guidance for references).

R.7.6.4 Evaluation of available information for reproductive toxicity

This section provides information on evaluation of the available data including aspects which influence the study designs. Both non-human (non-animal approaches and *in vivo* animal studies) and human data are considered. Under this section the studies required as standard information requirements are described as well as how to evaluate the conditions described in Column 2 to trigger a study or to adapt the study design. In addition, the evaluation of information from other internationally accepted *in vivo* studies are briefly described.

The generic guidance on the evaluation of available information gathered in the context of REACH Annexes VI-XI is provided in the *Guidance on IR&CSA, Chapter R.4: "Evaluation of available information"*. The information should be evaluated for its completeness and quality for the purpose of REACH to assess whether (see the detailed wording in Chapter R.4):

- It fulfils the information requirements;
- It is appropriate for hazard classification and risk assessment.

The evaluation process of data quality by judging and ranking the available data for its relevance, reliability and adequacy is provided in Chapter R.4. Chapter R.4 applies to all kinds of information; human, animal and non-animal sources and it is also applicable to information for reproductive toxicity endpoint. OECD guidance document 43 may be consulted for aid in the interpretation of reproductive and neurotoxicity results (see OECD GD 106 for histologic evaluation, OECD GD 57 and 207 for thyroid hormone modulation assays, and OECD retrospective performance assay for developmental neurotoxicity, No 89 (OECD, 2008)).

In the present document some additional scientific aspects relevant for reproductive toxicity have been highlighted in context of the relevant information sources.

The main principles for evaluation of non-human information (information from animal studies and non-animal approaches) are presented in REACH Annex I, 1.1 and it must be comprised of:

- Hazard identification for the effect based on all available non-human information;
- Establishment of the quantitative dose (concentration) response (effect) relationship.

Robust study summaries are necessary for key data on reproductive toxicity. If possible the information should be provided in the form of table(s) (see further details in REACH Annex I, 1.1.3.).

R.7.6.4.1 Non-animal data

For reproductive toxicity, a grouping and category approach and weight of evidence adaptation are the best fit-for-purpose tools for non-animal approaches for the time being to adapt the (standard) information requirements for reproductive toxicity. However, appropriate justification and documentation must be provided. In addition, non-animal approaches may be used for prioritisation and screening chemical inventories.

Information on the current developments of *in vitro* tests and methodology can be found on the ECVAM website (http://ihcp.jrc.ec.europa.eu/our_labs/eurl-ecvam) and other international centres for validation of alternative methods. ECHA's website is also updated with new internationally accepted non-animal approaches (http://www.echa.europa.eu/support/oecd-eu-test-guidelines). However, the regulatory acceptance of these studies and approaches to replace the animal testing for reproductive toxicity has not been achieved as they do not provide equivalent information and thus, cannot be used alone for classification and labelling and/or risk assessment. In spite of this, they may serve as elements in categories/read across and weight of evidence adaptation. They may also provide important information on mechanisms and modes of action, or preliminary screening information which can be used in planning further testing.

R.7.6.4.1.1 Physico-chemical properties

It may be possible to infer from the physico-chemical characteristics of a substance whether it is likely to be absorbed following exposure by a particular route and, furthermore, whether it (or an active metabolite) is likely to cross the placental, blood-brain or blood-testes barriers, or be secreted in milk. Information on the physico-chemical properties may contribute to a Column 2 adaptation (e.g. indicate concern on prolonged phase before reaching a steady state which is part of the conditions triggering extension of Cohort 1B in the extended one-generation reproductive toxicity study) or a weight of evidence adaptation according to REACH Annex XI, 1.2.

R.7.6.4.1.2 (Q)SAR

There are a large number of potential targets/mechanisms associated with reproductive toxicity which, on the basis of current knowledge, cannot normally be adequately covered by a battery of QSAR models. In principle QSAR models are potential adaptation possibilities according to REACH Annex XI, 1.3, but they should adequately cover the endpoint in question – all the key aspects/parameters should be covered.

QSAR models are usually trained (developed) to give binary results; the substance is predicted to have or not have a particular property, e.g. developmental toxicity. If the substance is predicted to have that property, the result of a QSAR prediction is considered as positive. Similarly, if the substance is predicted not to have a particular property, the result of the QSAR prediction is considered negative. QSAR approaches are currently not well fitted-for-purpose for reproductive toxicity and consequently no firm recommendations can be made concerning their routine use in a testing strategy in this area. A particular challenge for this endpoint is the complexity and amount of information needed from various functions and parameters to evaluate the effects on reproduction. Not all necessary aspects can be covered by a QSAR prediction. Therefore, a negative result from current QSAR models predicting that the substance has not a particular property, cannot be interpreted as demonstrating the absence of a reproductive hazard unless there is other supporting evidence. Another limitation of QSAR modelling is that dose response information, for example the N(L)OAEL, required for risk assessment is not provided.

However, a positive result from a reliable and relevant QSAR model with an appropriate applicability domain predicting that the substance has a particular property could provide a trigger for further testing beyond the standard information requirement (e.g. one element to trigger the extension of Cohort 1B in an extended one-generation reproductive toxicity study). For evaluation of the triggers see Appendix R.7.6–5 of this Guidance. Due to the limited confidence in this approach such a result would not normally be adequate for making a decision on classification on its own. It may, although not normally used, provide supportive information that can be used when concluding on the appropriate classification (see 3.7.2.5.4, Annex I, CLP).

Provided the applicability domain is appropriate, the results from using QSAR models may be used in a weight of evidence analysis where such data are considered alongside other relevant data (for classification and labelling and as one element for weight of evidence adaptation approach according to REACH Annex XI, 1.2). Also, the results from using QSAR models can be used as supporting evidence when assessing the toxicological properties by read-across in a grouping approach, providing the applicability domain is appropriate. Both positive and negative QSAR modelling prediction results concerning the existence or non-existence of a particular property, respectively, may be of value in supporting a read-across assessment.

R.7.6.4.1.3 *In vitro* data and Adverse Outcome Pathways (AOPs)

The design of alternatives to *in vivo* testing for reproductive toxicity is especially challenging in view of the complexity of the reproductive process and large number of potential targets/mechanisms associated with this broad area of toxicity. In addition, many *in vitro* approaches do not include elements of biotransformation which, in addition, may differ depending on organ.

Currently there are only three officially adopted EU test methods or OECD test guidelines for *in vitro* tests of relevance to modes of action for reproductive toxicity: two measuring oestrogenicity (OECD TG 455 and OECD TG 457) and the other measuring steroidogenesis (EU B.57, OECD TG 456). Most assays under development and international validation are focusing on agonist/antagonistic properties measured by binding and activating or blocking a steroid (or a thyroid) hormone receptor.

Three *in vitro* embryotoxicity tests to predict developmental toxicity have been validated but have not been accepted for regulatory use (Genschow *et al.,* 2002, Piersma *et al.,* 2004, Spielmann *et al.*, 2004 and 2006). These three tests, the embryonic stem cell test, the limb bud micromass culture and the whole embryo culture, showed high predictivity for certain strongly embryotoxic chemicals. However, due to the nature of the methods and limitations in their predictivity, they may be used only as supporting information along with other more reliable data to predict the developmental toxicity. The value of these validated methods could be increased by incorporating molecular based markers through the application of proteomic and toxicogenomic approaches (Piersma, 2006; van Dartel *et al.*, 2010). The embryonic stem cell method may be combined with Physiologically Based Biokinetic modelling in order to derive quantitative points of departure *in vitro*, which are then extrapolated to *in vivo* points of departure for use in risk assessment (Worth *et al.*, 2014).

The combination of assays in a tiered and/or battery approach may improve predictivity, but the *in vivo* situation remains more than the sum of the areas modelled by a series of *in vitro* assays (see Piersma, 2006 for review). Therefore, a negative result predicting absence of a particular property for a substance with no supporting information cannot be interpreted as demonstrating the absence of a reproductive hazard with the same confidence as an animal study. Another limitation of *in vitro* tests is that an N(L)OAEL and other dose-response information required for a risk assessment is not provided.

However, a positive result predicting a particular reproductive hazard in a validated *in vitro* test could provide a justification for the need of further testing beyond the standard information requirement, dependent on the effective concentration and taking account of what is known about the toxicokinetic profile of the substance. However, because of limited confidence in this approach at this time, such a result in isolation would not be adequate to support hazard classification.

Additionally, validated and non-validated *in vitro* tests, provided the applicability domain is appropriate, could be used with other data in a weight of evidence adaptation according to REACH Annex XI, 1.2 to gather information on hazardous properties. *In vitro* techniques can be used in mechanistic investigations, which can also provide support for regulatory decisions. Also, *in vitro* tests can be used as supporting evidence when assessing the toxicological properties by read-across within a substance grouping approach, providing the applicability domain is appropriate. Positive and negative *in vitro* test results may be of value in a read-across assessment and in category approach as one element.

Current developments on adverse outcome pathways (AOPs) to build a combination of studies and investigations to cover key events from an initiating molecular event to an adverse outcome may provide information on certain pathways, especially in developmental toxicity for certain malformations. Approaches may combine various different methods (e.g. *in vitro* tests, QSARs, *in chemico* assays etc). As these pathways do not cover all potential mechanisms/modes of action, negative results predicting absence of a particular property from those approaches do not provide enough confidence for regulatory decision making to demonstrate absence of a reproductive hazard. In addition, currently they do not provide an N(L)OAEL value or other dose-response information for risk assessment. However, they may provide necessary support for read across justification and categories and contribute to a weight of evidence adaptation according to REACH Annex XI, 1.2.

R.7.6.4.2 Animal data

In general, all findings on reproductive toxicity should be considered for classification purposes irrespective of the level of concurrent parental toxicity, see the *Guidance on the Application of the CLP criteria* (Setion 3.7.2.2.1, classification in the presence of parental toxicity).

For evaluation of the results of a reproductive toxicity study, it is important, where possible, to distinguish between a specific effect on reproduction (fertility and/or pre- and postnatal development) as a consequence of an intrinsic property of the substance and an adverse reproductive effect which is a secondary non-specific consequence to the general toxicity. Inclusion of additional parameters for general toxicity may enhance this interpretation. According to the criteria for classification, reproductive toxic effects should be considered if they occur in the absence of other (systemic) toxic effects or if they occur together with other toxic effects, are considered not to be a secondary non-specific consequence of the other toxic effects (see 3.7.2, Annex I of CLP).

R.7.6.4.2.1 Reproduction/developmental toxicity screening test

The screening studies provide initial information of the effects on male and female reproductive performance as well as on developmental toxicity during and shortly after birth, as well as certain additional parameters for endocrine disrupting mode of action including anogenital distance, nipple/areola retention, thyroid hormone levels as given in the revised TGs [137] (2015). These screening tests are not meant to provide complete information on all aspects of reproduction and development. However, the screening test (OECD TGs 421 or 422) is a standard information requirement for reproductive toxicity at REACH Annex VIII level. Thus, a negative study result at REACH Annex VIII is considered adequate although the screening study does not provide similar confidence than more comprehensive studies on reproduction toxicity. An evaluation of the screening tests (OECD TGs 421 or 422) has confirmed that these

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[137] OECD TGs 421 and 422 are in the process of being revised: adoption and publication is expected by the end of 2015.

tests are useful for initial hazard assessment and can contribute to decisions on further test requirements (Reuter *et al.*, 2003, Gelbke *et al.*, 2004, Beekhuiisen *et al.*, 2014).

With regard to male and female fertility, the number of parameters investigated are less than in the more comprehensive generation study designs such as the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) or the two-generation reproductive toxicity study (EU B.35, OECD 416), and the statistical power is much lower due to a lower number of animals per dose group. Furthermore, the pre-mating exposure duration in these screening studies may not be sufficient to detect all effects on the spermatogenic cycle or folliculogenesis. The two weeks premating exposure duration used in this study is equivalent to the time for epididymal transit of maturing spermatozoa and thus allows for the detection of post-testicular effects on sperm at mating (during the final stages of spermiation and epididymal sperm maturation). The two weeks premating exposure duration for females, covers 2-3 oestrous cycles and effects on cyclicity may be detected. Thus, the full spermatogenesis and folliculogenesis are not covered at the time of mating or together before and after the mating, as they take 70 and 62 days in rats, respectively.

Because exposure during the full spermatogenic period and folliculogenesis are not covered at the time of mating, effects at earlier stages of spermatogenesis and folliculogeneiss cannot be reflected in the functional fertility examination. For instance, earlier stages of the spermatogenesis (spermatogonia) and/or specific cell types (Sertoli cell and Leydig cells), are sensitive to many chemicals (see e.g review by Bonde, 2010). With a two-week premating exposure, the effects on functional fertility of exposure to these early stages of developing spermatozoa will not be covered. In addition, steady state may not be reached in all organs (see also discussion in Appendix R.7.6–3 of this Guidance). Histopathological data will be limited because the duration of the study itself does not cover the full spermatogenesis or folliculogenesis. Depending on the tonnage level, results from the 90-day study may be available with investigations of histopathology of gonads, however sperm parameters or oestrous cycles are usually not investigated. Histopathology of gonads may be among the most sensitive parameters to detect adverse effects on male fertility and the most sensitive parameter may be used to derive the NOAEL. However, the clarity of the effects rather than the sensitivity of the effects observed, are important for classification and labelling and will affect the category into which the substance is classified. Thus, to address the fertility also for the classification and labelling purposes, including the categorisation, it is necessary to consider how well all the available parameters address the fertility endpoint.

Due to its limitations, a screening study cannot be used to fulfil the information requirement of the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443). It should also be noted that these screening studies do not provide relevant information on post-natal developmental toxicity like a one- or two-generation reproductive toxicity studies (EU B.34/OECD 415 or EU B.56/OECD TG 443 or EU B.35/OECD 416) because the screening studies are already terminated at an earlier developmental stage than those more comprehensive studies.

With regard to developmental toxicity, these screening tests do not provide sufficient information on prenatal developmental toxicity because the pups are not examined for external, skeletal and visceral anomalies as in the prenatal developmental toxicity study (EU B.31, OECD TG 414). In addition, the pups in the screening studies are delivered naturally and the dams may cannibalise malformed pups. In the prenatal developmental toxicity study caesarean section is performed to avoid any cannibalism and to allow an appropriate evaluation of the foetuses. In addition, the statistical power of the screening study is lower than that of the prenatal developmental toxicity study. Therefore, a screening study cannot be used to fulfil the standard information requirement of a prenatal developmental toxicity study (EU B.31, OECD TG 414).

Depending on the tonnage level or based on adaptations, a screening study might be the only available reproductive toxicity study. However, the screening studies were not designed as an alternative or a replacement of the higher tier reproductive toxicity studies (EU B.31, OECD TG 414 and EU B.56, OECD TG 443). Therefore, the results of a screening study should be interpreted with caution and even statistically non-significant effects may be indicators for an impairment of reproduction. A result showing no effects in a OECD TGs 421 or 422 screening test does not provide reassurance of the absence of any hazardous property for reproductive toxicity. Further information on reproduction toxicity may be available to assist the interpretation of the results.

The observation of clear evidence of adverse effects on reproduction or on reproductive organs in these tests may be sufficient to meet the information needs for classification and labelling and risk assessment (using an appropriate assessment factor), and may provide an N(L)OAEL from which a DNEL can be identified (by adding an additional assessment factor due to higher uncertainty involved than in more comprehensive studies).

Effects observed in the screening study may serve as triggers, leading to more comprehensive reproductive toxicity studies or they may constitute conditions which specify the study design of an extended one-generation reproductive toxicity study. For instance EU B.56 (OECD TG 443) may be triggered based on evidence indicating concern on reproductive toxicity, (see Section R.7.6.2.3.2, Stage 4.4 REACH Annex IX, extended one-generation reproductive toxicity study, of this Guidance). A screening study may provide useful information when considering dose level selection for an extended one-generation reproductive toxicity study.

R.7.6.4.2.2 Prenatal developmental toxicity study

The prenatal developmental toxicity study (EU B.31, OECD TG 414) provides a focused evaluation of potential effects on prenatal development, although only effects that are manifested before birth can be detected. Detailed information on external, skeletal and visceral malformations and variations and other developmental effects are provided. Cesarean section allows precise evaluation of the number of foetuses affected.

For a comprehensive assessment of prenatal developmental toxicity, information from two species, one rodent (usually the rat) and one non-rodent (usually the rabbit) is assessed. However, depending on the REACH tonnage level, there might only be a standard information requirement for a prenatal developmental toxicity in one species (REACH Annex IX) or for none (REACH Annex VII and VIII). Under such circumstances, it needs to be evaluated if testing beyond the standard information requirement is triggered. If both or one of the default species (the rat or the rabbit) are not suitable species for prenatal developmental toxicity testing, a more suitable species considering the human relevancy should be selected for testing. An adequate justification must be provided for other species other than the rat and the rabbit. The results from prenatal developmental toxicity studies are considered relevant to humans unless there is substance-specific toxicokinetic or toxicodynamic evidence showing otherwise.

For evaluation, developmental effects should be considered in relation to adverse effects occurring in the parents, for further information see the *Guidance on the Application of the CLP criteria* (Section 3.7).

It should be noted that a prenatal developmental toxicity study (EU B.31, OECD TG 414) does not provide information on postnatal development or sufficient information on female fertility. However, some findings might raise concerns; if exposure started on gestation day 0, effects on preimplantation or implantation could indicate effects on female fertility. Also effects on maintenance of pregnancy and potentially on gestation length may be identified if significantly affected.

If a study is conducted according to an old test method and thus uses a shorter administration period than current test methods, it is important that there is no indication challenging the exposure period used. Thus, if there is a concern suggesting that a longer exposure period would have revealed developmental toxicity or more profound findings affecting also lower dose levels that were not observed using shorter exposure duration, this should be addressed; for example, by using an additional assessment factor which lowers the NOAEL to the next lower dose level or divides it by two if there is no lower dose level; or if a serious concern, a new study with longer exposure duration should be proposed. These indications challenging the exposure duration used may stem from fertility studies such as screening studies (OECD TGs 421 or 422) or from an extended one-generation reproductive toxicity study or from information on mechanisms/modes of action or structurally similar substances. It is to be noted that screening studies (OECD TGs 421 or 422) or the extended one-generation reproductive toxicity study do not provide equivalent information on prenatal developmental toxicity to that from the prenatal developmental toxicity study. Thus, if the indication of challenging the exposure duration rises from other available data, the results from these fertility studies may not always, depending on the case, provide sufficient confidence to conclude that there is no prenatal developmental toxicity.

Prenatal developmental toxicity studies may provide triggers for further reproductive toxicity studies, for example, in the form of foetotoxicity or foetal findings. In addition, some findings, such as increased foetal weight or placental weight, considered in light of litter size, may indicate an endocrine disrupting mode of action. Although there is no toxicological need to differentiate endocrine disrupting modes of action from other modes of action for developmental toxicity, in REACH the reproductive effects may trigger an extended one-generation reproductive toxicity study at REACH Annex IX and the indication of endocrine disrupting modes of action are one element in triggering the extension of Cohort 1B in an extended one-generation reproductive toxicity study.

R.7.6.4.2.3 Extended one-generation reproductive toxicity study

Introduction

The test method of the extended one-generation reproductive toxicity study (EOGRTS, EU B.56, OECD TG 443) describes a flexible modular study design with several investigational options allowing each jurisdiction to decide on the study design required for the respective regulatory context. The study design for REACH is described in detail in Appendix R.7.6–2 of this Guidance.

The extended one-generation reproductive toxicity study allows evaluation of the effects of the test substance on the integrity and performance of the adult male and female reproductive system and offspring viability, health and some aspects of physical and functional development until adulthood. The extension of the Cohort 1B (to mate the F1 animals to produce the F2 generation) also provides information on the fertility of the offspring (F1 generation), thus addressing the potential effects after exposure of the most sensitive life stages (i.e. *in utero* and early postnatal period). Therefore, mating of the Cohort 1B animals will cover information on the complete reproductive cycle.

In REACH the standard information requirement only includes Cohorts 1A and 1B for reproductive toxicity (without extension to produce the F2 generation). Thus, the basic study design is a one-generation study providing information on the fertility of the parental animals (P0 or F0 animals) and extended postnatal development of F1 animals. In addition, for REACH purposes it is necessary that the study design allows the adequate assessment of possible effects on fertility for risk assessment and classification and labelling purposes, including categorisation. To ensure that the study design adequately addresses the fertility endpoint, the duration of premating exposure period and the selection of the highest dose level are key aspects to be considered, see Appendix R.7.6–3 of this Guidance for further details. Regarding the highest dose level, it is important to ensure that toxicity in both female and male animals is considered to ensure that reproductive toxicity in either gender is not overlooked.

If the Column 2 conditions at REACH Annex IX/X are met, (for further information see Appendix R.7.6–2 of this Guidance) Cohort 1B must be extended, which means that the F2 generation is produced by mating the Cohort 1B animals. This extension also provides information on the mating, fertility and reproductive performance of the F1 animals. F1 animals are exposed *in utero* and during the early postnatal period allowing a comprehensive assessment of effects induced during these sensitive life stages. Similarly developmental neurotoxicity (Cohorts 2A and 2B) and/or developmental immunotoxicity (Cohort 3) cohorts need to be conducted if the triggers for such expansion of the basic study design, (which are provided in Column 2 of REACH Annex IX/X, 8.7.3) are fulfilled. These cohorts provide information on neurotoxic or immunotoxic potency of substances after exposure during sensitive life stages. When there are triggers for developmental neurotoxicity, both the Cohorts 2A and 2B are to be conducted as they provide complementary information. Considerations for evaluation of developmental neurotoxicity and developmental immunotoxicity are provided later in this section (see Sections R.7.6.4.2.6 and R.7.6.4.2.7 of this Guidance).

It is recommended that results from a range-finding study (or range-finding studies) for an extended one-generation reproductive toxicity study are reported with the main study. This will support the justifications of the dose level selections and interpretation of the study results.

If a range-finding study indicates adverse effects on fertility but the effects do not meet the criteria for Reproductive toxicity Category 1B, it is recommended that the main study should be designed to confirm the findings from the range-finding study. However, if the results from the range-finding study meet the criteria for Reproductive toxicity Category 1B reproductive toxicants, the adaptation of Column 2 may apply and further studies (including the main study) may not be needed.

General considerations related to investigation of (developmental) neurotoxicity and/or immunotoxicity

If triggers for neurotoxicity or immunotoxicity are identified at REACH Annex VIII or IX level but an extended one-generation reproductive toxicity study is not triggered, a separate neurotoxicity or immunotoxicity study in the developing organism or in adults must be proposed in line with the Column 2 adaptation to Section 8.6.1 of REACH Annex VIII or Section 8.6.2 of REACH Annex IX[138]. Depending on the cases, inclusion of additional parameters to the repeated dose toxicity study (including screening study), if not yet conducted may be considered to further characterise the effect.

Whether the neurotoxic and/or immunotoxic properties should be investigated in adults or in the developing organisms at REACH Annex VIII or REACH Annex IX level if an extended one-generation reproductive toxicity study is not triggered, should be considered on a case by case basis taking into account the various aspects affecting the decision, for example, the target population, toxicokinetics and mode of action. Generally, a study in developing organisms is recommended as a more conservative approach.

At REACH Annex X, the extended one-generation reproductive toxicity study is a standard information requirement, and if there are triggers for the (developmental) neurotoxicity and/or (developmental) immunotoxicity meeting the triggers described in Column 2, Section 8.7.3, the registrant must propose Cohorts 2A and 2B to address the concern for developmental neurotoxicity or Cohort 3 to address the concern for developmental immunotoxicity. The general evaluation of triggers is presented in Appendix R.7.6–5 of this Guidance. Instead of these cohorts, the registrant may propose separate developmental toxicity studies to address these concerns, as explained below in this section under "Proposals for developmental neurotoxicity or immunotoxicity studies". Likewise at REACH Annex IX, if an extended one-generation reproductive toxicity study is triggered, these cohorts, or separate studies, must be proposed by the registrant to address the concern in question.

It should be noted that neurotoxicity and/or immunotoxicity observed in adult animals may trigger developmental neurotoxicity and/or developmental immunotoxicity cohorts in an

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138 Column 2 at Annex VIII, 8.6.1 and Annex IX, 8.6.2: "Further studies shall be proposed by the registrant or may be required by the Agency in accordance with Article 40 or 41 in case of: …- indications of an effect for which the available evidence is inadequate for toxicological and/or risk characterisation. In such cases it may also be more appropriate to perform specific toxicological studies that are designed to investigate these effects (e.g. immunotoxicity, neurotoxicity), …")

extended one-generation reproductive toxicity study or in separate studies unless substance specific information is provided why these effects or mode of action would not be relevant in a developing organism (for evaluation of triggers see Stage 3.2.1). In addition, if the classification criteria for STOT are met, based on studies in adults, this is not an adaptation rule allowing the omission of investigations on developmental neurotoxicity and/or developmental immunotoxicity. This is due to expected higher sensitivity of the developing organisms (see e.g. Dietert, 2014), which may lead to a lower DNEL. In addition, a classification to Repr. 1B or 2 may be necessary if the effects are considered to be of developmental origin, i.e. exposure during development. Sensitivity has been evaluated in animal studies for nine reviewed (immuno)toxicants and, according to the authors, the developing immune system was found to be at least as sensitive or more sensitive than the general (developmental) toxicity parameters (Hessel *et al*., 2015).

Proposals for developmental neurotoxicity or immunotoxicity studies

REACH specifies that "*Other studies on developmental neurotoxicity and/or developmental immunotoxicity instead of cohorts 2A/2B (developmental neurotoxicity) and/or cohort 3 (developmental immunotoxicity) of the Extended One-Generation Reproductive Toxicity Study may be proposed by the registrant in order to clarify the concern on developmental toxicity."*

The cohorts for developmental neurotoxicity and developmental immunotoxicity included in the extended one-generation reproductive toxicity study provide information on these endpoints. Information on developmental neurotoxicity and developmental immunotoxicity are not standard information requirements in REACH but they must be proposed when particular concerns as specified in Column 2 are met. An advantage of this approach is that fewer animals are needed compared to running three separate studies (reproductive toxicity study, developmental neurotoxicity and developmental immunotoxicity study).

Other studies on developmental neurotoxicity

The registrant has a choice to propose a separate developmental neurotoxicity study instead of Cohorts 2A and 2B if the conditions for a particular concern for developmental neurotoxicity are met. The concern should be related to developmental neurotoxicity specifically. The study design for developmental neurotoxicity should follow the EU B.53 (OECD TG 426) protocol. The selection between the choices should be based on scientific and substance specific considerations taking into account which method adequately addresses the scientific concern with least amount of animals and investigations. However, practical limitations in testing laboratories can also be a reason to propose separate studies. Some examples of aspects of these considerations are presented below.

The developmental neurotoxicity cohort integrated into an extended one-generation reproductive toxicity study contains no endpoints for social or cognitive dysfunctions (e.g. autism, attention deficient hyperactivity disorders, attenuated learning and/or memory), thus, if there are signs of behavioural disturbances from adult animal studies, the design of the developmental neurotoxicity cohort in an extended one-generation reproductive toxicity study might have to be adjusted. Optionally EU B.53 (OECD TG 426) may be the preferred study design.

It should be borne in mind that, when it comes to developmental neurotoxicity, the outcome of a developmental neurotoxicity study (OECD 426) may differ from that of the developmental neurotoxicity Cohorts 2A and 2B in an extended one-generation reproductive study, considering the different exposure scenarios. For example, recent publications point at the importance of a healthy immune system of the mother during pregnancy for brain development of her offspring (Smith *et al.*, 2007); in other words, the maternal impact in the cohort study on nervous system development may be larger than that in the OECD 426 study (exposure from gestation day 6 to PND 21) due to a longer exposure period and the extent of effect often is unknown.

If an extended one-generation reproductive toxicity study is not triggered or a standard information requirement is not met but there are triggers for neurotoxicity, separate studies must be proposed according to REACH Annex VIII, 8.6.1, REACH Annex IX, 8.6.2, or REACH Annex X, 8.6.4.

Other studies on developmental immunotoxicity

The registrant has a choice to propose a separate developmental immunotoxicity study instead of Cohort 3 if the conditions for a particular concern for developmental immunotoxicity are met. The concern should be related to developmental immunotoxicity specifically. For developmental immunotoxicity there is currently no available internationally accepted protocol and thus the registrant must include the proposed protocol in his testing proposal until internationally accepted methods are available. For references to study designs for developmental immunotoxicity see Section R.7.6.4.2.7 of this guidance. The selection between the choices should be based on scientific and substance specific considerations taking into account which method adequately addresses the scientific concern with least amount of animals and investigations. Some examples of aspects of these considerations are presented below.

The nature and/or severity of the triggers may provide guidance to select between a separate study or a cohort. Other aspects to consider may include statistical power and the investigations included. It should be considered whether the cohorts or a separate study best address the particular concern identified (see also Appendix R.7.6–5 of this Guidance).

The outcome of a separate developmental immunotoxicity study may differ from that of the developmental immunotoxicity Cohort 3 in an extended one-generation reproductive study, if the exposure scenarios and set ups are different.

If an extended one-generation reproductive toxicity study is not triggered or a standard information requirement is not met but there are trigger(s) for immunotoxicity, separate studies must be proposed according to REACH Annex VIII, 8.6.1, REACH Annex IX, 8.6.2, or REACH Annex X, 8.6.4.

Common to both developmental neurotoxicity and immunotoxicity studies

Conflicts may arise to decide on the dose levels and premating exposure duration in an extended one-generation reproductive toxicity study. The adequacy of the study design to assess the effects on fertility should be ensured. Thus, the dose level selection should be based upon the fertility endpoint with the developmental neurotoxicity/immunotoxicity being tested at the same dose levels. The fertility endpoint is the only endpoint where *in vivo* data are typically available to make decisions on selecting dose levels for an extended one-generation reproductive toxicity study.

Even if there are trigger(s) for developmental neurotoxicity/immunotoxicity, the dose level setting must not compromise an appropriate investigation of the fertility endpoint. The challenge in deciding the dose levels and length for the premating exposure duration is that there may be a risk that in reducing fertility not enough pups will be produced for example, at the highest dose level for the evaluation of the potential developmental neurotoxicity/immunotoxicity at all dose levels. However, results from lower dose levels can still be used. Another possibility is to add an additional dose level or to address the developmental neurotoxicity/immunotoxicity in (a) separate stud(y)ies.

Evaluation of findings from developmental neurotoxicity and developmental immunotoxicity cohorts

Currently there is not much experience on the interpretation of the results of developmental neurotoxicity (see some considerations under R.7.6.4.2.6 of this Guidance) and developmental immunotoxicity cohorts included in extended one-generation reproductive toxicity studies. Guidance will be developed after gathering more experience. Until further experience on these cohorts, experiences from existing protocols on developmental neurotoxicity and developmental immunotoxicity can be used although all of them may not be standardised and internationally acceptable protocols yet. For evaluation of the results from separate studies, see Sections R.7.6.4.2.6 for developmental neurotoxicity and R.7.6.4.2.7 for developmental immunotoxicity of this Guidance.

Further aspects

The OECD GD 151 provides guidance for conducting the extended one-generation reproductive toxicity study as agreed at OECD level (OECD 2013) but does not for example, define the study design or criteria for the extension of Cohort 1B or the inclusion of cohorts. Thus, the study design should be defined to meet the REACH requirements. OECD GD 117 includes the internal triggers for extension of the Cohort 1B, however, these triggers are not used in REACH as such. The registrant may expand the study based on new information indicating a concern which needs to be addressed. The justification for the expansion must be documented.

For REACH purposes, the focus of the study should be on assessment of the effects on fertility and thus, a ten-week premating exposure duration and dose level setting based on toxicity are required as a starting point as explained above. In addition, for REACH the conditions which specify the extension of the Cohort 1B and the inclusion of Cohorts 2A, 2B and 3 are listed in Column 2 of REACH Annex IX/X, 8.7.3. EU B.56 (OECD TG 443) and OECD GD 151 should be followed only in conducting the study modules. It is recommended that results from a range-finding study (or range-finding studies) for an extended one-generation reproductive toxicity study are reported with the main study. This should support the justifications of the dose level selections, duration of the premating exposure and interpretation of the study results.

The study design of EU B.56 (OECD TG 443) selected must be adequately justified and documented in all cases[139].

In general, all findings on reproductive toxicity should be considered for classification purposes irrespective of the level of concurrent parental toxicity, see the *Guidance on the Application of the CLP criteria* (Chapter 3.7).

Most of the parameters investigated in the 90-day study are also included in the extended one-generation reproductive toxicity study. However, the results obtained may not be equivalent for several reasons and it may not be adequate to adapt the information requirement of a 90-day study by information from an extended one-generation reproductive toxicity study. This is because the 90-day study and the extended one-generation study have different aims. A 90-day study is meant to provide relevant information on systemic and organ-specific toxicity after a subchronic exposure and relevant route especially considering exposure conditions and non-pregnant animals are to be used. Usually the dose level selection for a 90-day study is higher when based on toxicity than the dose levels which can be used in an extended one-generation reproductive toxicity study. This is because the exposure is longer and pregnant animals (and offspring) may be more sensitive than non-pregnant animals. In addition, haematological, clinical chemistry, urinary and histological samples may be collected after a shorter exposure period in an extended one-generation reproductive toxicity study (8-10 weeks) than in a 90-day study (13 weeks) and if conducted in F1 animals, the exposure history and the developmental stages of the animals are different from that in a separate 90-day study. A very careful evaluation is needed when considering whether the information from an extended one-generation reproductive toxicity study can be used to adapt the information requirement of a 90-day study. In certain cases with adequate exposure levels and durations the results from an extended one-generation reproductive toxicity study may support for example, an older but with somewhat limited results, 90-day study.

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[139] REACH Art 3(28): "robust study summary: means a detailed summary of the objectives, methods, results and conclusions of a full study report providing sufficient information to make an independent assessment of the study minimising the need to consult the full study report;"

Information from a 90-day study may be valuable in deciding the dose levels of an extended one-generation reproductive toxicity study.

The extended one-generation reproductive toxicity study provides information on peri-postnatal development but does not address the same parameters as those in the prenatal developmental toxicity study and thus does not provide equivalent information.

R.7.6.4.2.4 Two-generation reproductive toxicity study

Two-generation reproductive toxicity studies are no longer standard information requirements (EU B.35, OECD TG 416) in REACH but those studies initiated before 13 March 2015[140] are considered appropriate to address the standard information requirement for REACH Annex IX/X, 8.7.3. The two-generation reproductive toxicity study was the standard information requirement for REACH until the amendment of REACH Annexes IX and X[141]. Two-generation reproductive toxicity studies initiated before the date indicated above are considered appropriate to address the standard information requirement and therefore fulfil the Column 1 requirements, however they do not automatically meet the adaptation criteria described in Column 2. If the available information shows triggers for developmental neurotoxicity and/or developmental immunotoxicity according to Column 2, these particular concerns must be addressed by proposing a separate developmental neurotoxicity and/or a separate developmental immunotoxicity study, respectively (see Section R.7.6.4.2.3, under "Proposals for developmental neurotoxicity or immunotoxicity studies", and R.7.6.4.2.6, and R.7.6.4.2.7 of this Guidance).

Although the two-generation reproductive toxicity study may lack information on some parameters which are part of EU B.56 (OECD TG 443), it addresses the fertility endpoint in two-generations and is adequate for risk assessment and classification and labelling, including categorisation when conducted according to the EU B.35 (OECD TG 416).

From the legal text it is clear that two-generation reproductive toxicity studies initiated after the date indicated in the legislation are not considered appropriate to address the standard information requirement at REACH Annex IX/X, 8.7.3, including the study design adaptation described in Column 2. This means that testing proposals for two-generation reproductive toxicity studies to fulfil the (standard) information requirement at REACH Annex IX/X, 8.7.3 cannot be accepted. If the study already exists and was initiated after March 13, 2015, the registrant may explore the possibilities to adapt the information requirement by substance specific justifications according to REACH Annex XI adaptation rules.

When considering the relevance of old non-guideline compliant two(multi)-generation reproductive toxicity studies to address the fertility endpoint (REACH Annex IX/X, 8.7.3), these studies will be assessed in line with REACH Annex XI, 1.1.2 adaptation rules for existing information. Thus, old existing non-guideline studies may fulfil the Column 1 standard information requirement or may serve as elements in a weight of evidence adaptation according to REACH Annex XI, 1.2 to identify hazardous properties or support a category approach.

R.7.6.4.2.5 One-generation reproductive toxicity study

The one-generation reproductive toxicity study (EU B.34, OECD TG 415) is not an appropriate study to fulfil the information requirement for an extended one-generation reproductive

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140 Commission Regulation (EU) 2015/282, Recital (11) and Article 2.

141 Commission Regulation (EU) 2015/282 of 20 February 2015 amending Annexes VIII, IX and X to Regulation (EC) No 1907/2006 of the European parliament and of the Council on the Registration, evaluation, Authorisation and Restriction of Chemicals (REACH).

toxicity study because of limited postnatal exposure duration and inadequate coverage of key aspects/parameters (REACH Annex XI, 1.1.2).

This study does not correspond to any REACH standard information requirement but could potentially be enhanced with certain parameters to fulfil the information requirement of the screening study. Compared to the screening study it has a higher statistical power, it addresses the functional fertility by covering the spermatogenesis and folliculogenesis before mating and reproductive performance until weaning. However, the test method lacks requirements of various important parameters as compared with the extended one-generation reproductive toxicity study. Existing studies may be used as one element in a weight of evidence approach according to REACH Annex XI, 1.2 to adapt the standard information requirement of REACH Annex IX/X, 8.7.3 together with other information or to support a category approach.

Existing studies according to a modified one-generation study protocol may also provide adaptation possibilities. A modified one-generation study is a flexible study design developed by NTP (National Toxicology Program, U.S. Department of Health and Human Services) which has no respective OECD Test guideline or EU Test method available. This study design provides information on reproductive toxicity after exposure from gestation day 6 of the parental animals up to mid gestation of the F1 generation.

R.7.6.4.2.6 Developmental neurotoxicity studies

Developmental neurotoxicity studies are not standard information requirements but may be triggered by REACH Annex VIII point 8.6.1 or REACH Annex XI point 8.6.2 or REACH Annex X point 8.6.4 based on Column 2 adaptation rules[142]. There, the Column 2 adaptation requires the registrant to propose further studies if there are indications of an effect for which the available evidence is inadequate for toxicological evaluation and/or risk characterisation. A separate developmental neurotoxicity study may also be proposed by the registrant instead of the developmental neurotoxicity cohorts (Cohorts 2A and 2B) in an extended one-generation reproductive toxicity study, if these cohorts are triggered.

Developmental neurotoxicity studies (e.g. EU B.53, OECD TG 426) are designed to provide information on the potential functional and morphological hazards of the nervous system arising in the offspring from exposure of the mother during pregnancy and lactation. These studies investigate changes in structure and function of the central nervous system (CNS) and the peripheral nervous system (PNS) using extensive neuropathology (structure) and behavioural (function) surveys. Advanced neuropathology may be assessed including quantitative structural measures as changes in cell structures related to for example, delayed development which may be of quantitative rather than qualitative nature. Such quantitative changes may be significant, but may still go unrecognised without quantification (De Groot *et al.*, 2005). To investigate behaviour a range of parameters, such as a behavioural test battery addressing different functions (domains) of the nervous system, motor activity and more advanced tests addressing cognitive behaviour, are performed. As behaviour may also be affected by the function of other organs such as liver, kidneys and the endocrine system, toxic effects on these organs in the offspring may also be reflected in general changes in behaviour. No single behaviour is able to reflect the entire complex and intricate function of behaviour and so integration of findings of different tests is deemed relevant to evaluate the relevance of the

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[142] Column 2 at Annex VIII, 8.6.1, Annex IX, 8.6.2, and Annex X, 8.6.4: "Further studies shall be proposed by the registrant or may be required by the Agency in accordance with Article 40 or 41 in case of: …- indications of an effect for which the available evidence is inadequate for toxicological and/or risk characterisation. In such cases it may also be more appropriate to perform specific toxicological studies that are designed to investigate these effects (e.g. immunotoxicity, neurotoxicity), …")

results on substance exposure. Likewise, it may be helpful for the interpretation to review behavioural (functional) changes in light of the neuropathology (structural) findings.

The severity and nature of the effect should be considered. Generally a pattern of effects (e.g. impaired learning during several consecutive trials) is more persuasive evidence of developmental neurotoxicity than one or a few unrelated changes. The reversibility of effects should be considered too. Important to mention in this context is that 'development' of an organism *a priori* goes with 'normal' structural and functional changes. Under toxic or pathologic circumstances a substance or disease may disturb 'normal' development and 'toxic' changes are built on top of 'normal' developmental changes. The nervous and immune systems are still under development up to and after birth. Moreover, different time-windows have been recognised for speed of developmental growth which in turn, may differ for different parts and structures of the developing nervous and immune systems. As a consequence, the vulnerability of these organ-systems differs during different time-windows of exposure. The nervous system possesses reserve capacity for repairing. We may for example, find the nervous system impaired during puberty, whereas the adult nervous system seems intact. In such a case, however, one should still realise that not only the trajectory from birth to puberty differed between control and substance-exposed individuals, but the trajectory from puberty to adulthood also differed. So even when a developmental neurotoxicant may not show adverse effects in the adult, the trajectories towards adulthood have been affected and the consequences of this are so far unknown. The nervous system may compensate for damage but the resulting reduction in reserve capacity is of concern and neurotoxicity occurring during development should be regarded as an adverse effect. If developmental neurotoxicity is only observed during part of the lifespan then compensation should be suspected. Also, effects observed for example during the beginning of a learning task but not at the end, should not be interpreted as reversible effects; rather the results may indicate that the speed of learning is decreased.

The experience of offspring especially during infancy may affect their later behaviour. For example, frequent handling of rats during infancy may alter the physiological response to stress and the behaviour in tests for emotionality and learning. In order to control environmental experiences, the conditions under which the offspring are reared should be standardised within experiments with respect to variables such as noise level, handling and cage cleaning. The performance of the animals during the behavioural testing may be influenced by for example, the time of day, and the stress level of the animals. Therefore, the most reliable data are obtained in studies where control and treated animals are tested alternatively and environmental conditions are standardised.

In interpreting the results, maternal toxicity should be taken into account as the development of pups may be affected by maternal toxicity. During early postnatal period pups are dependent of maternal care and maternal toxicity for example, in way of CNS depression, may compromise the survival and development of the pups. In addition, dams and pups should not be separated other than for very short periods of time during the first five postnatal days (e.g. for dose administration) and also later dams should not be moved from cages more than necessary (e.g. for inhalation exposure). In practise this would mean than for inhalation exposure, a whole-body exposure may be considered instead of nose-only exposure.

Adverse effects observed in a development neurotoxicity study will be relevant to hazard classification and the human health risk assessment, providing an N(L)OAEL, unless there is information to show that effects seen in these studies could not occur in humans. Due to a complexity of the endpoint, adversity should preferably be based on a holistic analysis of data by grouping similar parameters.

For more detailed reviews of how to interpret the developmental neurotoxicity results see OECD TG 426, OECD GD 43 and Tyl *et al.* (2008).

R.7.6.4.2.7 Developmental immunotoxicity studies

Developmental immunotoxicity studies are not standard information requirements but may be triggered by REACH Annex VIII point 8.6.1 or REACH Annex IX point 8.6.2 or REACH Annex X, point 8.6.4 based on Column 2 adaptation rules[143]. There, the Column 2 adaptation requires the registrant to propose further studies if there are indications of an effect for which the available evidence is inadequate for toxicological evaluation and/or risk characterisation. A separate developmental immunotoxicity study may be proposed by the registrant instead of the developmental immunotoxicity cohort (Cohort 3) in an extended one-generation reproductive toxicity study, if these cohorts are triggered.

Developmental immunotoxicity studies are designed to provide information on the potential functional and morphological hazards to the immune system arising in the offspring from exposure of the mother during pregnancy and lactation. Currently there is no OECD test guideline for developmental immunotoxicity testing. Recent reviews provide information on the available approaches and considerations (Gupta (2011), page 219-225; WHO, 2012; De Jong and Van Loveren 2007; DeWitt *et al*., 2012a and 2012b; Dietert and DeWitt, 2010; Dietert and Holsapple, 2007; Holsapple *et al*., 2005; Rooney *et al*., 2009; Boverhof *et al*., 2014).

These studies investigate changes in immune response due to effects on the innate or acquired immune system. As immune response may also be affected by the function of other organs such as liver, kidneys and the endocrine system, toxic effects on these organs in offspring may also be reflected in changes in immune response. No single immune parameter is able to reflect the entire complex and intricate function of immune system and so, integration of findings of different tests is relevant to evaluate the relevance of the results on substance exposure.

Effects considered as adverse will be relevant to hazard classification and the human health risk assessment, providing an N(L)OAEL, unless there is information to show that effects seen in these studies could not occur in humans. Due to a complexity of the endpoint, adversity should preferably be based on a holistic analysis of data by grouping similar parameters.

R.7.6.4.2.8 Repeated-dose toxicity studies

Although not aimed directly at investigating reproductive toxicity, repeated-dose toxicity studies are standard information requirements (e.g. the 28-day study EU B.7, OECD TG 407 or the 90-day study EU B.26, OECD TG 408) and may reveal clear effects on reproductive organs in adult animals. In addition to histopathology of reproductive organs and changes in organ weights, parameters evaluated, such as sperm analysis and measurements of oestrous cycle, may provide relevant information for reproductive toxicity or indicate a concern (trigger(s)). However, no observed effects in measured parameters predicting fertility in repeated dose toxicity studies do not rule out the possibility that the substance may have the capacity to affect fertility. At REACH Annex IX level, triggers for reproductive toxicity from repeated dose toxicity studies trigger an extended one generation reproductive toxicity study (EU B.56, OECD TG 443). At REACH Annex VIII level the registrant may consider proposing an extended one-generation reproductive toxicity study instead of a screening study, based on triggers from a 28-day study.

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143 Column 2 at Annex VIII, 8.6.1, Annex IX, 8.6.2, and Annex X, 8.6.4: "Further studies shall be proposed by the registrant or may be required by the Agency in accordance with Article 40 or 41 in case of: …- indications of an effect for which the available evidence is inadequate for toxicological and/or risk characterisation. In such cases it may also be more appropriate to perform specific toxicological studies that are designed to investigate these effects (e.g. immunotoxicity, neurotoxicity), …").

The observation of effects on reproductive organs in repeated-dose toxicity studies may also be sufficient to be used for classification and labelling and for identifying an N(L)OAEL for use in the risk assessment. It should, however, be noted that the sensitivity of repeated-dose toxicity studies for detecting effects on reproductive organs may be less than reproductive toxicity studies because of the lower number of animals per group (lower statistical power). In addition, a number of cases have demonstrated that effects on the reproductive system may occur at lower doses when animals are exposed during the development or as young animals rather than as adults. Consequently, if there are adverse effects on the reproductive organs in adult animals in the absence of reproductive toxicity studies, an increased assessment factor may be considered in the risk assessment process at REACH Annex VII-VIII levels. An extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) may be triggered based on findings from a repeated dose toxicity study at lower tonnage REACH Annexes, and must be proposed at REACH Annex IX.

The adversity of some effects seen in repeated dose toxicity studies may be difficult to interpret, for example changes in sex hormone levels, and may need to be investigated further as part of studies that may be required to meet standard REACH information requirements (for example EU B.26 (OECD TG 408) or other repeated-dose toxicity studies), rather than serve as a trigger for the immediate conduct of an extended one-generation reproductive toxicity study. Whether or not a finding will serve as a trigger depends on the reliability of the finding and if it can be considered as adverse (see discussions in Appendix R.7.6–5 of this Guidance). It may be considered that statistically significant changes from relevant studies can be considered as triggers; however, sometimes a statistically non-significant change can be also considered as biologically relevant if not contradicting to other available information.

Repeated-dose toxicity studies may also provide indications of a particular concern to evaluate the need to investigate developmental neurotoxicity or developmental immunotoxicity endpoints. The potential triggers for these cohorts in an extended one-generation reproductive toxicity study or separate studies are described in the context of the extended one-generation reproductive toxicity study (Section R.7.6.4.2.3 of this Guidance).

R.7.6.4.2.9 *In vivo* assays for endocrine disruption mode of action

The endocrine system has a critical role in the control of all aspects of the reproductive cycle and therefore endocrine disruption is a potential mechanism for reproductive toxicity. None of the available *in vivo* assays only focusing on identification of endocrine disrupting potency, such as Uterotrophic assay (EU B.54, OECD TG 440) and Herschberger assay (EU B.55, OECD TG 441), correspond to standard REACH information requirements. These studies involve dosing of immature or ovarectomised/castrated animals, and the weighing of oestrogen/androgen dependent tissues (e.g. uterus or prostate). The methods can be used to identify (anti)oestrogenic or (anti)androgenic modes of action and the results may serve as triggers for further studies in certain cases. These animal models are sensitive to detect the hormonal mode of action. However, only investigation in intact animals proves if the mode of action is relevant in non-manipulated conditions. A comprehensive collection of screening tests and tests for endocrine disrupting chemicals are presented in OECD GD 150 and are included within the "*OECD Conceptual Framework for the Screening and Testing of Endocrine Disrupting Chemicals*".

A result in the uterotrophic assay in a well conducted dose-response study showing no effect indicates that the test substance is not an oestrogen receptor (ER)-ligand in those *in vivo* conditions. Equally, a result in the Hershberger assay showing no effect indicates that the test substance is neither an androgen receptor (AR)-ligand nor a 5-alpha reductase inhibitor in those *in vivo* conditions. A test substance not causing effect in these assays may, however, still have endocrine disrupting properties as well as a potential for reproductive toxicity mediated through other mechanisms. The uterotrophic and Hershberger assays may be used to provide NOEL/LOELs for these endocrine disruption modes of action only if immature (intact) animals are used. The results may also support findings from other studies or serve as triggers for further studies and examinations.

A number of assays in experimental animals may provide information on the ability of a substance to act on the production of steroids and the pubertal assays and the intact male assay may provide information about the endocrine disruption potency of the substance *in vivo* (OECD GD 150). Effects on the various endpoints included in these assays may be considered adverse and/or as representing an effect on a mechanism relevant for humans and serve as triggers for further studies and examinations.

In summary, while these *in vivo* assays in intact animals may be considered predictive for adverse effects on reproduction, they do not provide adequate information on reproductive toxicity for risk assessment and classification and labelling. The repeated dose 28-day oral toxicity study (EU B.7, OECD TG 407) has been updated (2008) to include parameters aiming to identify substances acting through (anti)oestrogenic, (anti)androgenic and (anti)thyroid mechanisms. Validation studies indicate that enhanced design can reliably identify substances with strong potential to act through endocrine modes of action on the gonads and thyroid. A result suggesting no effects in such a study up to the highest dose tested provides some evidence of the absence of potent endocrine activity. However, effects induced by a lower endocrine disrupting potency cannot be ruled out and therefore a result showing no effects does not provide reassurance of the absence of the capability to cause reproductive toxicity via the mechanism of endocrine disruption. Notably in this context, prolongation of exposure from 28 days up to 90 days is unlikely to improve the detectability of endocrine effects (Gelbke *et al*., 2006). Evidence of effects on reproductive organs potentially via endocrine disrupting mode of action seen in a repeated-dose toxicity study provides a trigger for the conduct of a more comprehensive study, i.e. the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) at REACH Annex IX.

The potential triggers related to endocrine disrupting modes of action to be used to define the study design of an extended one-generation reproductive toxicity study are presented along with other triggers in Appendix R.7.6–2 of this Guidance.

The screening studies (OECD TGs 421 or 422) may be adopted [144] with additional parameters for endocrine disrupting modes of action, including measurements of anogential distance, nipple/areolae retention, and thyroid hormone) levels These parameters indicate endocrine disrupting mode of action and may be predictive for adverse effects on reproduction. A statistically significant change in anogenital distance that cannot be explained by the body weight/size of the animal indicates an antiandrogenic mode of action and should be used for setting the NOAEL. To support the adversity of this parameter an association with reduced human reproduction has been reported (Jain and Singal, 2013; Eisenberg *et al*., 2011 and 2012; Mendiola *et al*., 2011). A statistically significant change in nipple/areolae retention indicates also an antiandrogenic mode of action but likely via other spectrum of mechanisms than that of anogenital distance. Due to the difference in biology in controlling the final number of nipples between male rats and human, it is not possible to study the association between nipple/areolae retention findings in rats and adversity in humans as for anogenital distance. However, as the assumed mode of action (antiandrogenicity) and potential underlining mechanisms affecting nipple/areolae retention in rats are also relevant to humans, although not causing similar effects, this finding can be considered likely to predict an adverse effect and used to set the NOAEL. Nipple/areolae retention measures the same mode of action (antiandrogenicity) as anogenital distance but due to different tissue specific underlining mechanisms and possibly toxicokinetic differences nipple/areolae retention may be more or less sensitive than anogenital distance. It is recommended that these endpoints are evaluated together.

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[144] OECD TGs 421 and 422 are in the process of being revised: adoption and publication is expected by the end of 2015.

As the extended one-generation reproductive toxicity study is a more comprehensive reproductive toxicity study which includes certain parameters to detect endocrine disrupting modes of action, it may be possible a) to identify an endocrine disrupting mode of action, b) to identify an adverse effect on reproduction, and c) for both (a) and (b) not necessarily indicating a causal relationship. If an endocrine disrupting mode of action is identified without an adverse effect on reproduction (e.g. reduced thyroid hormone level in pups), further studies or actions may be considered. If the findings on reproduction meet the classification criteria to Category 1B reproductive toxicant, irrespective indications of an endocrine disrupting mode of action, the substance should be classified accordingly.

R.7.6.4.3 Human data on reproductive toxicity

Epidemiological data require a detailed critical appraisal that includes an assessment of the adequacy of controls, the quality of the health effects and exposure assessments, and of the influence of bias and confounding factors. Epidemiological studies can generally only provide associations, not causality because, although it may be possible to show the link and estimate the likelihood of the causality,it cannot give a final proof.

Epidemiological studies, case reports and clinical data may provide sufficient hazard and dose-response evidence for classification of chemicals as reproductive toxicants in Category 1A and for risk assessment, including the identification of a NAEL or LAEL. In such cases, there will not normally be a need to test the substance. However, convincing human evidence of reproductive toxicity for a specific substance is rarely available because it is often impossible to identify a population suitable to study that is/was exposed only to the substance of interest. Human data may provide limited evidence of reproductive toxicity that indicates a need for further studies of the substance; the test method selected should be based on the potential effect suspected.

When evidence of a reproductive hazard has been derived from animal studies it is unlikely that the absence of evidence of this hazard in an exposed human population will negate the concerns raised by the animal model. This is because there will usually be methodological and statistical limitations to the human data. For example, statistical power calculations indicate that a prospective study with well-defined exposure during the first trimester with 300 pregnancies could identify only those developmental toxins that caused at least a 10-fold increase in the overall frequency of malformations; a study with around 1000 pregnancies would have power to identify only those developmental toxins that caused at least a 2-fold increase (EMA/CHMP Guideline, 2006). Extensive, high quality and preferable prospective, data are necessary to support a conclusion that there is no risk from exposure to the chemical. Thus, the absence of effects in humans at a dose level below the dose levels inducing reproductive toxicity in animals, will not negate the concerns raised by the animal model.

R.7.6.4.4 Derivation of DNELs and DMELs

Identification of DNEL(s) are referred to in REACH Annex I, 1.4. Depending on the available information and the exposure scenario(s), it may be necessary to identify different DNELs for each relevant human population (consumers, professional, workers, humans exposed indirectly via the environment and certain vulnerable subpopulations (children, pregnant woman) and for different routes of exposure and all routes combined. In certain cases exposure from various sources may need to be considered. For reproductive toxicity endpoints it is especially relevant to consider deriving the different DNELs for vulnerable subpopulations.

Generally, effects on reproduction have been considered as effects having a threshold and thus allowing derivation of a DNEL. However, in certain cases, the possibility for a non-threshold mode of action may need to be considered (e.g. if a substance has (anti)hormonal activity similar to a hormone having a primary biological control role and there is a concern of lack of body’s regulation capacity). For these cases derivation of DMEL may need to be considered.

In order to be suitable for CSA appropriate DNELs (a DNEL for fertility and a DNEL for development) have to be established for each exposure scenario and each population exposed. Typically, the derivation of the DNEL takes into account a dose descriptor, modification of the starting point and application of assessment factors – see the *Guidance on IR&CSA*, *Chapter R.8 Characterisation of dose [concentration]-response for human health* (Section R.8.4 and Appendix R.8-12) and Section R.7.6.4.3 of this Guidance. Appendix R.8-12 Reproductive toxicity provides specific advice for reproductive toxicity studies.

R.7.6.5 Classification and labelling

Guidance on classification and labelling is given in the *Guidance on the Application of the CLP criteria* (Section 3.7) and specifically for parental toxicity see Section 3.7.2.2.1 *Classification in the presence of parental toxicity*.

R.7.6.6 Conclusions on reproductive toxicity

Reproductive toxicity endpoints should be considered separately for establishing the relevant endpoint(s) and NOAEL(s) to be used in risk assessment (for fertility and developmental toxicity endpoints) and for classification (for sexual function and fertility; developmental toxicity; and lactation). The study or studies giving rise to the highest concern must normally be used to establish the DNEL(s) (see REACH Annex I, 1.2.4). If another study / other studies are used, an acceptable justification for this exception needs to be provided. Derivation of DMEL needs to be considered if adverse effects are likely to be induced via a non-threshold mode of action.

Risk assessment and determination of classification involves the consideration of all data that is available and may be relevant to reproductive toxicity (see Section R.7.6.3 of this Guidance for different data sources). There can be no firm rules on how to conduct the risk assessment and determination of classification for hazards as this process involves expert judgment and also because the mix and reliability of information available for a particular substance will probably be unique. Also data resulting from studies on other hazards, for example, repeated dose toxicity, can be relevant for consideration in the risk assessment and determination of classification of reproductive toxicity.

In order to conclude on a hazard classification and category, all the available information needs to be taken into account, and compared with the criteria in Annex I of the CLP Regulation (see also the *Guidance on the Application of the CLP criteria*). If the information is not adequate to decide on classification and labelling, the registrant must indicate and justify the action or decision he has taken as a result of inadequate data (REACH Annex VI, 4.1 and REACH Annex VI, 1.3.2).

If the substance has an EU harmonised classification for Reproductive toxicity (included in Annex VI, CLP) or meets the classification criteria and is subject to self-classification, exposure scenarios should be established and the risk characterisation ratio (RCR) calculated to indicate the safe use of the substance.

R.7.6.7 Integrated Testing Strategy (ITS) for reproductive toxicity

Section R.7.6.2 of this Guidance, includes guidance on how to define and generate relevant information on substances in order to meet the information requirements and address the concerns related to intrinsic properties of substances related to reproductive health.

An integrated testing strategy (ITS) may be defined as an approach which combines one or more non-animal methods with animal studies to fulfil the information requirements or could only include non-animal methods which covered all key aspects of reproductive toxicity. Thus, REACH Annex XI adaptations (with the exception of Section 3.2.a – substance tailored exposure-driven testing) play an important role in ITSs for reproductive toxicity. An ITS must produce information usable for a robust risk assessment and/or for classification and labelling. A definition for ITS is given by Blaauboer *et al*., (1999)[145]. The ITS concept is similar to that of IATA, Integrated Approaches to Testing and Assessment. In principle, ITS and IATA are approaches where information is collected, evaluated and weighted with the aim to provide a sufficient amount of information by development of the weight of evidence. ITS and IATA could be used with a view to generate information in a step-wise approach, allowing for justifying an adaptation of one or more standard information requirements according to REACH REACH Annex XI, 1.2. (weight of evidence) taking into account that REACH Annex XI, 1.2 is a hazard-based approach and exposure and risk-based consideration cannot be used.

A comprehensive use of ITS for reproductive toxicity endpoint requires knowledge on all different mechanistic steps and processes involved in the outcome of a possible adverse effect. Reproductive toxicity relates to a number of potential target tissues and comprises a very large number of interacting processes, which are not even known in their entirety and which at present are far from being fully understood in their complexity. Another particular challenge in the identification of reproductive toxicity effects relates to the potential impact of systemic toxicity on the fertility and maternal toxicity on the development of the offspring. The existence of windows of particular sensitivity during the development of the embryo is another characteristic feature of reproductive toxicity. However, currently adverse outcome pathways (AOPs) are under development each covering one specific effect for example, vasculogenesis and cleft palates. It is to be noted that also the specific effects like clefts can be formed via several different mechanisms and AOPs increasing the complexity. AOPs may form a basis for ITS/IATA in describing the key events in toxicity pathways that need to be addressed by and ITS/IATA.

Combined approaches including various methods may be used as preliminary steps only because they do not provide equivalent information on the standard information requirements. In addition they may be elements in a weight of evidence adaptation according to REACH Annex XI, 1.2 approach or supporting categories and read across according to REACH Annex XI, 1.5 approach. However, as these combined approaches include more uncertainty due to missing parts of information; this should be addressed when such approaches are proposed. As all the potential molecular mechanisms and regulatory mechanisms are not covered these approaches may not be appropriate to prove the absence of an effect. Currently derivation of a NOAEL is not possible with these methods.

R.7.6.8 References on reproductive toxicity

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[145] "An Integrated Testing Strategy is any approach to the evaluation of the hazard which serves to reduce, refine or replace an existing animal procedure, and which is based on the use of two or more of the following: physicochemical data, in vitro data, human data (for example, epidemiological, clinical case reports), animal data (where unavoidable), computational methods (such as quantitative structure activity relationships (QSARs) and biokinetic models" (Blaauboer *et al*., 1999).

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Appendices R.7.6-1 to 5 to Section R.7.6

NOTE to the reader: The references cited in the Appendices are given in Section R.7.6.8 References on Reproductive Toxicity

Appendix R.7.6–1 A check list for information that contributes to EOGRTS design

This is a "check list" for information (sources) that should be checked in order to establish the existence or the nonexistence of the triggers and conditions specifying the study design of an extended one-generation reproductive toxicity study for REACH. Please note that this is not advice on how to conduct an evaluation of the data.

The information is expected to be derived from the substance itself but if it is a surrogate, such as a component of a multiconstituent substance, the triggers from all the components and metabolites must be considered and justified.

More details and examples of triggers are provided in Appendix R.7.6–2 (EOGRTS study design) of this Guidance and length of the premating exposure duration is discussed in Appendix R.7.6–3 of this Guidance.

Condition/trigger	Where to find the information to decide on the existence or nonexistence of the triggers and conditions
E1: Uses leading to significant exposure of consumers or professional, taking into account inter alia consumer exposure from articles	Consumer and/or professional uses (one very wide uses or several limited uses) of a substance as neat(concentrate), in a mixture, in an article with intended release, or in an article with unintended migration from the matrix. The registrant must record and justify the existence or nonexistence of any of the conditions above. If any of these exist together with any of the other three conditions below (E2, E3 or E4), fulfilling the criteria detailed in Appendix R.7.6–2 of this Guidance, then the extension of the Cohort 1 B must be proposed.
E2: Genotoxicity potentially meeting classification criteria to Mutagen Category 2	Results from *in vivo* mutagenicity studies (if one of the *in vitro* tests is positive, then an *in vivo* somatic cell mutagenicity test must have been conducted). The registrant must record the findings and justify the existence or nonexistence of the condition.
E3: Extended exposure is needed to reach the steady state kinetics.	Indications on the exposure duration needed to reach the steady state can be obtained from various sources. • toxicokinetic studies in animals • human data, e.g. substance or metabolite(s) level(s) in blood or organs. • existing *in vivo* studies with long exposure duration showing unexpected severity or occurrence of findings compared with studies with short exposure. • Any other indication of potential to accumulate, such as prediction from log Pow, non-animal approaches (QSAR predictions), information from eco-toxicity (elevated levels in biota, high levels at the top of food chain, very slow

	depuration, bioaccumulation potency (B or vB, or similar concern), biomagnifications) All the components and metabolites of the multicomponent substance (multiconstituent or UVCB substances) must be considered and justified. The registrant must record the findings and justify the existence or nonexistence of the condition.
E4: Indications of modes of action related to endocrine disruption from *in vivo* or non-animal approaches	Repeated dose toxicity studies and reproductive toxicity studies may provide indication of endocrine disrupting modes of action. Check the parameters related to endocrine modes of action. Check *in vivo* assays for endocrine (disrupting) modes of action. Check the non-animal approaches for prediction to endocrine (disrupting) modes of action. Check data from eco-toxicity testing for predicting endocrine (disrupting) modes of action The registrant must record the findings and justify the existence or nonexistence of the condition.
N1: Information on neurotoxicity from *in vivo* studies or non-animal approaches.	*In vivo* toxicity studies may provide information on neurotoxicity. Check all the parameters related to nervous system. Check the non-animal approaches for prediction of (developmental) neurotoxicity. The registrant must record the findings and justify the existence or nonexistence of the triggers and particular concern for developmental neurotoxicity.
N2: Specific mechanism/modes of action with association to (developmental) neurotoxicity.	Some studies may include measurements which reveal the mechanism, or there may be specific mechanistical studies (*in vivo* or *in vitro*) available. The registrant must record the findings and justify the existence or nonexistence of the triggers and particular concern for developmental neurotoxicity.
N3: Existing information on (developmental) neurotoxicity from structurally analogous substances	Structurally analogous substances should be identified and existing information on effects showing (developmental) neurotoxicity must be checked. The registrant must record the findings and justify the existence or nonexistence of the triggers and particular concern for developmental neurotoxicity.

I1: Information on immunotoxicity from *in vivo* studies or non-animal approaches.	*In vivo* toxicity studies may provide information on immunotoxicity. Check all the parameters related to immune system. Check the non-animal approaches for prediction of (developmental) immunotoxicity. The registrant must record the findings and justify the existence or nonexistence of the triggers and particular concern for developmental neurotoxicity.
I2: Specific mechanism/modes of action with association to (developmental) immunotoxicity.	Some studies may include measurements which reveal the mechanism or there may be specific mechanistical studies (*in vivo* or *in vitro*) available. The registrant must record the findings and justify the existence or nonexistence of the triggers and particular concern for developmental immunotoxicity.
I3: Existing information on (developmental) immunotoxicity from structurally analogous substances	Structurally analogous substances should be identified and existing information on effects showing (developmental) immunotoxicity must be checked. The registrant must record the findings and justify the existence or nonexistence of the triggers and particular concern for developmental immunotoxicity.

Appendix R.7.6–2 EOGRTS Study Design

The registrant must propose the study design for an extended one-generation reproduction toxicity study with the following specifications. Relevant justifications are needed for the study design, including the existence or nonexistence of the conditions for extension of the Cohort 1B and trigger(s) for the Cohorts 2A and 2B, and Cohort 3.

Specifications for study designs in REACH are needed for the following aspects:

1) Premating exposure duration and dose level selection;
2) The need to extend the reproduction toxicity Cohort 1B and to define the termination time for F2;
3) The need to include the developmental neurotoxicity Cohorts 2A and 2B;
4) The need to include the developmental immunotoxicity Cohort 3.

In the following text the specifications and triggers (conditions) are presented for each study design. The Table in Appendix R.7.6–1 of this Guidance provides a check list for the registrants in order to provide a short list of studies/tests which could provide information on triggers to specify the study design of an extended one-generation reproductive toxicity study. The existence or the nonexistence of triggers (conditions) must be recorded in order to allow an independent evaluation.

The study design should be decided before the study is started. For REACH the in-study triggers are not recommended. However, the registrant may expand the study based on new information (that arises after the ECHA Evaluation decision has been issued) indicating a concern which needs to be addressed. The justification for the expansion must be documented.

The OECD guidance document GD 151 provides guidance for the conduct of cohorts of the extended one-generation reproductive toxicity study (OECD 2013) but the study design applicable for REACH and CLP is outlined in REACH Annexes IX and X and Recital (7) of Commission Regulation (EC) 2015/282 amending REACH and described in more detail in this guidance.

Specifications needed in testing proposals:

1) Premating exposure duration and dose level selection

Recital (7) of Commission Regulation (EC) No 2015/282 of 20 February 2015 amending REACH states that an extended one-generation reproductive toxicity study should allow adequate assessment of fertility and that premating exposure duration and dose levels should be appropriate to meet the risk assessment and classification and labelling purposes[146].

[146] Recital (7) of Commission Regulation (EU) No 2015/282 Of 20 February 2015 amending Annexes VIII, IX and X to Regulation (EC) No 1907/2006 of the European parliament and of the Council on the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) as regards the Extended one-generation reproductive toxicity study: *"It should be ensured that the reproductive toxicity study carried-out under point 8.7.3 of Annexes IX and X to Regulation (EC) No 1907/2007 will allow adequate assessment of possible effects on fertility. The premating exposure duration and dose selection should be appropriate to meet risk assessment and classification and labelling purposes as required by Regulation (EC) No 1907/2006 and Regulation (EC) No 1272/2008 of the European parliament and of the Council."*

Both the length of premating exposure duration and dose level setting are aspects which influence the possibility to adequately assess potential adverse effects on fertility. In order to adequately address the assessment of the fertility endpoint, the starting point for deciding on the length of premating exposure period should be ten weeks to cover the full spermatogenesis and folliculogenesis before the mating, allowing meaningful assessment of the effects on fertility. The exposure can be started when the animals are around 5 weeks old and mate them around 15 weeks of age. However, based on substance specific justifications a shorter premating exposure duration may be proposed, but it should not be shorter than two weeks. Further discussion on premating exposure duration is provided in Appendix R.7.6–3 of this Guidance. If the registrant prefers another length of premating exposure duration, an acceptable substance-specific scientific justification must be provided.

The highest dose for an extended one-generation reproductive toxicity study should be selected with the aim to induce some toxicity (or to use the limit dose of 1000 mg/kg bw/day if humans are not exposed to higher dose levels), in order to allow a conclusion on whether effects on reproduction are considered to be secondary, non-specific consequence of other toxic effects seen (see also the dose level selection under Section R.7.6.2.3.2, Stage 4.1(6) of this Guidance). Only in this way is it possible to assess if the substance is a reproductive toxicant and/or if the effects on reproduction are potentially associated with systemic toxicity and to what extent.

The possibility to select the highest dose level, based on the toxicokinetic data as mentioned in EU B.56 (OECD TG 443) and in the OECD GD 151, may not allow comparison of adverse effects on fertility with systemic toxicity and, thus, does not support production of data for classification and labelling purposes, including categorisation. Regarding the highest dose level, it is important to ensure that toxicity in both female and male animals is considered to ensure that reproductive toxicity in either gender is not overlooked.

Both the ten weeks premating exposure duration and the highest dose level meeting the requirement of inducing toxicity, should allow conclusion on classification and labelling, including categorisation, for the hazard endpoint for sexual function and for fertility according to CLP.

2) Extension of Cohort 1B and termination time for F2

REACH specifies that the extension of cohort 1B to include the F2 generation shall be proposed by the registrant or may be required by the Agency if:

a*) "the substance has uses leading to significant exposure of consumers or professionals, taking into account, inter alia, consumer exposure from articles, and*

b) any of the following conditions are met:

- *the substance displays genotoxic effects in somatic cell mutagenicity tests in vivo which could lead to classifying it as Mutagen Category 2, or*
- *there are indications that the internal dose for the substance and/or any of its metabolites will reach a steady state in the test animals only after an extended exposure, or*
- *there are indications of one or more relevant modes of action related to endocrine disruption from available in vivo studies or non-animal approaches."*

In the following lists, examples are provided for the criteria when the registrant must propose the extension of Cohort 1B to mate the Cohort 1B animals to produce a F2 generation:

Guidance for uses leading to significant exposure:

- If the substance is intended to be used[147] in the EU by consumers (i.e. members of the public) or professionals, either neat or in a chemical mixture and there is one very wide use or several limited uses potentially affecting many consumers and/or professionals, then this is considered as meeting the criterion;
- If the substance is in an article used by consumers or professionals in the EU the criterion would be met if the substance is intended to be released from the article during use of the article by the consumers or professionals and there is one very wide use or several limited uses potentially affecting many consumers and/or professionals;
- Use of a substance in consumer articles exhibiting significant migration from the matrix and for which dermal absorption is relevant.

Guidance for substance specific toxicity conditions to be used together with criteria for uses leading to significant exposure:

(i) "*The substance displays genotoxic effects in somatic cell mutagenicity tests in vivo which could lead to classifying it as Mutagen Category* 2":

- Genotoxicity/mutagenicity observed *in vivo* potentially meeting the classification criteria to Mutagen Category 2:
  - Note: If the substance meets the criteria to Mutagen Category 1A/1B and the adequate risk management measures are in place then the reproductive toxicity studies need not to be conducted (according to adaptation possibilities in REACH Annex IX/X, point 8.7, Column 2);
  - *An in vivo* mutagenicity study should be available if one of the *in vitro* mutagenicity studies is positive (i.e. predicts mutagenicity). If one of the *in vitro* mutagenicity studies is positive, an *in vivo* mutagenicity study should be conducted before deciding on the study design of an extended one-generation reproductive toxicity study, if the other criteria for extending the Cohort 1B are not met.

(ii) "*There are indications that the internal dose for the substance and/or any of its metabolites will reach a steady state in the test animals only after an extended exposure*":

Extended time to reach the steady state may be indicated by available toxicokinetic information, physico-chemical properties and information from (eco)toxicological data. The effect of sex and life stages could be also considered[148]. Information can be obtained from:

- Assessment of toxicokinetic behaviour of the substance:
  - Generally, duration of longer than a week to reach the steady state may be considered as extended (in practise a steady state can be considered to be achieved after 4 to 6 half-lives)[149];
  - Attention needs to be also given to indications of very slow clearance (e.g. perfluorooctanoic acid (PFOA) which is a Category 1B reproductive toxicant).

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[147] Registrant to provide data to support his registration.

[148] See e.g. Blagojević, J *et al.*, Age Differences in Bioaccumulation of Heavy Metals in Populations of the Black-Striped Field Mouse, *Apodemusagrarius* (Rodentia, Mammalia) *Int. J. Environ. Res., 6(4):1045-1052, Autumn 2012).*

[149] Steady state is achieved when the rate of elimination equals the rate of administration. Accumulation factor is 2 for a substance given once every half-live. Accumulation can be expected for a substance with slow elimination; e.g. with high octanol-water coefficient and no predicted hydrophilic metabolites. For lipophilic substances excretion may be impossible if there is no metabolism.

- Physico-chemical properties of the substance:
    - An octanol-water partition coefficient (log Kow) value (e.g. above 4.5) indicates (bio)accumulative potential (determined experimentally or estimated by QSAR models) of the substance and/or its metabolites unless the substance is fully metabolised to hydrophilic metabolites.
- Indications on (bio)accumulation in animals or from human biomonitoring data:
    - High levels of substance/metabolites in human body fluids or tissues, such as blood, milk or fat are indicative of a concern on accumulation and persistence. Substances of purely endogenous origin and high levels only due to high exposure are excluded;
    - Bioaccumulation potency, for example if the substance properties meet the bioaccumulation screening criteria described in Table C.4-1 of the *Guidance on IR&CSA - Part C: PBT/vPvB assessment*. The assessment approach is described further in Section R.11.4.1.2 of the *Guidance on IR&CSA - Chapter R.11: PBT/vPvB assessment*;
    - If the substance fulfils the bioaccumulation criterion (B or vB) described in REACH Annex XIII;
    - Indications of biomagnifications (high levels of the substance in biota or terrestrial animals in the top of food chains, resulting from the effective accumulation of the substance in organisms and the slow elimination (not from high releases). This is further discussed under 'Field data and biomagnification', page 52, Section R.11.4.1.2 of the *Guidance on IR&CSA - Chapter R.11 PBT/vPvB assessment*.
- Indications from existing *in vivo* studies that after longer exposure duration the effects are more severe/occurring at lower dose than would be expected based on assessment factors generally used to extrapolate the dose descriptor between studies with different exposure duration:
    - e.g. if the NOAEL/LOAEL of a subchronic study (90-day) is more than 3 times lower than the NOAEL/LOAEL from a subacute study (28-day), taking the dose level selection and other differences into account;
    - Effects observed only at a later time point in chronic studies, thus indicating a need to have a longer exposure time to cause the toxicity likely, due to accumulation of a substance or its metabolites.

(iii) "*There are indications of one or more relevant modes of action related to endocrine disruption from available in vivo studies or non-animal approaches*".

Indications of endocrine disrupting mode(s) of action[150] such as (anti)oestrogenicity, (anti)androgenicity or influence on thyroid hormone activity or other modes of action related to endocrine disrupting properties relevant to reproductive toxicity. These modes of action have been associated with adverse effects on fertility, reproductive performance or development of offspring. See Appendix R.7.6–5 of this Guidance for evaluation of triggers:

- Endocrine disrupting modes of action may be indicated from *in vivo* studies by 1) changes in organ weight sensitive to endocrine disrupting activity (intact and/non-

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[150] A comprehensive collection of screens and tests for endocrine disrupting chemicals are presented in OECD GD 150, covering the oestrogen receptor, androgen receptor and thyroid hormone mediated and steroidogenesis interference modalities. Both the test results for toxicity and ecotoxicity may be relevant.

intact animals), 2) (increased) body weight, 3) measurements of hormone levels, or 4) effects on reproduction associated to endocrine (disrupting) modes of action;

- Repeated dose toxicity studies, especially the 28-day repeated dose toxicity study (EU B.7, OECD TG 407) updated in 2008, may provide indication of endocrine (disrupting) modes of action. Check the parameters related to endocrine modes of action; e.g.:
    - Changes in reproductive organs and other endocrine organs (e.g. ovaries, testes, uterus, cervix, epididymides, seminal vesicles, coagulating glands, prostate, vagina, pituitary, mammary gland, thyroid and adrenal gland);
    - Changes in body weight (increase);
    - Alterations in oestrus cycle;
    - Changes in relevant hormone levels.
- Reproductive toxicity studies (e.g. a screening study) may provide indication of endocrine modes of action. Check the parameters related to endocrine modes of action; e.g.:
    - Changes in reproductive organs and other endocrine organs (see above);
    - Changes in anogenital distance, nipple retention, mammary gland histopathology or in any indicators of hormonal modes of action;
    - Changes is oestrus cycle;
    - Changes in gestation length;
    - Changes in body weight (increase);
    - Changes in pup body weight (increase not secondary to reduced litter size);
    - Other effects showing a likely endocrine disrupting mode of action.
- Endocrine effects from ecotoxicology studies and tests predicting endocrine (disrupting) modes of action (especially thyroid, see OECD GD 150);
- Non-animal approaches and specific animal studies may provide mechanistic data, information on receptor binding, epigenetics or other regulatory mechanism for endocrine (disrupting) modes of action, e.g.:
    - Uterotrophic assay (EU B.54, OECD TG 440) ;
    - Hershberger assay (EU B.55, OECD TG 441);
    - Performance-based test guideline for stably transfected transactivation *in vitro* assays to detect oestrogen receptor agonists (OECD TG 455);
    - H295R steroidogenesis assay (OECD 456);
    - BG1Luc Estrogen receptor transactivation test method for identifying oestrogen receptor agonists and antagonists;
    - Yeast Estrogen Screening (YES) and Yeast Androgen Screening (YAS) Tests;
    - Androgen receptor binding study;
    - Aromatase assay;
    - Endocrine organ cultures;
    - QSAR and computational predictions considered adequately reliable to serve as trigger(s).

The identified triggers should not be contradicted by other findings in the available data. The relevance and quality of triggers from the *in vivo* studies and non-animal approaches used must be adequately documented and justified. Case by case considerations are needed in evaluating trigger(s); evaluation is discussed in Appendix R.7.6–5 of this Guidance.

Further aspects to consider related to extension of the Cohort 1B and termination time for F2:

An extension of Cohort 1B to F2 is considered relevant in the context for classification and labelling and categorisation especially if the effect in P0 parental/F1 offspring is significant but not meeting classification criteria to Repr. 1B and more severe effects are seen in the F1 mating pairs/F2 offspring, thus affecting both P0 parental/F1 offspring and F1 mating pairs/F2 offspring but being more prominent or with a broader/different spectrum in F1 mating pairs/F2 offspring This could lead to a change in the classification from Repr. 2 to Repr. 1B.

Substances meeting the classification to Mutagen Category 2 are considered to have properties which increase the concern for reproductive toxicity and especially to the vitality and health of the second generation. The substance may have adverse effects on primordial germ cell development, proliferation and migration during *in utero* development, which may then be observed as reduced fertility in the F1 animals. Many genotoxic substances are also reproductive toxicants.

The test method for the extended one-generation reproductive toxicity study provides the possibility to terminate the F2 generation on postnatal day (PND) 4 based on a weight of evidence approach (integrated evaluation of the existing data). A weight of evidence adaptation approach according to REACH Annex XI, 1.2 could be usedfor example, if the results already meet the classification criteria to Repr 1B and it is highly likely that results from the rest of the lactation period (PND 5-21) would not lead to a lower NOAEL value. To cover the remaining uncertainty, an additional assessment factor may be applied.

The decision on whether or not to extend the Cohort 1B to F2 generation is/should be done before starting the study when the specified conditions are met. The testing proposal submitted by the registrant must include the study design proposed with justifications. During conduct of the experimental study the registrant is responsible for implementing the overall design of the study as requested, conduct of the study and interpretation of the results in order to meet the regulatory requirements and to insure the scientific integrity of the study in line with the test method.

So called internal triggers or in-study triggers for mating the Cohort 1B animals to produce the F2 generation (as those described in OECD TG 117) are not recommended to be used as such in REACH. However, the registrant may expand the study based on new information indicating a concern which needs to be addressed. The justification for the expansion must be documented.

3) Inclusion of Cohorts 2A and 2B

The main concepts of the triggers (conditions) for Cohort 2 (developmental neurotoxicity, DNT) are based on a particular concern for (developmental)[151] neurotoxicity[152]. A particular concern

[151] Both particular concerns for neurotoxicity as well as for developmental neurotoxicity may be addressed. See discussion in Section R.7.6.4.2.3 of this Guidance, under "General considerations related to investigation of (developmental) neurotoxicity and/or immunotoxicity" and "Proposals for developmental neurotoxicity or immunotoxicity studies".

[152] (Nielsen *et al*., 2008) "Signs of neurotoxicity in standard acute or repeated dose toxicity tests may be secondary to other systemic toxicity or to discomfort from physical effects such as a distended or blocked gastrointestinal tract. Nervous system effects seen at dose levels near or above those causing lethality should not be considered, in isolation, to be evidence of neurotoxicity. In acute toxicity studies where high doses are administered, clinical signs are often observed which are suggestive of effects on the nervous system (e.g. observations of lethargy, postural or behavioural changes), and a distinction should be made between specific and non-specific signs of neurotoxicity." "A consistent pattern of neurotoxic findings rather than a single or a few unrelated effects should be taken as persuasive evidence of neurotoxicity."

means that the concern should be specific to (developmental) neurotoxicity but also that the concern needs to reach a certain level of severity. Based on text inREACH Annex VIII, 8.6.1 for example, it can be understood that a particular concern may be indicated, such as by serious or severe effects[153]. The examples provided in the legal text at REACH Annex IX/X, 8.7.3, Column 2 also provide guidance on the "severity level" of triggers for a particular concern with words such as "evidence of adverse effects" and findings "associated to adverse effects". There should be sufficient evidence, weighing all the information, to raise a reasonable expectation that the substance could be a developmental neurotoxicant (see Appendix R.7.6–5 of this Guidance for evaluation of triggers).

REACH specifies that an extended one-generation reproductive toxicity study including Cohorts 2A and 2B (developmental neurotoxicity cohorts) shall be proposed by the registrant or may be required by ECHA if a particular concern on (developmental) neurotoxicity.

Conditions for a particular concern for developmental neurotoxicity:

- existing information on the substance itself derived from relevant available *in vivo* or non-animal approaches, or
- specific mechanisms/modes of action of the substance with an association to (developmental) neurotoxicity, or
- existing information on effects caused by substances structurally analogous to the substance being studied, suggesting such effects or mechanisms/modes of action.

For the precise legal text see REACH regulation, Annexes IX and X, 8.7.3. The registrant must record the findings and justify the existence or nonexistence of the trigger(s) for the need to include the Cohorts 2A and 2B.

Examples of substance specific findings which may indicate a particular concern justifying inclusion of the developmental neurotoxicity cohort:

- abnormalities observed in the central nervous system or nerves
  - changes in brain weight or in specific neural areas not secondary to body weight
  - changes in brain volume or specific neural areas, obtained e.g. from morphometry/stereology measurements
  - (histo)pathological findings in brain, spinal cord and/or nerves (e.g. sciatic nerve)
- any signs of behavioural or functional adverse effects on the nervous system in adult studies e.g. repeated-dose and acute toxicity studies and neurotoxicity studies, not likely to be secondary to general toxicity.
  - clinical and/or behavioural signs (such as abnormal gait, narcosis, seizures or any other altered activity) if seen in absence of general toxicity
- specific mechanism/mode of action that has been closely linked to (developmental) neurotoxic effects (see Gupta RC (2011), pages 835-862),
  - (adult) brain cholinesterase inhibition (by 20%);
  - relevant changes in thyroid hormone levels or signs of thyroid toxicity indicating such changes,

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[153] A serious or severe effect is an effect which has regulatory consequences, i.e. leads to a NOAEL values and/or contributes to hazard classification. Thus, a particular concern is an expectation that the substance has (developmental) neurotoxic properties contributing to the regulatory decision making. This also means that they are not secondary to other systemic toxicity.

        - information on specific hormonal mechanisms/modes of action with clear association with the developing nervous system, such as oestrogenicity (Fryer *et al.*, 2012) and antiandrogenicity (Pallarés *et al*,. 2014)

- Information from (validated) non-animal approaches, such as from an *in vitro* developmental neurotoxicity test (see de Groot, 2013), predicting developmental neurotoxicity, e.g.:
    - Any sign of adverse neuronal differentiation *in vitro* e.g.:
        - Neurite outgrowth
        - Neural stem cell proliferation
        - Gene expression (mRNA and protein) biomarkers that are linked to neuronal differentiation, synaptogenesis and other neurodevelopmental differentiation
    - Functional endpoints, e.g. cell membrane potential, excitability, electrical activity
    - Specific modes of action that are linked to neurotoxic effects *in vivo* can be indicated *in vitro* by non-validated assays, e.g. cholinesterase inhibition, neuropathy target (neurotoxic) esterase inhibition.
- structurally analogue substances show (developmental) neurotoxic effects in *in vivo* or *in vitro* studies suggesting that similar effects or similar mechanisms/modes of action are likely to apply also for the registered substance (see the examples above for substance specific findings)
    - adequacy of an approach to use the trigger(s) from an analogous substance must be justified

The identified triggers should not be contradicted by other findings in the available data. The relevance and quality of triggers from the *in vivo* studies and non-animal approaches used must be adequately documented and justified. Evaluation of triggers is described in Appendix R.7.6–5 of this Guidance.

Further consideration related to adults vs developmental neurotoxicity is provided in Section R.7.6.4.2.3, of this Guidance under "General considerations related to investigation of (developmental) neurotoxicity and/or immunotoxicity".

4) Inclusion of Cohort 3

The main concepts of the triggers (conditions) for Cohort 3 (developmental immunotoxicity, DIT) are based on a particular concern for (developmental) immunotoxicity[154]. A particular concern means that the concern should be specific to (developmental) immunotoxicity but also that the concern needs to reach a certain level of severity. Based on text in REACH Annex VIII, 8.6.1 for example, it can be understood that a particular concern is indicated, such as by serious or severe effects[155]. The examples provided in the legal text at REACH Annex IX/X, 8.7.3, Column 2 provides also guide on the "severity level" of triggers for a particular concern

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[154] Both particular concerns for immunotoxicity as well as for developmental immunotoxicity may be addressed. See discussion in Section R.7.6.4.2.3 of this Guidance, under "General considerations related to investigation of (developmental) neurotoxicity and/or immunotoxicity" and "Proposals for developmental neurotoxicity or immunotoxicity studies".

[155] A serious or severe effect is an effect which has regulatory consequences, i.e. leads to a NOAEL values and/or contributes to hazard classification. Thus, a particular concern is an expectation that the substance has (developmental) immunotoxic properties contributing to the regulatory decision making. This also means that they are not secondary to other systemic toxicity.

with wordings such as “evidence of adverse effects” and findings “associated to adverse effects”. There should be sufficient evidence, weighing all the information, to raise a reasonable expectation that the substance could be a developmental immunotoxicant (see Appendix R.7.6–5 of this Guidance for evaluation of triggers).

REACH specifies that an extended one-generation reproductive toxicity study including Cohort 3 (developmental immunotoxicity cohort) shall be proposed by the registrant or may be required by ECHA if a particular concern on (developmental) immunotoxicity.

Conditions for particular concern for developmental immunotoxicity:

- existing information on the substance itself derived from relevant available *in vivo* or non-animal approaches, or
- specific mechanisms/modes of action of the substance with an association to (developmental) immunotoxicity, or
- existing information on effects caused by substances structurally analogous to the substance being studied, suggesting such effects or mechanisms/modes of action.

For the precise legal text see REACH regulation, Annexes IX and X, 8.7.3. The registrant must record the findings and justify the existence or nonexistence of the trigger(s) for the need to include the Cohort 3.

Examples of substance specific findings which may indicate a particular concern justifying inclusion of the developmental immunotoxicity cohort:

- Combination of at least two (statistically significant and) biologically meaningful changes in haematology/clinical chemistry and/or organ weight associated with immunotoxicity, e.g. reduced leucocyte count in combination with reduced spleen weight.
- One severe (see footnote 43) statistically and/or biologically significant organ weight or histopathological finding related to an immunology organ, e.g. thymus atrophy.
- (respiratory) sensitisation (as a supportive factor only)
- Information on changes in immune function involving innate (e.g. NK-cell function, phagocytosis and oxidative burst) or acquired immunity (e.g. generation of immunological memory, cytotoxic T-cells and antibody production)
- Information on hormonal mechanisms/modes of action with clear association with the immune system, such as oestrogenicity (Adori *et al.*, 2010) and androgenicity (Trigunaite *et al.*, 2015) .
- Structural similarity with a substance causing structural or functional immunotoxicity or suggesting a similar mechanism/mode of action (see the examples above for substance specific findings)
    - adequacy of an approach to use the trigger(s) from an analogous substance must be justified

WHO Guidance document for immunotoxicity provides further examples of potential triggers for immunotoxicity testing (WHO, 2012). All effects on any immune-parameters found either *in vivo* (adult animals), or predicted *in vitro* or *in silico* may have impact on the developing immune system. These effects could be defined as quantitative or qualitative changes in cell counts or histopathology studying immune-specific organs or cell-populations in peripheral blood but may also include functional end-points such as antibody-production, delayed-type hypersensitivity test (to investigate cytotoxic T-cell activity), cytokine production, lymphocyte proliferation, NK-cell-function, phagocytosis, and oxidative burst.

The identified triggers should not be contradicted by other findings in the available data. The relevance and quality of triggers from the *in vivo* studies and non-animal approaches used must be adequately documented and justified. Evaluation of triggers is described in Appendix R.7.6–5 of this Guidance.

Appendix R.7.6–3 Premating exposure duration in the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443)

1. Importance of the premating exposure duration

The two main aspects in a reproductive toxicity study influencing how well fertility parameters and thus, the potential adverse effects on fertility can be evaluated are the length of the premating exposure duration and dose level setting.

The fertility part of the reproductive toxicity study should be capable of providing information on fertility that is adequate for both risk assessment and classification, including categorisation. For the classification purpose, it is important to produce and evaluate the full spectrum of effects on fertility. Just to detect a most sensitive effect may not be enough for deciding on classification categorisation because full information on magnitudes, incidences, severity and types of all effects (MIST information) should be evaluated together to assist the decision.

If the registrant applies ten weeks premating exposure duration in an extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) no justification for premating exposure duration is needed. Substance specific justifications should be provided substantiated with data if shorter than ten weeks premating exposure duration is proposed.

Further insight to female and male reproduction toxicity can be obtained in relevant chapters of books for reproductive toxicology (such as Korach (1998) Reproductive and developmental toxicology; Gupta (2011) Reproductive and developmental toxicology).

1.1 Main parameters for evaluating effects on fertility

Mating/fertility

Mating and fertility are functional parameters which include effects on mating behaviour and fertility outcome. Parameters such as precoital interval, mating index, fertility index, preimplantation loss, post-implantation loss, number of corpora luteae, number of implantations, number of resorptions, dead foetuses, abortions, gestation length, litters size, and number of live pups are measuring effects on fertility (some of these parameters may also reflect developmental toxicity).

The length of the premating exposure may influence the mating and fertility parameters if the substance 1) causes adverse effects on primordial germ cell development, their migration and/or proliferation in embryo/foetus, 2) causes adverse effects on sperm development and maturation, 3) causes adverse effects on follicle development and/or development of ovum, 4) causes adverse effects on brain sexual development, 5) causes effects on hypothalamus-pituitary-gonad axis or other effects on hormonal control mechanisms.

The primordial germ cells already develop, migrate and proliferate during embryonic development. In addition to histopathological analysis of gonads, organ weight measurements and sperm parameter analysis, adverse effects on germ cell development/migration/proliferation during these early stages, as well as the other effects listed above, can be fully evaluated only by exposing the animals already *in utero* and then until adulthood and mating them. This full evaluation is possible if the mating and littering of the Cohort 1B animals is triggered in an extended one-generation reproductive toxicity study (EU B.56, OECD TG 443).

An effect on fertility may be due to exposure *in utero,* postnatal period or during adulthood. In some cases it may be possible to conclude that effects on fertility are of developmental origin. For instance if there is information on fertility in both the parental animals and their offspring and effects on fertility are only seen in the mature offspring.

Sperm parameter analysis

Sperm parameter analysis includes for example, total cauda epididymal sperm number, percent progressively motile sperm, percent morphologically normal sperm and potentially percent of sperm with each identified abnormality for animals. The sperm count is measured by counting the number of sperm in cauda epididymis (sometimes also from testis as homogenisation resistant spermatid counts).

In the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) these parameters are to be reported for both the P and F1 males at termination. Other studies required in REACH as standard information requirements do not normally report results from sperm parameter analysis.

Sperm parameter analysis informs on the number of cauda epididymal sperm and their quality in terms of motility and morphological normality. The results from a sperm parameter analysis reflects the effects during the spermatogenic cycle in testes and during the epididymal maturation, if the exposure is long enough to cover both of these periods. The ability of sperm to fertilise eggs and produce alive and healthy offspring is examined in the reproductive toxicity studies by mating the animals and letting them litter. If the measurement of sperm parameters coincides close to mating, it assists and supports the evaluation of effects on fertility with the same exposure history through the same life stages. Sperm parameters may provide important information because in humans even a slight reduction in sperm quality/count may be critical for fertility.

Oestrous cycle

Oestrous cycle measurements reflect the normality of the hormonal level changes affecting the responsiveness of females. Direct measurements on function of hypothalamus-pituitary-gonad axis are not generally done in reproductive toxicity studies. In an extended one-generation reproductive toxicity study it is important to measure the oestrous cycle before mating and also after sexual maturity.

Organs weights of gonads and accessory sex organs

Organ weights of gonads and accessory sex organs, together with other parameters, can predict effects on fertility. These measurements can only be done at termination. Thus, this information can be obtained from the P males soon after mating but from the P females only after weaning. However, the measurements should be done as close as possible with the other information because information from various sources after the same exposure history allows combined and meaningful evaluation of effects on fertility based on all the data.

Histopathology of gonads and accessory sex organs

Histopathology of testes, ovaries and accessory sex organs can only be done at termination. Histopathological evaluation of testes allows assessment of the structural normality of testes including Leydig cells, Sertoli cells and seminiferous tubules with various developmental stages of sperm (e.g. Russell *et al*., 1990). The information is generally qualitative, and quantitative measurements are not made and not required in test methods. Thus, it may not be possible to judge the amount of various cell types including the amount of various developmental stages of sperm. There may be a reduction of sperm at one developmental stage but it may be difficult to evaluate. Histopathological evaluation should reveal if multinuclear cells are present or another effect on sperm development if a significant reduction in the amount of certain cell types or their developmental stages is present. The information obtained is related to the morphological normality of testes but does not inform on the functional fertility and ability of the sperm to fertilise the eggs.

Histopathological evaluation of ovaries is complicated. The structure of an ovary is not organised and follicles at various developmental stages are distributed throughout the organ without a clear system. Thus, to count the number of follicles at different developmental stages requires several slices for histopathological examination. Quantitative evaluation of various cell types (e.g. granulosa cells and theca cells), indicative of toxicity is not generally done or required in the test methods. The number of primordial follicles (which can be combined with small growing follicles) is counted in the extended one-generation reproductive toxicity study in F1 animals which reflects the number of potential ova for future ovulations. The number of primordial and small growing follicles does not inform on actual functional fertility of the females but if the follicle number is reduced, it is a clear indication of gonad toxicity and should be taken into account in assessing effects on fertility.

Histopathology of accessory sex organs provides valuable information on how these organs have been developed and their morphological normality. The information should be evaluated together with other fertility findings.

It is important to be able to analyse the histopathological findings after the same exposure length and history as the other effects, including mating, to be able to understand a full picture of the spectrum of effects. Information on morphology is one important parameter in evaluation but as a stand alone measurement, to focus on morphology is too limited in order to provide a comprehensive picture of all the relevant aspects of fertility. However, it may be sufficient for classification (e.g. findings in histopathology alone from a repeated dose toxicity study may meet the classification criteria to category 1B for reproductive toxicity).

1.2 Ten weeks premating exposure duration

The full spermatogenesis, without sperm maturation, and folliculogenesis take 48-53 and 62 days in rats, respectively (e.g. Kerr *et al*., 2006; McGee and Hsueh, 2000). In addition to spermatogenesis, sperm maturation in rats takes around two weeks in epididymides. When the exposure is long enough, it covers both the sperm and follicle development through all the stages. Ten weeks premating exposure duration covers the full spermatogenesis and maturation meaning that the full cycle of development of sperm from spermatogonia into mature sperm is exposed. Thus, ten weeks premating exposure duration allows an assessment of the adverse effects on fertility by combining the information from all possible parameters in males evaluated at the same time. Similarly, the folliculogenesis, which lasts around 62 days, is fully covered only after a long exposure period, such as ten weeks. It is important to expose all the developmental stages of the sperm and follicles before the mating in order to be able to evaluate any potential adverse effect on fertility. Earlier stages of the spermatogenesis have been reported to be generally more sensitive than later stages to chemical and radiation exposure (Sjöblom *et al*., 1995) which also support that the exposure should cover all the stages before the mating.

For a comprehensive assessment of effects on fertility, which is often needed when deciding on classification for fertility effects, evaluation of the full spectrum of effects on fertility is necessary. Information from a limited number of parameters does not allow a conclusion on the absence of effects on fertility. The best outcome can be obtained when mating is allowed after an exposure covering one full spermatogenic cycle (including sperm maturation) and folliculogenesis, and an analysis of sperm parameters, organ weights and histopathology of gonads and accessory sex organs are conducted around the same time after the same exposure history.

In the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443), ten weeks premating exposure duration together with sperm parameter analysis, organ weights and histopathology of testis and accessory sex organs with the same exposure history is achievable for males. For females the organ weight measurements and histopathological analysis of gonads and accessory sex organs can only be made later and not near to mating. However, it is considered that the most important aspect is that the exposure duration for the female gonads covers the folliculogenesis before mating.

Organ weights (e.g. Bailey *et al*., 2004; Sellers *et al*., 2007; Hood *et al*., 2011)) and/or histopathology (e.g. Jacobson-Kram and Keller, 2006) of gonads may be among the most sensitive parameters for male fertility. For instance, testicular weight is quite a stable parameter because generally it is not influenced by small or moderate changes in body weight. Several studies have not established a correlation between testes-to-body weight and testes-to-brain weight (Bailey *et al*., 2004). Therefore, it could be concluded that variations on testicular weight will be linked to direct effects within the testes.

The most sensitive parameter showing an adverse effect is used to derive the NOAEL. However, the findings from the most sensitive parameters may not be sufficient for deciding on classification, including categorisation because the value of the NOAEL is not predictive for classification and (other) effects may be more relevant for classification purposes than the effect leading to a NOAEL. It is the clarity and the spectrum of the effects observed which counts for the classification and labelling. Thus, to address the fertility also for the classification and labelling purposes, including the categorisation, it is necessary to consider how well all the available parameters address the fertility endpoint. Information on magnitude, incidence, severity and type of all effects (MIST) influence on the classification, including categorisation. Evaluation of various parameters after the exposure length covering the critical reproductive aspects and after the same exposure history improves the quality of the assessment.

Environmental factors, such as chemical substances, pesticides, high temperatures and radiation have been associated with a reduction of sperm DNA integrity in infertile men (Evgeni *et al.*, 2014). It is to be noted that some effects on sperm, such as DNA fragmentation, may affect fertility and cannot be examined by routine gonadal histopathology (morphology) or sperm analysis. Several studies have attempted to investigate the possible correlation between human sperm DNA fragmentation and conventional sperm parameters. Most of them found an inverse correlation between DNA fragmentation rate and sperm quality (Evgeni *et al.*, 2014). In contrast, several authors have failed in finding a correlation between DNA fragmentation and standard sperm parameters, such as sperm concentration, motility and morphology (Evgeni *et al.*, 2014). The extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) does not contemplate sperm DNA damage assessment and consequently would not identify those cases where a reduction of sperm DNA integrity is not manifested in routine histopathology or in sperm parameter analysis.

The blood-testis barrier prevents free exchange of large proteins and some xenobiotics between the blood and the fluid within the seminiferous tubules (see Gupta, 2011, page 14). This may prolong time needed before the substance reaches the developing sperm supporting a long premating exposure duration. Thus, the blood level measurements of a substance may not reflect the exposure levels within the seminiferous tubules at a given time. All the different cell types representing various developmental stages of spermatogenesis are available in the testis at the same time and may allow detecting an adverse effect for a specific developmental stage or stages. However, a potential cumulative effect requiring exposure through several sequential stages cannot be detected with limited exposure duration.

In summary, the ten weeks premating exposure duration is one of the elements together with the appropriate dose level selection which allow production of data for an informed decision making for classification and labelling, including categorisation, for the hazard endpoint for sexual function and fertility according to CLP Regulation and for risk assessment.

1.3 Shorter than ten weeks premating exposure duration

Shorter than ten weeks premating exposure duration may be used based on substance specific justifications, but not shorter than 2 weeks. It is important to consider and document the reasoning why it is assumed that a longer premating exposure duration will not induce more or more severe effects.

A two weeks premating exposure duration is equivalent to the time for epidymal transit of maturing spermatozoa and thus, allows only the detection of post-testicular effects on sperm at mating (during the final stages of spermiation and epididymal sperm maturation). With a two-week premating exposure, the effects on functional fertility of exposure to the early stages of developing spermatozoa will not be covered as described above under heading 1.2.

The two weeks premating exposure duration is considered adequate to detect most of the male reproductive toxicants according to OECD GD 151. For females, two weeks premating exposure duration covers 2-3 oestrous cycles and effects on cyclicity may be detected. The detection of an effect may be adequate for NOAEL derivation but for classification and labelling purposes, including categorisation, information on magnitude, incidence, severity and all type of effects, i.e. full spectrum of effects is important (see text under heading 1.2).

Exposure during the full spermatogenic period and ovarian folliculogenesis are not covered at the time of mating, therefore, if only two weeks premating exposure duration has been selected, effects at earlier stages of spermatogenesis and folliculogenesis cannot be reflected in the functional fertility examination. This is a disadvantage and limited information may not allow adequate evaluation, including categorisation for classification, of potential adverse effects on fertility. It is to be noted that for the screening study (OECD TGs 421 or 422) the histopathological data will be limited also due to the limited duration of the whole study and limited statistical power as compared to the more comprehensive reproductive toxicity study such as the extended one-generation reproductive toxicity study.

A two-week premating period may be too short to produce results appropriate to conclude whether the substance meets the criteria for a category 1B reproductive toxicant, and thus may not be sufficient for classification and labelling purposes. Under point 2 below some considerations are presented on when a shorter than ten weeks premating exposure duration could be applied. In these cases substance specific justifications must be provided.

2. Considerations to be made in deciding if shorter than a ten weeks premating exposure duration could be adequate

2.1 Starting Point

To adequately assess the fertility endpoint, the best place to start considering the length of the premating exposure period should be ten weeks. Ten weeks cover the full spermatogenesis, sperm maturation and folliculogenesis before the mating allowing a meaningful assessment with the full spectrum of the effects after the same exposure history.

Based on substance specific justifications a shorter premating exposure duration may be proposed, but it should not be shorter than two weeks and sufficiently long to reach a steady-state (in reproductive organs) if such kinetic information is available.

2.2 Examples of cases where the existing information may support shorter than ten weeks premating exposure duration

If the registrant prefers another length of premating exposure duration than ten weeks, an acceptable substance-specific scientific justification substantiated with adequate data should be provided.

Such reasoning could be that *effects on fertility are already adequately addressed* and the extended one-generation reproductive toxicity study is used to address developmental toxicity. (It is, however, to be noted that the extended one-generation reproductive toxicity study does not provide equivalent information to the prenatal developmental toxicity study and thus cannot replace a prenatal development toxicity study)

There may be existing information from a good quality one-generation reproductive toxicity study (EU B.34, OECD TG 415) or similar addressing the fertility parameters. If information on a good quality one-generation reproductive toxicity study is available, then the fertility parameters are normally covered with adequate statistical power and the premating exposure duration may be shorter in a planned extended one-generation reproductive toxicity study.

An extended one-generation reproductive toxicity study is normally still needed to address the standard information requirement in REACH Annex IX/X, 8.7.3 because a one-generation reproductive toxicity study (EU B.34, OECD TG 415) does not cover the extended exposure period of F1 animals and the same parameters (e.g. sexual maturity and hormonal activity). In addition, the column 2 provisions of REACH Annex IX/X, 8.7.3 are not covered by one-generation reproductive toxicity study if triggered (for further details see Appendix R.7.6–2 of this Guidance for the extended one-generation reproductive toxicity study and Section R.7.6.4.2.3 of this Guidance, under "Proposals for developmental neurotoxicity or immunotoxicity studies" for separate developmental neurotoxicity and separate immunotoxicity studies).

There may be existing information from a good quality two-generation reproductive toxicity study (EU B.35, OECD TG 416) addressing the fertility parameters. If information on a good quality two-generation reproductive toxicity study is available, then the standard information requirement in REACH Annex IX/X, 8.7.3 is covered and an extended one-generation reproductive toxicity study may not be needed. However, the registrant must fulfil the column 2 provisions regarding developmental neurotoxicity and/or developmental immunotoxicity if the triggers are met. In these cases the registrant may consider fulfilling the adaptation requirements by proposing separate developmental neurotoxicity and/or developmental immunotoxicity study rather than an extended one-generation reproductive toxicity study. Similarly, if there are concerns related to the endocrine disrupting modes of action/properties not assessed in an existing two-generation reproductive toxicity study but which would have been measured in an extended one-generation reproductive toxicity study, the registrant may consider addressing these concerns in separate studies or add relevant parameters to other studies to be conducted (for further details see Appendix R.7.6–2 of this Guidance for the extended one-generation reproductive toxicity study and Section R.7.6.4.2.3, of this Guidance under "Proposals for developmental neurotoxicity or immunotoxicity studies" for separate developmental neurotoxicity and separate immunotoxicity studies).

There may also be cases where the fertility effects based on the existing information do meet the criteria for Reproductive toxicity Category 1B, but the column 2 adaptation (REACH Annex IX/X, 8.7) is not applicable due to further concerns on developmental toxicity. This information on fertility effects may stem for example, from good quality repeated dose toxicity studies (sex organ weights, histopathology of gonads and/or accessory sex organs, sperm parameters analysis), screening studies (OECD TGs 421 or 422; e.g. reduced fertility, litter size) or equivalent. In these cases, as the fertility is already addressed, shorter premating exposure duration in an extended one-generation reproductive toxicity study, if conducted to address the developmental toxicity, may be considered. If the effects from these studies only meet the classification criteria for Reproductive toxicity Category 2 for fertility, those should not be used as an argumentation to reduce the premating exposure length as the findings should be confirmed in a more comprehensive reproductive toxicity study (EU B.56, OECD TG 443).

There may be good quality information from existing repeated dose toxicity 90-day studies showing no effects in organ weights or histopathology of reproductive organs, and covering also the spermatogenesis and folliculogenesis. However, this information alone, or with the results from a screening study (OECD TGs 421 or 422) may not provide adequate confidence to shorten the premating exposure duration from ten weeks. This is because the information on mating and fertility from a screening study as well as the data from the repeated dose toxicity study is limited. Mating and fertility data from screening studies (OECD TGs 421 or 422) is after two weeks premating exposure duration not covering the full spermatogenesis and folliculogenesis and may also not be adequately long enough for detecting toxicity in hypothalamus-pituitary-gonad axis. In addition, the statistical power is low in these studies as they are not meant to provide comprehensive information on reproductive toxicity. Repeated dose toxicity 90-day studies may provide information on organ weights and histopathology but no mating data. The statistical power in the 90-day study is lower than that in the extended one-generation reproductive toxicity study also considering the data for histopathology. In addition, the exposure duration and exposure history are different in screening studies (OECD TGs 421 or 422) and 90-day studies. Thus, it may be difficult to conclude based on this information that a two weeks premating exposure duration is sufficient for a substance in question. However, the registrant may have additional information that may provide elements which together may support the justification, such as very low general toxicity (no effects up to the limit does of 1000 mg/kg bw/day in any of the existing studies), fast elimination, no distribution to sex organs, accessory sex organs and brain, and no concern on germ cell toxicity/mutagenicity (no effect in germ cell mutagenicity test). The substance specific justifications should be substantiated with adequate data.

Results showing no effects or some effects in reproductive organ weights and histopathology from a 28-day repeated dose toxicity study generally do not provide conclusive information to justify a shorter than ten weeks premating exposure duration. First of all, the length of the study is only 28-days and not covering the full spermatogenesis and folliculogenesis and the statistical power is low due to low number of animals.

Finally, if animals of Cohort 1B in an extended one-generation reproductive toxicity study are mated to produce the F2 generation, then the premating exposure duration will be ten weeks for these Cohort 1B animals and the fertility parameters will be covered allowing an evaluation of the full spectrum of effects on fertility. In these cases, shorter premating exposure duration for parental (P) animals may be considered. The consideration should take into account whether the findings from P animals (such as clinical signs, clinical chemistry, haematology) after a longer premating exposure would provide important information for interpretation of the findings in F1 animals, for example, when considering the potential developmental origin of such findings. It is to be noted that the results of the hazard class classification may differ depending on the interpretations of the origin of the results (differences in classification for specific target organ toxicity and developmental toxicity).

3. Summary

To fully evaluate effects on fertility, effects on all critical aspects and development stages should be covered; this can be done only by exposing the animals already *in utero* and then until adulthood and mating them. This full evaluation is possible if the extension (mating) of the Cohort 1B animals is triggered in an extended one-generation reproductive toxicity study (EU B.56, OECD TG 443). The premating exposure duration of ten weeks is also covered in mated Cohort 1B animals.

If the extension of the Cohort 1B animals is not triggered, a ten -week premating exposure duration should be the starting point. This allows for assessing the consequences of early effects on the sex organs (spermatogenesis and folliculogenesis) assessor sex organs, hypothalamus-pituitary-gonad axis, and for example, prolonged distribution or any accumulation to relevant organs and tissues.

The registrant may prefer another length of premating exposure duration and substance specific justifications are needed to support shortened premating exposure duration.

Appendix R.7.6–4 Procedure for testing approaches and adaptation; Stage 3 - Stages 3.1.1 – 3.1.8

General adaptation rules of REACH Annex XI and certain specific adaptation rules in Column 2 provide possibilities for omitting the testing. These rules, except for those already passed at Stage 1, are presented here and the possibilities to omit the testing according to Stages 3.1.1 – 3.1.8 should be explored before conducting (REACH Annex VIII level test) or proposing (REACH Annexes IX and X level tests) the test.

Stage 3.1.1 Adaptation based on existing information not carried out according to GLP or the test methods indicated in the test method regulation (REACH Annex XI, 1.1.2)

Although the REACH standard information requirements refer to a specific series of reproductive studies, it is recognised that there may be other studies already performed that could address some of the endpoints covered by these standard protocols, reducing the need for new animal testing (adaptation according to REACH, Annex XI 1.1.2). The available data should be evaluated to assess their suitability for use, taking account of the robustness of design, and quality as outlined in Chapter R4 (*Guidance on IR&CSA, Chapter R4: "Evaluation of available information"*). The data from these studies (one or several together) are considered to be equivalent to data generated by the REACH standard test methods if the conditions of REACH Annex XI, Section 1.1.2 are met. An illustrative summary of these conditions is given below:

1) adequate for classification and labelling and/or risk assessment;
2) adequate and reliable coverage of key parameters;
3) exposure duration comparable or longer, if exposure duration is a relevant parameter;
4) adequate and reliable documentation;

a. adequate and reliable reporting of study design including dose levels tested.

Examples of other studies include: old studies conducted in other than preferred species; an NTP[156] modified one-generation study; non-GLP studies; or non-guideline investigations such as the NTP continuous breeding study (Chapin and Sloane, 1997).Such studies may be available and should be evaluated for fulfilling the criteria in REACH Annex XI, Section 1.1.2, in order to conclude that the information provided is equivalent to that foreseen to be the information provided by the EU test method. In addition, a study conducted according to a new test method not yet internationally acceptable may be valid and provide equivalent information.

It is to be noted that existing information on the two-generation reproductive toxicity study (EU B.35, OECD TG 416) is considered to fulfil the standard information requirement for REACH Annex IX/X, 8.7.3 (EU B.56, OECD TG 443), because this was the previous standard information requirement before the revision of the REACH Annexes to require an extended one-generation reproductive toxicity study. For further details see Section R.7.6.4.2.4 of this Guidance on the two-generation reproductive toxicity study (EU B.35, OECD TG 416).

Tests carried out according to old methods are evaluated case by case taking into account the toxicological properties of the substance. If the old study has for example, shorter exposure duration than the current test method, the registrant should justify using substance-specific arguments why the study with shorter exposure duration does not cause concern; for an example see Section R.7.6.4.2.2 of this Guidance. Similarly, if not all the key parameters are measured, but there are adequate substances-specific justifications to

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show that the missing information is of no concern, the old study may be acceptable. If the conditions summarised above for REACH Annex XI, 1.1.2 are not met, the study or test could still be of use for example, under REACH Annex XI, 1.2 as one element for weight of evidence adaptation.

Stage 3.1.2 Adaptation based on existing historical human data (REACH Annex XI, 1.1.3)

Epidemiological studies, conducted in the general population or in occupational cohorts, may provide information on possible associations between exposure to a chemical and adverse effects on reproduction. Clinical data and case reports (e.g. biomonitoring after accidental substance release) may also be available.

The criteria for assessing the adequacy of historical human data are listed in REACH Annex XI, Section 1.1.3. In exceptional cases human data may meet the classification criteria to Reproductive toxicity Category 1A and provide adequate information for risk assessment.

Stage 3.1.3 Adaptation based on existing information in a weight of evidence approach (REACH Annex XI, 1.2)

There are two possibilities to use the weight of evidence adaptation:

1) sufficient evidence from several independent sources of information; or
2) sufficient evidence from the use of newly developed test methods

leading to the conclusion that a substance has or has not a particular hazardous property.

It is to be noted that the weight of evidence approach described in REACH Annex XI, Section 1.2 needs to be substance and case specific and address the relevant standard information requirements of REACH Annex VII to X. Furthermore, it is hazard-based and therefore it has to be shown whether a substance has or has not a particular hazardous property. Because the weight of evidence approach is hazard-based, it means that exposure conditions or risk considerations are not part of the approach. To address the particular hazardous property of a substance, the key aspects/parameters of the study of the (standard) information requirement for which a weight of evidence approach is proposed need to be addressed to a sufficient extent.

In any case, adequate and reliable documentation of the information needs to be provided.

Adequate reporting of a weight of evidence approach is explained in the ECHA Practical Guide 2 (http://echa.europa.eu/documents/10162/13655/pg_report_weight_of_evidence_en.pdf).

Elements of a weight of evidence adaptation approach according to this adaptation rule for reproductive toxicity could be available from experimental studies addressing reproductive toxicity endpoints, reproductive toxicity studies performed with structurally similar substances and non-animal approaches, such as suitable validated *in vitro* methods, valid qualitative and quantitative structure-activity relationship models ((Q)SARs) or adverse outcome pathways (AOPs) (for further information on non-animal approaches see Stages 3.1.4 and 3.1.5).

Stage 3.1.4 Adaptation based on non-animal approaches such as QSAR approaches and in vitro methods (REACH Annex XI, 1.3 and 1.4)

REACH Annex XI, Sections 1.3 "Qualitative or Quantitative structure-activity relationship (QSAR) and Section 1.4 "*in vitro* methods" are potential adaptation possibilities. However, the available methods are currently not sufficient to address the complex endpoints on reproductive toxicity to replace an animal test. QSAR and *in vitro* methods may be used to support grouping and read-across approaches and may have a role in weight-of-evidence approach. For further details see Section R.7.6.4.1.1 of this Guidance.

Stage 3.1.5 Adaptation based on grouping and read-across (REACH Annex XI, 1.5)

The grouping of substances and read-across offer a possibility for adaptation of the standard information requirements of the REACH Regulation. If the read-across approach is adequate, unnecessary testing can be avoided. A read-across approach can also support a conclusion for a REACH endpoint using a weight of evidence approach.

The application of the grouping concept means that REACH information requirements for physicochemical properties, human health effects and/or environmental effects may be predicted from tests conducted on reference substance(s) within the group, referred to as source substance(s), by interpolation (extrapolation is generally not recommended for grouping) to other substances in the group, referred to as target substance(s) and this is called read-across.

The read-across approach has to be considered on an endpoint-by-endpoint basis due to the different complexities (e.g. key parameters, biological targets) of each endpoint. This means that read across (and category approach) is endpoint specific.

The term analogue approach is used when read-across is employed within a group of a very limited number of substances.

Read-across must, in all cases, be justified scientifically and documented thoroughly. There may be several lines of evidence used to justify the read-across, with the aim of strengthening the case.

Guidance on read-across is provided in the *Guidance on IR&CSA, Chapter R.6 "QSAR and grouping of chemicals"*. Further guidance can be found following this link: http://echa.europa.eu/support/grouping-of-substances-and-read-across.

Stage 3.1.6 Testing is technically not possible (REACH Annex XI, Section 2)

Tests do not need to be performed if it is not technically possible to do so. It may be that it is not possible to administer the substance for a particular reason. For example, the substance may be flammable in air or degrades explosively. It may also not be possible to produce sufficiently high enough exposure levels due to technical reasons. Justification for not performing tests is required and must be documented.

Stage 3.1.7 Substance-tailored exposure-driven testing (REACH Annex XI, Section 3)

The information requirements for reproductive toxicity at REACH Annex VIII, IX, and X levels may be omitted *if relevant human exposure can be excluded*. This clause states that tests may be omitted based on exposure scenarios developed in the Chemical Safety Report. The criteria defines three alternative sets of conditions that can, when justified and demonstrated, lead to an adaptation of standard information requirements (REACH Annex XI, 3.2.(a), (b) or (c)).

The adaptation according to REACH Annex XI Section 3.2.(a) is usually not applicable for REACH Annex IX and X reproductive toxicity studies as a DNEL derived from a reproduction/developmental toxicity screening test must not be considered appropriate to omit prenatal developmental toxicity study or an extended one-generation reproductive toxicity study (see REACH Annex XI, 3.2(a)(ii)footnote).

At REACH Annex IX level, the triggered prenatal developmental toxicity study on a second species may not need to be conducted based on a case-by-case justification. Such a justification may include the observation that triggers for the study on a second species are only at very high exposure levels compared with the identified and documented human exposure and that there are substance specific justifications that the second species would not be more sensitive/relevant to humans than the first species used. In such cases the DNEL derived based on the results from the first species may suffice although there were triggers for the study on second species.

For substances following strictly controlled conditions as described in REACH Annex XI, 3.2(b) or for substances rigorously permanently incorporated in an article according to REACH Annex XI, 3.2(c), the use of substance-tailored exposure-driven waiving may be possible.

In all cases, adequate justification and documentation must be provided (see REACH Annex XI, 3.2).

Stage 3.1.8 Adaptation based on column 2 rules others than CMR classification

(a) REACH Annex VIII (applicable for any registration of 10 tonnes or more per year)

The screening test for reproductive/developmental toxicity does not need to be conducted if a prenatal developmental toxicity study (OECD TG 414), an extended one-generation reproductive toxicity study (B.56, OECD TG 443) or a two-generation reproductive toxicity study (B.35, OECD TG 416) is available.

The screening test for reproductive/developmental toxicity provides initial information on reproduction toxicity. An extended one-generation reproductive toxicity study or a two-generation reproductive toxicity study provides more comprehensive information on the same and further key parameters with a higher statistical power. Thus, it is clear that these studies can cover the key parameters of the screening study and are superior to the screening study. However, if the prenatal developmental toxicity study is available, it provides information on embryonic and foetal development and the ability of the dam to maintain pregnancy, but not on fertility (or postnatal development). Thus, even though a prenatal developmental toxicity study is available, it is strongly recommended that the conduct of the screening study should be considered to obtain preliminary information on the fertility endpoint[157] and peri/early postnatal development.

(b) REACH Annexes IX and X (applicable for any registration of 100 tonnes or more per year)

The reproductive toxicity studies (prenatal developmental toxicity study(ies) and an extended one-generation reproductive toxicity study) do not need to be conducted if the following criteria are met:

1. The substance is of low toxicological activity (no evidence of toxicity seen in any of the tests available) <u>and</u>
2. It can be proven from toxicokinetic data that no systemic absorption occurs via relevant routes of exposure (e.g. plasma/blood concentrations below detection limit

[157] This position is supported by a relevant Ombudsman Case: "Hence it is strongly recommended in accordance with the endpoint specific REACH Guidance on information requirements and chemical safety assessment R.7, more specifically, paragraph 7.6.6.3 for reproductive toxicity that you consider conducting a screening reproductive/development toxicity study (OECD TGs 421 or 422) in addition to the pre-natal developmental toxicity study."

using a sensitive method and absence of the substance and of metabolites of the substance in urine, bile or exhaled air) and

3. There is no or no significant human exposure[158].

It is necessary that all three criteria are fulfilled. The starting assumption is that substances with low toxicological activity may be less likely to be reproductive toxicants. The likelihood of the lack of reproductive toxicity potential is further increased and strengthened by requiring information proving no systemic absorption. When the substance has in addition no significant human exposure, it is considered safe to waive the reproductive toxicity study at REACH Annex IX and REACH Annex X levels.

158 "No significant human exposure" must be considered in relation to the toxicity and amount and quality of available information.

Appendix R.7.6–5 Evaluation of triggers

Most of the triggers lead to information needs beyond the standard information requirements. For reproductive toxicity, the only standard information requirement (Column 1 requirement) which is triggered by toxicity and not only by a tonnage level is an extended one-generation reproductive toxicity which is triggered as indicated in Column 1 of REACH Annex IX, 8.7.3.

In this Appendix various aspects of triggers are discussed.

What is a trigger?

Triggers are findings which challenge the existing toxicity database. This means that due to existing triggers it is not possible to conclude on the potential for adverse health effects for a substance, and to address the concern, further information may be needed or is needed, depending on the condition. Before the concern is addressed with adequate information, the concern should be covered by applying (adequate) risk management measures.

In this document a general term of trigger is used. It is used instead of all the various possible terms used in the REACH Regulation or other places, such as an alert, condition, indication, indication of concern, serious concern, a particular concern.

A trigger is any factor present in the existing toxicological database, whether based on theoretical substance specific scientific considerations or from experimental or observational data that raises concerns that a substance may cause toxicity but information is not comprehensive enough to allow a conclusion to be drawn. It helps identifying where testing may need to go beyond the applicable standard information requirements. Where a standard information requirement applies, testing is required, unless an adaptation can be justified, irrespective of triggers. Case by case considerations are needed in evaluating triggers.

What needs to be done if there are triggers?

The term triggers is used as a general term. It depends also if there is legal text specifying what are the following actions needed. For example, if the legal text states "*if the conditions are met, the registrant shall…*" it means that in the existence of a trigger (condition) registrant must act accordingly. On the other hand, if the legal text states that the registrant may propose a test based on an indication or concern, then the registrant may act.

In the REACH Annex text for the information requirements the following terms are used as triggers:

1) **Condition**: if the conditions are met, the registrant **must** act. Condition may be e.g. **an (adverse)** effect, **an indication**, or **other relevant** existing information; thus, it may be e.g.:
    a. an effect which has (had) a regulatory consequence (NOAEL, classification; e.g. Muta 2), or
    b. a non-adverse effect (e.g. change in hormone level, *in vitro* results), other information (e.g. toxicokinetics), or
    c. indications of an effect inadequate for toxicological evaluation, or
    d. indications of modes of action from *in vivo* studies or non-animal approaches
    e. a combination of two or several indications (e.g. for a mode of action)
    f. a result of weighing all the relevant data for an endpoint (e.g. genotoxicity data)
2) **A particular concern**: if there is a particular concern, the registrant **must** act. A particular concern may be e.g. **serious/severe** effects, **adverse** effects, focused on a **specific type** of effects, or **other relevant** existing information; thus, it may be e.g.:
    a. an effect which has (had) a regulatory consequence (NOAEL, classification; e.g. STOT 1 or 2), or

b. existing information from non-animal approaches

c. specific mechanisms/modes of action

d. existing information on effects from various different data sources (in some cases also from structurally analogous substances)

e. information from one source may be sufficient when severe or considered adverse

f. a combination of two or several indications (e.g. for a mode of action)

g. a result of weighing all the relevant data (e.g. (developmental) neurotoxicity)

An exception: At REACH Annex VIII, 8.7.1, Column 2, based on a serious concern the registrant may act

3) Indications: may be

a. A condition

b. Adverse effects

c. Non-adverse effects, e.g. hormonal change

d. Mechanism/modes of action

e. From animal studies

f. From non-animal approaches

g. Indications are not the same as a particular concern, but may still require an action from the registrant, depending on the context

Sources for triggers

Triggers may stem from various sources of information including non-animal approaches, mechanistic studies, structurally analogous substances and *in vivo* studies and information from humans.

Findings observed in non-intact animals should generally be used as triggers unless there is evidence that the findings would not be also relevant for intact animals and/or humans. Experiments with non-intact animals may include animals with removal of an endocrine organ, such as ovary (ovariectomy). Another possibility is hormonal manipulation, for example, causing decrease or increase of organ weight. These animal models may be very sensitive to detect a change in for example, hormonal response; however, it should be considered whether the same applies in intact animals.

Classification and triggers

Adverse effects meeting the classification criteria for Category 1A or 1B reproductive toxicant are not triggers for further studies because they trigger the self-classification or harmonised classification and may allow omitting further reproductive toxicity studies according to REACH Annex VIII-X, point 8.7, Column 2 adaptation rules. However, effects meeting classification criteria for Category 2 reproductive toxicant may be triggers because they can raise concern that classification criteria for a higher category may be met.

Adverse effects not meeting classification criteria may be triggers. Whether findings which are considered non-adverse may serve as triggers depends on the parameter(s) in question and this is discussed below. The relevance and quality of triggers from the *in vivo* studies and non-animal approaches used should be adequately documented and justified.

Standard information requirements and triggers for further studies

The full (standard) information requirement in REACH Annex X, i.e. the extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) (or a two-generation reproductive toxicity study initiated before 15 March 2015; EU B.35, OECD TG 416), and prenatal development toxicity studies (EU B.31, OECD TG 414) performed in two species, when adequately conducted, should normally provide reliable information for conclusion on reproductive toxicity properties. If no conclusion can be drawn from the (standard) information requirement at the respective REACH Annex level, the registrant should address the remaining concern by proposing further studies to clarify the uncertainty over the reproductive potential of the substance.

For certain studies (e.g. the extended one-generation reproductive toxicity study, the study design is to be defined based on the existence/non-existence of the conditions/triggers.

Quality and relevance of the triggers

The generic guidance on the evaluation of available information gathered in the context of REACH Annexes VI-XI is provided in the *Guidance on IR&CSA, Chapter R.4: "Evaluation of available information*".

Chapter R.4 applies for all kind of information; human, animal and non-animal sources and it is applicable also for information for reproductive toxicity endpoint. Principles described in Chapter R.4 apply to some extent also to the evaluation of triggers, although it is to be noted that a trigger is an indication of concern which challenges the available data as indicated in the definition of a trigger above and does not necessarily allow for conclusion on the hazardous properties to reproductive health – conclusion on classification or NOAEL values.

Certain general important aspects to assist the evaluation of triggers are presented below.

Consistency

It is important that the identified triggers are not contradicted by other findings in the available data. Consideration should be given to the statistical power and overall quality of the available data. Sometimes when the data is scarce it may not be possible to evaluate the consistency more than by noting if other data is contradicting with the potential trigger(s) or not.

When evaluating the consistency, differences in the existing studies must be taken into account. Apparent inconsistencies may be due to species/strain differences, different route and/or dose levels, different exposure duration, differences in methodology in measuring parameters, etc. Thus, whether the inconsistencies are likely due to methodological differences or differences in statistical power and not real inconsistencies in results, those must be analysed prior to weighing the results and deciding on the existence/non-existence of triggers.

Statistical significance and biological relevance

Dose responsiveness would provide more confidence and be more indicative of a chemically mediated effect rather than just a statistically significant finding in one dose group. The statistical power of the results from screening studies (OECD TGs 421 or 422) or 28-day study is quite low and there it may be more important to look at the ranges rather than statistical significances. It should also be remembered that statistical significance is not the same as biological relevance. There may be for example, 20% change in a parameter with biological relevance but without statistical significance. On the other hand there may be a statistically significant finding without a biological relevance. If the statistical power is high and biological variation is low for a parameter, the biological relevance of a change is high. It is necessary to evaluate if the statistical power is adequate in respect to the biological variation of a parameter. Historical data may provide guide for normal ranges but the control group of the study should generally be the main source of information in deciding on normal values and variation.

It should be also considered, case-by-case, the possibility of a non-monotonic dose-response curve.

Deciding on biological relevance of information from non-animal approaches may be challenging. Generally these predictive methods provide indication(s) and triggers rather than conclusions on hazardous properties of substances. If the non-animal approach is not reliable or the results are observed at extreme conditions (e.g. over 100x higher concentrations than the biologically plausible maximal concentration), the validity and relevance of such a single test result should be confirmed before conclusion. In best conditions results from two or more non-animal approaches are available supporting each other.

Human relevance

In the absence of further knowledge and proof, it is assumed that biologically relevant findings in animals are also relevant to humans. To justify that findings/modes of action/mechanisms of action are not relevant to human, information on humans is needed. It is not enough to state that there are no indications of the same findings/modes of action/mechanisms of action in humans than in animals, if the issue has not been adequately investigated.

Relationship of triggers with systemic toxicity

Clear triggers occur at dose levels without (other) systemic toxicity. However, the triggers have to be considered case-by-case as the relationship with the systemic toxicity may not be always clear although they may occur at the same dose level as the triggers. Generally triggers should be considered relevant even if observed at the same dose level than the (other) systemic toxicity findings if it cannot be justified why the triggers are secondary to (other) systemic toxicity.

Quality of the studies and tests

The quality of the studies or the reliability of the information should be considered. For example, triggers from *in vivo* and *in vitro* tests should have been tested with the biologically relevant material, in a robust system, and the data should be determined to be of adequate quality. Many non-animal approaches, for example, *in vitro* tests are not validated yet, but the result from them may be used if considered to be reliable case by case. For example, no *in vitro* tests for neuronal differentiation are validated but as triggers for motivating evaluation of developmental neurotoxicity, results from scientifically evaluated (peer reviewed) publications and reports may be used as triggers when considered relevant. The same goes for *in vitro* tests for other triggers such as for developmental immunotoxicity and endocrine disrupting modes of action/mechanisms.

When evaluating the results from non-animal approaches the predictivity and applicability domain and potential other limitations of the approaches need to be considered. Triggers from non-animal approaches such as QSAR predictions may be challenging to interpret especially when various methods show diverging results. Generally, consistent results from more than one non-animal approach are needed to increase the confidence of the existence or nonexistence of a trigger.

Triggers from structurally analogous substances

Triggers may also stem from structurally analogous substances. In that case, the adequacy to use the information as triggers should be considered and justified.

Evaluation of data for identification of triggers:

As part of the Stage 3.2.1 data review the following questions should be asked:

- Are there triggers for further studies/investigations specified in Column 2?
- Are there triggers for reproductive toxicity not specified in Column 2? (Considering also structurally analogous substances)

- Is there any knowledge of the substance, chemical groups or categories that would indicate special features related to reproductive toxicity to be included in the study design? If so, which?
- Are there triggers for mechanisms/modes of action relevant for reproductive toxicity? (Considering also structurally analogue substances)
- If Column 2 specific adaptation rules and REACH Annex XI general adaptation rules apply and the data is adequate for assessing and concluding the classification and labelling and risk assessment, evaluation of triggers is not needed. This means e.g. that if a substance meets the classification criteria for Category 1 for any of the CMR properties as defined at Stage 1 in Section R.7.6.2.3.2 of this Guidance and fulfils the adaptation criteria described in Column 2, then evaluation of triggers for further reproductive toxicity studies is not needed.

From a scientific perspective, it is not possible to generate an exhaustive and rigid list of triggers that would automatically trigger a particular study or have clearly defined implications for classification and risk assessment. However, certain conditions are specified in REACH Annexes and, when met, require a particular study or study design to be proposed.

A trigger (or triggers) may trigger:

- a study, which would fulfil a standard information requirement, which otherwise only applies at a higher tonnage level,; or
- a certain study design (or a particular independent study) when specified conditions are met (e.g. extension of Cohort 1B to include F2 or inclusion of Cohort 2 and/or 3 in an extended one-generation reproductive toxicity study); or
- inclusion of certain selected additional investigational parameters to a range-finding study or a study required in the (standard) information requirement (e.g. selected parameters for immunotoxicity under conditions where the trigger(s) need(s) to be confirmed before considering the need for further studies to address the concern; or
- special investigational studies/tests, e.g. studies on mechanisms/modes of action.

The following triggers are referred to in REACH Annex IX 8.7.3 and trigger the information requirement:

- **At REACH Annex IX level**, an extended one-generation reproductive toxicity study may be triggered by triggers from repeated dose toxicity studies (including screening studies) according to description in Column 1 (see further details in this Guidance, Section R.7.6.2.3.2, Stage 4.4 (iii) of this Guidance.

The following triggers are referred to in Column 2 adaptation rules for reproductive toxicity/developmental neurotoxicity/developmental immunotoxicity:

- **At REACH Annex VIII level**, based on trigger(s) for reproductive toxicity, either for developmental toxicity or for fertility, causing serious concern[159], the registrant **may** propose a prenatal developmental toxicity study or an extended one-generation reproductive toxicity study instead of a "screening for reproduction/developmental toxicity" test, as appropriate. The appropriate study depends on whether the concern is on prenatal developmental toxicity, prenatal developmental toxicity manifested

[159] Serious concern reflects a high likelihood for adverse effects on reproductive health.

postnatally, postnatal developmental toxicity or on fertility[160]. The triggers may stem for example from relevant non-animal approaches[161] or *in vivo* studies e.g. from 28-day repeated dose toxicity study which is required at this REACH Annex level or respective other information. A testing proposal is required for REACH Annex IX/X level studies.

- **At REACH Annex IX level**, trigger(s) for prenatal developmental toxicity should trigger a prenatal developmental toxicity study on a second species as a Column 2 requirement. Examples of triggers for this study are shown under Section R.7.6.2.3.2, Stage 4.4 (ii), prenatal developmental toxicity study of this Guidance.
- At REACH Annex IX level, if an extended one-generation reproductive toxicity study is triggered, triggers for extending the Cohort 1B, including Cohorts 2 and/or 3 are given in Column 2. The study design of an extended one-generation reproductive toxicity study and triggers to expand the study are described in Appendix R.7.6–2 of this Guidance.
- At the same REACH Annex level, an extended one-generation reproductive toxicity study on a second species or strain may be triggered at this REACH Annex (REACH Annex IX) or the next REACH Annex level (REACH Annex X). Examples of triggers are presented under Section R.7.6.2.3.2, Stage 4.4 (iii), extended one-generation reproductive toxicity study of this Guidance.
- **At REACH Annex X level**, an extended one-generation reproductive toxicity study is a (standard) information requirement. The triggers for extending the Cohort 1B, including Cohorts 2 and/or 3 are given in Column 2. The study design of the extended one-generation reproductive toxicity study and triggers to expand the study are described in Appendix R.7.6–2 of this Guidance.
- At REACH Annex X level, the full information requirements i.e. an extended one-generation reproductive toxicity study (EU B.56, OECD TG 443) (or a two-generation reproductive toxicity study initiated before 15 March 2015; EU B.35, OECD TG 416), and prenatal development toxicity studies (EU B.31, OECD TG 414) performed in two species, when adequately conducted, should normally provide reliable information for conclusion on reproductive toxicity properties as indicated above.

If no conclusion can be drawn from the (standard) information requirement, the registrant should address the remaining concern by proposing further studies to clarify the uncertainty over the reproductive potential of the substance.

Exposure triggers/conditions upgrading testing requirements

- Guidance on exposure-based adaptation and triggering of information requirements is provided in Section R.5.1 in the *Guidance on IR&CSA*, *Chapter R.5: Adaptation of information requirements.*
- The use pattern and the exposure to a substance may indicate a concern with the need for additional information requirements, on a case-by-case basis. For example,

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160 However, in case of proposing a prenatal developmental toxicity study it is strongly recommended that the registrant should consider conducting a screening study because a prenatal developmental toxicity study does not address the effects on the fertility endpoint and developmental toxicity manifested shortly after birth.

161 In order to be considered providing "*serious concern*", information from non-animal approaches should be reliable, relevant and from validated studies with appropriate applicability domain (for QSAR models a formal validation process is not required). Based on case-by-case scientific justification results from non-validated and non-guideline tests may be acceptable. Generally several information sources may be needed.

there may be serious concerns that human exposure, particularly to consumers, is close to the levels at which human health effects might be expected. Such concerns for human health need to be addressed by producing additional information on hazard. In very exceptional cases such concerns may be satisfactorily addressed by improved risk management measures.

Documentation and addressing the triggers/conditions

If the triggers for reproductive toxicity or the conditions described in Column 1 or 2 are met for further investigations, they must be described in the dossier as well as how they are addressed at the respective endpoint section.

R.7.7 Mutagenicity and carcinogenicity

R.7.7.1 Mutagenicity

R.7.7.1.1 Definition of mutagenicity

Mutagenicity refers to the induction of permanent transmissible changes in the amount or structure of the genetic material of cells or organisms. These changes may involve a single gene or gene segment, a block of genes or chromosomes. The term clastogenicity is used for agents giving rise to structural chromosome aberrations. A clastogen can cause breaks in chromosomes that result in the loss or rearrangements of chromosome segments. Aneugenicity (aneuploidy induction) refers to the effects of agents that give rise to a change (gain or loss) in chromosome number in cells. An aneugen can cause loss or gain of chromosomes resulting in cells that have not an exact multiple of the haploid number. For example, three number 21 chromosomes or trisomy 21 (characteristic of Down syndrome) is a form of aneuploidy.

Genotoxicity is a broader term and refers to processes which alter the structure, information content or segregation of DNA and are not necessarily associated with mutagenicity. Thus, tests for genotoxicity include tests which provide an indication of induced damage to DNA (but not direct evidence of mutation) *via* effects such as DNA strandbreaks, unscheduled DNA synthesis (UDS), sister chromatid exchange (SCE), DNA adduct formation or mitotic recombination, as well as tests for mutagenicity.

The chemical and structural complexity of the chromosomal DNA and associated proteins of mammalian cells, and the multiplicity of ways in which changes to the genetic material can be effected make it difficult to give more precise, discrete definitions.

In the risk assessment of substances it is necessary to address the potential effect of *mutagenicity*. It can be expected that some of the available data will have been derived from tests conducted to investigate potentially harmful effects on genetic material (*genotoxicity*). Hence, both the terms *mutagenicity* and *genotoxicity* are used in this document.

R.7.7.1.2 Objective of the guidance on mutagenicity

The aims of testing for genotoxicity are to assess the potential of substances to induce genotoxic effects which may lead to cancer or cause heritable damage in humans. Genotoxicity data are used in risk characterisation and classification of substances. Genotoxicity data are useful for the determination of the general mode of action of a substance (*i.e.* type(s) of genotoxic damage induced) and can provide some indication on the dose (concentration)-response relationship and on whether the observed effect can be reasonably assumed to have a threshold or not. Genotoxicity data are thus useful in deciding the best approach to use for the risk assessement. Expert judgement is necessary at each stage of the testing strategy to decide on the relevance of a result based on the data available for each endpoint.

Alterations to the genetic material of cells may occur spontaneously endogenously or be induced as a result of exposure to ionising or ultraviolet radiation, or genotoxic substances. In principle, human exposure to substances that are mutagens may result in increased frequencies of mutations above background.

Mutations in somatic cells may be lethal or may be transferred to daughter cells with deleterious consequences for the affected organism (*e.g.* cancer may result when they occur in proto-oncogenes, tumour suppressor genes and/or DNA repair genes) ranging from trivial to detrimental or lethal.

Heritable damage to the offspring, and possibly to subsequent generations, of parents exposed to substances that are mutagens may follow if mutations are induced in parental germ cells. To date, all known germ cell mutagens are also mutagenic in somatic cells *in vivo.* Substances that are mutagenic in somatic cells may produce heritable effects if they, or their active metabolites, have the ability to interact with the genetic material of germ cells. Conversely, substances that do not induce mutations in somatic cells *in vivo* would not be expected to be germ cell mutagens.

There is considerable evidence of a positive correlation between the mutagenicity of substances *in vivo* and their carcinogenicity in long-term studies with animals. Genotoxic carcinogens are substances for which the most plausible mechanism of carcinogenic action involves genotoxicity.

R.7.7.2 Information requirements on mutagenicity

The information requirements on mutagenicity are described by REACH Annexes VI-XI, that specify the information that must be submitted for registration and evaluation purposes. The information is thus required for substances produced or imported in quantities of >1 t/y (tons per annum). When a higher tonnage level is reached, the requirements of the corresponding Annex have to be considered. However, factors including not only production volume but also pre-existing toxicity data, information about the identified use of the substance and exposure of humans to the substance will influence the precise information requirements. The REACH Annexes must thus be considered as a whole, and in conjunction with the overall requirements of registration, evaluation and the duty of care.

Column 1 of REACH Annexes VII-X informs on the standard information requirements for substances produced or imported in quantities of >1 t/y, >10 t/y, >100 t/y, and >1000 t/y, respectively.

Column 2 of REACH Annexes VII-X lists specific rules according to which the required standard information may be omitted, replaced by other information, provided at a different stage or adapted in another way. If the conditions are met under which column 2 of these Annexes allows adaptations, the fact and the reasons for each adaptation should be clearly indicated in the registration dossier.

The standard information requirements for mutagenicity and the specific rules for adaptation of these requirements are presented in Table R.7.7–1.

Table R.7.7–1 REACH information requirements for mutagenicity

COLUMN 1 STANDARD INFORMATION REQUIRED	COLUMN 2 SPECIFIC RULES FOR ADAPTATION FROM COLUMN 1
Annex VII: 1. *In vitro* gene mutation study in bacteria.	Further mutagenicity studies shall be considered in case of a positive result.
Annex VIII: 1. *In vitro* cytogenicity study in mammalian cells or *in vitro* micronucleus study.	1. The study does not usually need to be conducted • if adequate data from an *in vivo* cytogenicity test are available or • the substance is known to be carcinogenic category 1A or 1B or germ cell mutagenic category 1A, 1B or 2.
2. *In vitro* gene mutation study in mammalian cells, if a negative result in Annex VII, 1 and Annex VIII, 1.	2. The study does not usually need to be conducted if adequate data from a reliable *in vivo* mammalian gene mutation test are available. Appropriate *in vivo* mutagenicity studies shall be considered in case of a positive result in any of the genotoxicity studies in Annex VII or VIII.
Annex IX:	If there is a positive result in any of the *in vitro* genotoxicity studies in Annex VII or VIII and there are no results available from an *in vivo* study already, an appropriate *in vivo* somatic cell genotoxicity study shall be proposed by the registrant. If there is a positive result from an *in vivo* somatic cell study available, the potential for germ cell mutagenicity should be considered on the basis of all available data, including toxicokinetic evidence. If no clear conclusions about germ cell mutagenicity can be made, additional investigations shall be considered.
Annex X:	If there is a positive result in any of the *in vitro* genotoxicity studies in Annex VII or VIII, a second *in vivo* somatic cell test may be necessary, depending on the quality and relevance of all the available data. If there is a positive result from an *in vivo* somatic cell study available, the potential for germ cell mutagenicity should be considered on the basis of all available data, including toxicokinetic evidence. If no clear conclusions about germ cell mutagenicity can be made, additional investigations shall be considered.

In addition to these specific rules, the required standard information set may be adapted according to the general rules contained in Annex XI. In this case as well, the fact and the reasons for each adaptation should be clearly indicated in the registration.

In some cases, the rules set out in Annex VII to XI may require certain tests to be undertaken earlier than or in addition to the tonnage-triggered requirements. Registrants should note that a testing proposal must be submitted for a test mentioned in Annex IX or X, independently from the registered tonnage. Following examination of such a testing proposal ECHA has to approve the test in its evaluation decision before it can be undertaken. See Section R.7.7.6 of this Guidance for further guidance on testing requirements.

R.7.7.3 Information sources on mutagenicity

To be able to evaluate the mutagenic potential of a substance in a comprehensive way, information is required on its capability to induce gene mutations, structural chromosome aberrations (clastogenicity) and numerical chromosome aberrations (aneugenicity). Many test methods are available by which such information can be obtained. Non-testing methods, such as SAR, QSAR and read-across approaches, may also provide information on the mutagenic potential of a substance.

Typically, *in vitro* tests are performed with cultured bacterial cells, human or other mammalian cells. The sensitivity and specificity of tests will vary with different classes of substances and, if adequate data are available for the class of substance to be tested, these data can guide the selection of the most appropriate test systems to be used. In order to detect mutagenic effects also of substances that need to be metabolically activated to become mutagenic, an exogenous metabolic activation system is usually added in *in vitro* tests. For this purpose the post-mitochondrial 9000 x *g* supernatant (S-9 fraction) of whole liver tissue homogenate containing a high concentration of metabolising enzymes and extracted from animals that have been induced to raise the oxidative P450 levels is most commonly employed. In the case when information is required on the mutagenic potential of a substance *in vivo,* several test methods are available. In *in vivo* tests whole animals are used, in which metabolism and toxicokinetic mechanisms in general exist as natural components of the test animal. It should be noted that species-specific differences in metabolism are known. Therefore, different genotoxic responses may be obtained. Some *in vivo* genotoxicity tests such as the Transgenic rodent (TGR) somatic and germ cell gene mutation assays and the comet assay employ methods by which any tissue (containing nucleated cells) of an animal can in theory be examined for effects on the genetic material. This gives the possibility to examine target tissues (including germ cells) and site-of-contact tissues (*i.e.* skin, epithelium of the respiratory or gastro-intestinal tract). However differences can exist regarding the number and type of tissues for which the use a specific test has been scientifically validated. For instance, the TGR assays can be used to examine germ cells whereas the comet assay as described in the OECD test guideline (TG) is, at present, not recommended for that purpose.

Some test methods, but not all, have an officially adopted EU and/or OECD TG for the testing procedure. In cases where no adopted EU or OECD TG is available for a test method, rigorous and robust protocols should be followed, such as those defined by internationally recognised groups of experts like the International Workshop on Genotoxicity Testing (IWGT) under the umbrella of the International Association of Environmental Mutagen Societies. Furthermore, modifications to OECD TGs have been developed for some classes of substances and may serve to enhance the accuracy of test results. Use of such modified protocols is a matter of expert judgement and will vary as a function of the chemical and physical properties of the substance to be evaluated. Similarly, use of standard test methods for the testing of tissue(s) not covered by those standard test methods should be scientifically justified and validity of the results will depend on the appropriateness of the acceptability criteria, which should have been specifically developed for this (these) tissue(s) based on sufficient experience and historical data.

R.7.7.3.1 Non-human data on mutagenicity

Non-testing data on mutagenicity

Non-test information about the mutagenicity of a substance can be derived in a variety of ways, ranging from simple inspection of the chemical structure through various read-across techniques, the use of expert systems, metabolic simulators, to *global* or *local* (Q)SARs. The usefulness of such techniques varies with the amount and nature of information available, as well as with the specific regulatory questions under consideration.

Regarding substances for which testing data exist, non-test information can be used in the *Weight of Evidence* approach, to help confirm results obtained in specific tests, or to help develop a better understanding of mutagenicity mechanisms. The information may be useful in deciding if, or what, additional testing is required. At the other extreme, where no testing data are available, similar alternative sources of information may assist in setting test priorities. In cases where no testing is likely to be done (low exposure, <1 t/y) they may be the only options available to establish a hazard profile.

Weight of Evidence approaches that use expert judgement to include test results for close chemical analogues are ways of strengthening regulatory positions on the mutagenicity of a substance. Methods that identify general *structural alerts* for genotoxicity such as the Ashby-Tennant super-mutagen molecule (Ashby and Tennant, 1988) may also be useful.

Prediction models for mutagenicity

There are hundreds of (Q)SAR models available in the literature for predicting test results for genotoxic endpoints for closely related structures (Naven *et al.*, 2012; Bakhtyari *et al.*, 2013). These are known as *local* (Q)SARs. When essential features of the information domain are clearly represented, these models may constitute the best predictive tools for estimating a number of mutagenic/genotoxic endpoints. However, quality of reporting varies from model to model and predictivity must be assessed case-by-case on the basis of clear documentation. Use of harmonised templates, such as the QSAR Model Reporting Format (QMRF) and the QSAR Prediction Reporting Format (QPRF) developed by the Joint Research Centre (JRC) of the European Commission (http://ihcp.jrc.ec.europa.eu/our_labs/predictive_toxicology/qsar_tools/QRF), can help ensure consistency in summarising and reporting key information on (Q)SAR models and substance-specific predictions generated by (Q)SAR models. The JRC website also hosts the JRC (Q)SAR Model Inventory, which is an inventory of information on the validity of (Q)SAR models that have been submitted to the JRC (http://ihcp.jrc.ec.europa.eu/our_databases/jrc-qsar-inventory).

Generally, (Q)SAR models that contain putative mechanistic descriptors are preferred; however many models use purely structural descriptors. While such models may be highly predictive, they rely on statistical methods and the toxicological significance of the descriptors may be obscure.

(Q)SAR models for mutagenicity can apply to a limited set of congeneric substances (local models) or to a wide variety of non-congeneric substances (global models). Global (Q)SARs are usually implemented in computer programs and may comprise a set of local models; these global models first categorise the input molecule into the chemical domain it belongs to, and then apply the corresponding local prediction model. These are known as expert systems. Other global models apply the same mathematical algorithm on all input molecules without prior separation. It is generally observed that the concept of applicability domain is a useful one and the endpoints for substances inside the applicability domains of the models are better predicted than for substances falling outside.

Many global models for mutagenicity are commercial and some of the suppliers of these global models consider the data in their modelling sets to be proprietary. Proprietary means that the training set data used to develop the (Q)SAR model is hidden from the user. In other cases it means that it may not be distributed beyond use by regulatory authorities. The models do not always equal the software incorporating them, and the software often has flexible options for expert uses. Thus, the level of information available, from both (Q)SAR models and compiled databases, should be adequate for the intended purpose.

A list of the available (free and commercial) predictive software for ecotoxicological, toxicological and environmental endpoints, including mutagenicity models, has been compiled within the frame of the EU project Antares (http://www.antares-life.eu/).

The most common genotoxicity endpoint for global models has been to predict results of the Ames test. Some models for this endpoint include a metabolic simulator.

There are models for many other mutagenicity endpoints. For example, the Danish EPA and the Danish QSAR group at DTU Food (National Food Institute at the Technical University of Denmark) have developed a (Q)SAR database that contains predictions from a number of mutagenicity models. In addition to assorted Ames models, the database contains predictions of the following *in vitro* endpoints: chromosomal aberrations (CHO and CHL cells), mouse lymphoma/*tk*, CHO/hprt gene-mutation assays and UDS (rat hepatocytes); and the following *in vivo* endpoints: *Drosophila* SLRL, mouse micronucleus, rodent dominant lethal, mouse SCE in bone marrow and mouse comet assay data. The database is freely accessible *via* http://qsar.food.dtu.dk. The online database contains predictions for over 166,000 substances and includes a flexible system for chemical structure and parameter searching. A user manual with information on the individual models including training set information and validation results is available at the website. The database is also integrated into the OECD (Q)SAR Toolbox. A major update of the database with consensus predictions by use of different QSAR models for each of the modelled endpoints for more than 600,000 structures, including over 70,000 REACH pre-registered substances, and with an improved user interface is scheduled for the beginning of 2015.

Another example of a database with predictions on mutagenicity is the Enhanced NCI Database Browser (http://cactus.nci.nih.gov) sponsored by the U.S. National Cancer Institute. It contains predictions for over 250,000 substances for mutagenicity as well as other non-mutagenic endpoints, some of which may provide valuable mechanistic information (for example alkylating ability or microtubule formation inhibition). It is also searchable by a wide range of parameters and structure combinations.

Neither of these two examples is perfect, but they illustrate a trend towards predictions of multiple endpoints and may assist those making *Weight of Evidence* decisions regarding the mutagenic potential of untested substances. More detailed information on the strengths and limitations of the different (Q)SAR models can be found elsewhere (Serafimova *et al.*, 2010).

OECD QSAR Toolbox

To increase the regulatory acceptance of (Q)SAR models, the OECD has started the development of a QSAR Toolbox to make (Q)SAR technology readily accessible, transparent and less demanding in terms of infrastructure costs (http://www.qsartoolbox.org/). The OECD QSAR Toolbox facilitates the practical application of grouping and read-across approaches to fill gaps in (eco-)toxicity data, including genotoxicity and genotoxic carcinogenicity, for chemical hazard assessment. In particular, the OECD QSAR Toolbox covers the *in vitro* gene mutation (Ames test), *in vitro* chromosomal aberration, *in vivo* chromosomal aberration (micronucleus test), and genotoxic carcinogenicity endpoints. The predictions are based on the implementation of a range of profilers connected with genotoxicity and carcinogenicity (to quickly evaluate substances for common mechanisms or modes of action), and the incorporation of numerous databases with results from experimental studies (to support read-across and trend analysis) into a logical workflow. The Toolbox and guidance on its use are freely available. A user manual "Strategies for chemicals to fill data gaps to assess genetic toxicity and genotoxic carcinogenicity" and various tutorials for categorisation of substances by use of the Toolbox in relation to protein- and DNA- binding and Ames test mutagenicity are also available on the OECD QSAR Toolbox web site.

The Guidance on IR&CSA *Chapter R.6: QSARs and grouping of chemicals* explains basic concepts of (Q)SARs and gives generic guidance on validation, adequacy and documentation for regulatory purposes. It also describes a stepwise approach for the use of read-across/grouping and (Q)SARs. Further information on the category formation and read-across approach for the prediction of toxicity can be found in Enoch (2010).

Testing data on mutagenicity

Test methods preferred for use are listed in Table R.7.7–2, Table R.7.7–3 and Table R.7.7–4. The introduction to the OECD TGs on genetic toxicity testing as well as some of the related OECD TGs are currently being revised under the OECD Test Guidelines Programme (TGP). In addition, an OECD Guidance Document on the selection and application of the assays for genetic toxicity is being developed. For further information, please see http://www.oecd.org/env/testguidelines.

In vitro data

Table R.7.7–2 *In vitro* test methods

Test method	GENOTOXIC ENDPOINTS measured/ PRINCIPLE OF THE TEST METHOD	EU/OECD guideline[a]
Bacterial reverse mutation test	Gene mutations / The test uses amino-acid requiring strains of bacteria to detect (reverse) gene mutations (point mutations and frameshifts).	EU: B.13/14 OECD: 471
In vitro mammalian cell gene mutation test – *hprt* test	Gene mutations / The test identifies substances that induce gene mutations in the *hprt* gene of established cell lines.	EU: B.17 OECD: 476[b]
In vitro mammalian cell gene mutation test – Mouse lymphoma assay	Gene mutations and structural chromosome aberrations / The test identifies substances that induce gene mutations in the *tk* gene of the L5178Y mouse lymphoma cell line. If colonies in a *tk* mutation test are scored using the criteria of normal growth (large) and slow growth (small) colonies, gross structural chromosome aberrations (*i.e.* clastogenic effect) may be measured, since mutant cells that have suffered damage to both the *tk* gene and growth genes situated close to the *tk* gene have prolonged doubling times and are more likely to form small colonies.	EU: B.17 OECD: 476[b]
In vitro mammalian chromosome aberration test	Structural and numerical chromosome aberrations / The test identifies substances that induce structural chromosome aberrations in cultured mammalian established cell lines, cell strains or primary cell cultures. An increase in polyploidy may indicate that a substance has the potential to induce numerical chromosome aberrations, but this test is not optimal to measure numerical aberrations and is not routinely used for that purpose. Accordingly, this test guideline is not designed to measure numerical aberrations.	EU: B.10 OECD: 473[b]
In vitro micronucleus test	Structural and numerical chromosome aberrations / The test identifies substances that induce micronuclei in the cytoplasm of interphase cells. These micronuclei may originate from acentric fragments or whole chromosomes, and the test thus has the potential to detect both clastogenic and aneugenic substances.	EU: B.49 OECD: 487[b]

[a] For EU guidelines, see Regulation (EC) No 440/2008 (http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32008R0440:en:NOT) / for OECD guidelines see http://www.oecd.org/env/testguidelines
[b] OECD TGs 473, 476 and 487 are currently being revised (see http://www.oecd.org/env/testguidelines)

As noted earlier, accepted modifications to the standard test guidelines/methods have been developed to enhance test sensitivity to specific classes of substances. Expert judgement should be applied to judge whether any of these are appropriate for a given substance being registered. For example, protocol modifications for the Ames test might be appropriate for substances such as gases, volatile liquids, azo-dyes, diazo compounds, glycosides, and petroleum oil derived products, which should be regarded as special cases.

Animal data

- Somatic cells

Table R.7.7–3 *In vivo* test methods, somatic cells

Test method	GENOTOXIC ENDPOINTS measured/ PRINCIPLE OF THE TEST METHOD	EU/OECD guideline[a]
In vivo mammalian bone marrow chromosome aberration test	Structural and numerical chromosome aberrations / The test identifies substances that induce structural chromosome aberrations in the bone-marrow cells of animals, usually rodents. An increase in polyploidy may indicate that a substance has the potential to induce numerical chromosome aberrations, but this test is not optimal to measure numerical aberrations and is not routinely used for that purpose. Accordingly, this test guideline is not designed to measure numerical aberrations.	EU: B.11 OECD: 475[b]
In vivo mammalian erythrocyte micronucleus test	Structural and numerical chromosome aberrations / The test identifies substances that cause micronuclei in erythroblasts sampled from bone marrow and/or peripheral blood cells of animals, usually rodents. These micronuclei originate from acentric fragments or whole chromosomes, and the test thus has the potential to detect both clastogenic and aneugenic substances.	EU: B.12 OECD: 474[b]
Unscheduled DNA synthesis (UDS) test with mammalian liver cells *in vivo*	DNA repair / The test identifies substances that induce DNA damage followed by DNA repair (measured as unscheduled "DNA" synthesis) in liver cells of animals, commonly rats. The test is usually based on the incorporation of tritium labelled thymidine into the DNA by repair synthesis after excision and removal of a stretch of DNA containing a region of damage.	EU: B.39 OECD: 486
Transgenic rodent (TGR) somatic and germ cell gene mutation assays	Gene mutations and chromosomal rearrangements (the latter specifically in the plasmid and Spi- assay models) / Since the transgenes are transmitted by the germ cells, they are present in every cell. Therefore, gene mutations and/or chromosomal rearrangements can be detected in virtually all tissues of an animal, including target tissues and specific site of contact tissues.	EU: B.58 OECD: 488
In vivo alkaline single-cell gel electrophoresis assay for DNA strand breaks (comet assay)	DNA strand breaks / The DNA strand breaks may result from direct interactions with DNA, alkali labile sites or as a consequence of incomplete excision repair. Therefore, the alkaline comet assay recognises primary DNA damage that would lead to gene mutations and/or chromosome aberrations, but will also detect DNA damage that may be effectively repaired or lead to cell death. The comet assay can be applied to almost every tissue of an animal from which single cell or nuclei suspensions can be made, including specific site of contact tissues.	EU: none OECD: 489

[a] For EU guidelines, see Regulation (EC) No 440/2008 (http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32008R0440:en:NOT) / for OECD guidelines see http://www.oecd.org/env/testguidelines
[b] OECD TGs 474 and 475 are currently being revised (see http://www.oecd.org/env/testguidelines)

A detailed review of transgenic animal model assays, including recommendations on how to perform such assays in somatic cells, has been produced for the OECD (Lambert *et al.,* 2005; OECD, 2009).

Validation studies and recommendations have been published in recent years, identifying experimental factors which are of importance for improved harmonisation of data obtained in the alkaline single-cell gel electrophoresis assay for DNA strand breaks (comet assay) (Ersson *et al*., 2013; Azqueta *et al.*, 2013; Forchhammer *et al*., 2012; Azqueta *et al*., 2011a; Azqueta *et al*., 2011b; Forchhammer *et al*., 2009; Collins *et al*., 2008). Specifically, various international groups have proposed protocols and recommendations for performing the *in vivo* alkaline comet assay (Tice *et al.,* 2000; Hartmann *et al.,* 2003; McKelvey-Martin *et al.*, 1993; Brendler-Schwaab *et al.*, 2005; Burlinson et al., 2007; Smith *et al.*, 2008; Rothfuss *et al.*, 2010; Burlinson, 2012; Vasquez, 2012; Johansson *et al.*, 2010; Kirkland and Speit, 2008; EFSA, 2012). An international validation study on the *in vivo* alkaline single-cell gel electrophoresis assay was coordinated by the Japanese Centre for the Validation of Alternative Methods (JaCVAM) from 2006 to 2012. The validation study report was peer reviewed by the OECD and an OECD expert group drafted the comet OECD TG, which was approved by the OECD Working Group of National Coordinators of the Test Guidelines Programme (WNT) in April 2014. While awaiting the adoption of the comet OECD TG 489, the minimum criteria for acceptance of the comet assay published by EFSA (2012) can be used.

- Germ cells

Testing in germ cells has in the past been conducted only on very rare occasions (see Section R.7.7.6 of this Guidance).

Table R.7.7–4 *In vivo* test methods, germ cells

Test method	GENOTOXIC ENDPOINTS measured/ PRINCIPLE OF THE TEST METHOD	EU/OECD guideline[a]
Mammalian spermatogonial chromosome aberration test	Structural and numerical chromosome aberrations / The test identifies substances that induce structural chromosome aberrations in mammalian, usually rodent, spermatogonial cells and is, therefore, expected to be predictive of induction of heritable mutations in germ cells. An increase in polyploidy may indicate that a substance has the potential to induce numerical chromosome aberrations, but this test is not optimal to measure numerical aberrations and is not routinely used for that purpose. Accordingly, this test guideline is not designed to measure numerical aberrations.	EU: B.23 OECD: 483[b]
Rodent dominant lethal test	Structural and numerical chromosome aberrations / The test identifies substances that induce dominant lethal effects causing embryonic or foetal death resulting from inherited dominant lethal mutations induced in germ cells of an exposed parent, usually the male. It is generally accepted that dominant lethals are due to structural and numerical chromosome aberrations. Rats or mice are recommended as the test species.	EU: B.22 OECD: 478[b]
Transgenic rodent (TGR) somatic and germ cell gene mutation assays	Gene mutations and chromosomal rearrangements (the latter specifically in the plasmid and Spi- assay models) / Since the transgenes are transmitted by the germ cells, they are present in every cell. Therefore, gene mutations and/or chromosomal rearrangements can be detected in virtually all tissues of an animal including specific site of contact tissues and germ cells. Delayed sampling times may need to be considered in order to detect mutations in different stages of spermatogenesis.	EU: none OECD: 488

[a] For EU guidelines, see Regulation (EC) No 440/2008 (http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32008R0440:en:NOT) / for OECD guidelines see http://www.oecd.org/env/testguidelines
[b] OECD TGs 478 and 483 are currently being revised (see http://www.oecd.org/env/testguidelines)

A detailed review of transgenic animal model assays, including recommendations on how to perform such assays in germ cells, has been produced for the OECD (Lambert *et al.,* 2005; OECD, 2009). The ability to include sampling of somatic and germ cells in a single study significantly reduces the need to perform additional studies to obtain such information, thereby conforming to the 3Rs principles. As specified in the OECD TG 488, additional sampling times may be needed to cover for the all the stages of spermatogenesis. The test can also be used to investigate transmission of mutations to the offspring since treatment of transgenic male mice can result in offspring carrying mutations (Barnett *et al*., 2002). An example of mutagenicity investigation in epididymal spermatozoa using a transgenic mouse model has been published (Olsen *et al.,* 2010).

The applicability of the standard alkaline comet assay to germ cells has been discussed by the OECD. The assay as described in the OECD TG 489 (see http://www.oecd.org/env/testguidelines) is not considered appropriate to measure DNA strand breaks in mature germ cells. Since high and variable background levels in DNA damage were reported in a literature review on the use of the comet assay for germ cell genotoxicity (Speit *et al.*, 2009), protocol modifications together with improved standardization and validation trials are deemed necessary before the comet assay on mature germ cells (*e.g.* sperm) can be included in the test guideline. In addition, the recommended exposure regimen described in this guideline is not optimal and longer exposures or sampling times would be necessary for a meaningful analysis of DNA strand breaks in mature sperm. Genotoxic effects as measured by the comet assay in testicular cells at different stages of differentiation have been described in the literature (Zheng *et al.*, 1997; Cordelli *et al.*, 2003). However, it should be noted that gonads contain a mixture of somatic and germ cells. For this reason, positive results in whole gonad (testis) are not necessarily reflective of germ cell damage, nevertheless, they suggest that tested chemicals have reached the gonad.

Databases with experimental data

There are several open-source databases with experimental information on mutagenicity and carcinogenicity (the two endpoints can often not easily be separated). A review of these databases can be found in Serafimova *et al.* (2010).

R.7.7.3.2 Human data on mutagenicity

Occasionally, studies of genotoxic effects in humans exposed by, for example, accident, occupation or participation in clinical studies (*e.g.* from case reports or epidemiological studies) may be available. Generally, cells circulating in blood are investigated for the occurrence of various types of genetic alterations.

R.7.7.4 Evaluation of available information on mutagenicity

Genotoxicity is a complex endpoint and requires evaluation by expert judgement. For both steps of the effects assessment, *i.e.* hazard identification and dose (concentration)-response (effect) assessment, it is very important to evaluate the data with regard to their adequacy and completeness. The evaluation of adequacy should address the reliability and relevance of the data in a way as outlined in the introductory chapter. The completeness of the data refers to the conclusion on the comparison between the available adequate information and the information that is required under the REACH provisions for the applicable tonnage level of the substance. Such a conclusion relies on *Weight of Evidence* approaches, which categorise available information based on the methods used: *guideline tests*, *non-guideline tests*, and other types of information which may justify adaptation of the standard testing regime. Such a *Weight of Evidence* approach also includes an evaluation of the available data as a whole, *i.e.* both *over and across* toxicological endpoints (for example, consideration of existing carcinogenicity data, repeated dose toxicity data and genotoxicity data all together can help understand whether a substance could be a genotoxic or non-genotoxic carcinogen).

This approach provides a basis to decide whether further information is needed on endpoints for which specific data appear inadequate or not available, or whether the requirements are fulfilled.

R.7.7.4.1 Non-human data on mutagenicity

Non-testing data for mutagenicity

In a more formal approach, documentation can include reference to a related substance or group of substances that leads to the conclusion of concern or lack of concern. This can either be presented according to scientific logic (read-across) or sometimes as a mathematical relationship of chemical similarity.

If well-documented and applicable (Q)SAR data are available, they should be used to help reach the decision points described in the section below. In many cases the accuracy of such methods will be sufficient to help, or allow either a testing or a specific regulatory decision to be made. In other cases the uncertainty may be unacceptable due to the severe consequences of a possible error. This may be driven by many factors including high exposure potential or toxicological concerns.

Substances for which no test-data exist or for which testing is technically not possible represent a special case in which reliance on non-testing data may be absolute. Many factors will dictate the acceptability of non-testing methods in reaching a conclusion based on no tests at all. It may be discussed whether *Weight of Evidence* decisions based on multiple genotoxicity and carcinogenicity estimates can equal or exceed those obtained by one or two *in vitro* tests, and whether general rules for adaptation of the standard testing regime as described in Annex XI to REACH may be invoked based on such estimates. This must be considered on a case-by-case basis.

Testing data on mutagenicity

Evaluation of genotoxicity test data should be made with care.

Regarding ***positive* findings**, particular points should be taken into account:

- are the testing conditions (*e.g.* pH, osmolality, precipitates) in *in vitro* mammalian cell assays relevant to the conditions *in vivo*?
- for studies *in vitro*, factors known to influence the specificity of mammalian cell assays such as the cell line used, the top concentration tested, the toxicity measure used or the metabolic activation system used, should be taken into consideration
- responses generated only at highly toxic/cytotoxic doses or concentrations should be interpreted with caution (*i.e.* taking into account the criteria defined in OECD guidelines)
- the presence or absence of a dose (concentration)-response relationship should be considered

Particular points to take into account when evaluating ***negative* test results** include:

- the doses or concentrations of test substance used (were they high enough? For studies *in vivo*, was a sufficienlty high dose level inducing signs of toxicity used? For studies *in vitro*, was a sufficient level of cytotoxicity reached?)
- was the test system used sensitive to the nature of the genotoxic changes that might have been expected? For example, some *in vitro* test systems will be sensitive to point mutations and small deletions but not to mutagenic events that create large deletions
- the volatility of the test substance (were concentrations maintained in tests conducted *in vitro*?)

- for studies *in vitro*, the possibility of metabolism not being appropriate in the test system including studies in extra-hepatic organs
- was the test substance taken up by the test system used for *in vitro* studies?
- were sufficient cells scored/sampled for studies *in vitro*? Has the appropriate number of samples/technical replicates been scored to support statistical significance of the putative negative result?
- for studies *in vivo*, did the substance reach the target organ? Or was the substance only in a position to act at the site of contact due to its high reactivity or insufficient systemic availability (taking also toxicokinetic data into consideration, *e.g.* rate of hydrolysis and electrophilicity may be factors that need to be considered)?
- for studies *in vivo*, was sampling appropriate? (Was a sufficient number of animals used? Were sufficient sampling times used? Was a sufficient number of cells scored/sampled?)

Different results between different test systems should be evaluated with respect to their individual significance. Examples of points to be considered are as follows:

- different results obtained in non-mammalian systems and in mammalian cell tests may be addressed by considering possible differences in substance uptake and metabolism, or in genetic material organisation and ability to repair. Although the results of mammalian tests may be considered of higher significance, additional data may be needed to explain differences
- if the results of indicator tests detecting putative DNA lesions (*e.g.* DNA binding, DNA damage, DNA repair; SCE) are not in agreement with results obtained in tests for mutagenicity, the results of mutagenicity tests are generally of higher significance provided that appropriate mutagenicity tests have been conducted. This is subject to expert judgement.
- if different findings are obtained *in vitro* and *in vivo*, in general, the results of *in vivo* tests indicate a higher degree of reliability. However, for evaluation of *negative* results *in vivo*, it should be considered whether the most appropriate tissues were sampled and whether there is adequate evidence of target tissue exposure
- the sensitivity and specificity of different test systems vary for different classes of substances. If available testing data for other related substances permit assessment of the performance of different assays for the class of substance under evaluation, the result from the test system known to produce more accurate responses would be given higher priority

Different results may also be available from the same test, performed by different laboratories or on different occasions. In this case, expert judgement should be used to evaluate the data and reach an overall conclusion. In particular, the quality of each of the studies and of the data provided should be evaluated, with special consideration of the study design, reproducibility of data, dose (concentration)-effect relationships, and biological relevance of the findings. The identity and purity of the test substance may also be a factor to take into account. In the case where an EU/OECD guideline is available for a test method, the quality of a study using the method is regarded as being higher if it was conducted in compliance with the requirements stated in the guideline, unless convincing scientific evidence can be provided to justify certain deviations from the standard test guideline for the specific substance evaluated. Furthermore, compared to non GLP-studies, studies compliant with GLP for the same assay generally provide more documentation and details of the study, which are important factors to consider when assessing study reliability/quality.

When making an assessment of the potential mutagenicity of a substance, or considering the need for further testing, data from various tests and genotoxic endpoints may be found. Both the strength and the weight of the evidence should be taken into account. The strongest evidence will be provided by modern, well-conducted studies with internationally established test guidelines/methods. For each test type and each genotoxic endpoint, there should be a separate *Weight of Evidence* analysis. It is not unusual for positive evidence of mutagenicity to be found in just one test type or for only one endpoint. In such cases the positive and negative results for different endpoints are not conflicting, but illustrate the advantage of using test methods for a variety of genetic alterations to increase the probability of identifying substances with mutagenic potential. Hence, results from methods testing different genotoxic endpoints should not be combined in an overall *Weight of Evidence* analysis, but should be subjected to such analysis separately for each endpoint. Based on the whole data set one has to consider whether there are data gaps: if there are data gaps further testing should be considered, otherwise an appropriate conclusion/assessment can be made.

R.7.7.4.2 Human data on mutagenicity

Human data have to be assessed carefully on a case-by-case basis. The interpretation of such data requires considerable expertise. Attention should be paid especially to the adequacy of the exposure information, confounding factors, co-exposures and to sources of bias in the study design or incident. The statistical power of the test may also be considered. It may be mentioned that, to date, no germ cell mutagen has been identified based on human data.

R.7.7.4.3 Remaining uncertainty on mutagenicity

Reliable data can be generated from well-designed and conducted studies *in vitro* and *in vivo*. However, due to the lack of human data available and the degree of uncertainty which is always inherent in testing, a certain level of uncertainty remains when extrapolating these testing data to the effect in humans.

R.7.7.5 Conclusions on mutagenicity

R.7.7.5.1 Concluding on Classification and Labelling

In order to conclude on an appropriate classification and labelling position with regard to mutagenicity, the available data should be considered using the criteria according to Annex I to the CLP Regulation (EC) No 1272/2008 (See also Section 3.5 of the *Guidance on the Application of the CLP criteria*).

R.7.7.5.2 Concluding on suitability for Chemical Safety Assessment

Considerations on dose (concentration)-response shapes and mode of action of mutagenic substances in test systems

Considerations on the dose (concentration)-response relationship and on possible mechanisms of action are important components of a risk assessment. The default assumption for genotoxic substances has for long been that they have a linear dose (concentration)-response relationship. However, this assumption has recently been challenged by experimental evidence showing that both direct and indirect acting genotoxins can possess non-linear or thresholded dose (concentration)-response curves.

Examples of non-DNA reactive mechanisms that may be demonstrated to lead to genotoxicity *via* non-linear or thresholded dose (concentration)-response relationships include inhibition of DNA synthesis, alterations in DNA repair, overloading of defence mechanisms (anti-oxidants or metal homeostatic controls), interaction with microtubule assembly leading to aneuploidy, topoisomerase inhibition, high cytotoxicity, metabolic overload and physiological perturbations (*e.g.* induction of erythropoeisis). The mechanisms underlying non-linear or thresholded dose (concentration)-response relationships for some DNA reactive genotoxic substances like alkylating agents seem linked to DNA repair capacity.

Assessment of the significance to be assigned to genotoxic responses mediated by such mechanisms would include an assessment of whether the underlying mechanism can be induced at substance concentrations that can be expected to occur under relevant *in vivo* conditions.

In general, several concentrations/doses are tested in genotoxicity assays. At least three experimental concentrations/doses have to be tested as recommended in the OECD test guidelines for genotoxicity. Determination of experimental dose (concentration)-effect relationships is one of several pieces of experimental information that are important to assess the genotoxic potential of a substance, and may be used as indicated below. It should be recognised that not all of these considerations may be applicable to *in vivo* data.

- the OECD introduction to the genotoxicity test guidelines lists the relevant criteria for identification of clear positive findings: (i) the increase in genotoxic response is concentration- or dose-related, (ii) at least one of the data points exhibits a statistically significant increase compared to the concurrent negative control, and (iii) the statistically significant result is outside the distribution of the historical negative control data (*e.g.* 95% confidence interval). In practice, the criterion for dose (concentration)-related increase in genotoxicity will be most helpful for *in vitro* tests, but care is needed to check for cytotoxicity or cell cycle delay which may cause deviations from a dose (concentration)-response related effect in some experimental systems

- genotoxicity tests are not designed in order to derive no effect levels. However, the magnitude of the lowest dose with an observed effect (*i.e.* the Lowest Observed Effect Dose or LOED) may, on certain occasions, be a helpful tool in risk assessment. This is true specifically for genotoxic effects caused by thresholded mechanisms, like, *e.g.* aneugenicity. Further, it can give an indication of the mutagenic potency of the substance in the test at issue. Modified studies, with additional dose or concentration points and improved statistical power may be useful in this regard. The Benchmark dose (BMD) approach presents several advantages over the NOED/LOED approach and can be used as an alternative strategy for dose (concentration)-response assessment (see the *Guidance on IR&CSA, Chapter R.8*)

- unusual shapes of dose (concentration)-response curves may contribute to the identification of specific mechanisms of genotoxicity. For example, extremely steep increases suggest an indirect mode of action or metabolic switching which could be confirmed by further investigation.

Considerations on genetic risks associated with human exposure to mutagenic substances

There are no officially adopted methods for estimating health risks associated with (low) exposures of humans to mutagens. In fact, most – if not all tests used today – are developed and applied to identify mutagenic properties of the substance, *i.e.* identification of the mutagenic hazard *per se*. In today's regulatory practice, the assessment of human health risks from exposure to mutagenic substances is considered to be covered by assessing and regulating the carcinogenic risks of these agents. The reason for this is that mutagenic events underlie these carcinogenic effects. Therefore, mutagenicity data is not used for deriving dose descriptors for risk assessment purposes and the reader is referred to this aspect in Section R.7.7.8 (Carcinogenicity) for guidance on how to assess the chemical safety for mutagenic substances.

R.7.7.5.3 Information not adequate

A *Weight of Evidence* approach, comparing available adequate information with the tonnage-triggered information requirements by REACH, may result in the conclusion that the requirements are not fulfilled. In order to proceed in gathering further information, the following testing strategy can be adopted:

R.7.7.6 Integrated Testing Strategy (ITS) for mutagenicity

R.7.7.6.1 Objective / General principles

This testing strategy describes a flexible, stepwise approach for hazard identification with regard to the mutagenic potential of substances, so that sufficient data may be obtained for adequate risk characterisation including classification and labelling. It serves to help minimise the use of animals and costs as far as it is consistent with scientific rigour. A flow chart of the testing strategy is presented in Figure R.7.7–1 and recommendations on follow up procedures based on different testing data sets are given in Table R.7.7–5. As noted later in this section, deviations from this strategy may be considered if existing data for related substances indicate that alternate testing strategies yield results with greater sensitivity and specificity for mutagenicity *in vivo*.

The strategy defines a level of information that is considered sufficient to provide adequate reassurance about the potential mutagenicity of most substances. As described below, this level of information will be required for most substances at the Annex VIII tonnage level specified in REACH, although circumstances are described when the data may be required for substances at Annex VII.

For some substances, relevant data from other sources/tests may also be available (*e.g.* physico-chemical, toxicokinetic, and toxicodynamic parameters and other toxicity data; data on well-investigated, structurally similar, substances). These should be reviewed because, sometimes, they may indicate that either more or less genotoxicity studies are needed on the substance than defined by standard information requirements; *i.e.* they may allow tailored testing/selection of test systems. For example, bacterial mutagenesis assays of inorganic metal compounds are frequently negative due to limited capacity for uptake of metal ions and/or the induction of large DNA deletions by metals in bacteria potentially leading to an increased death rate in mutants. The high prevalence of false negatives for metal compounds might suggest that mutagenesis assays with mammalian cells, as opposed to bacterial cells, would be the preferred starting point for testing for this class of Annex VII substances.

In summary, a key concept of the strategy is that initial genotoxicity tests and testing guidelines/methods should be selected with due consideration to existing data that has established the most accurate testing strategy for the class of compound under evaluation. Even then, initial testing may not always give adequate information and further testing may sometimes be considered necessary in the light of all available relevant information on the substance, including its use pattern. Further testing will normally be required for substances which give rise to positive results in any of the *in vitro* tests.

If negative results are available from an adequate evaluation of genotoxicity from existing data in appropriate test systems, there may be no requirement to conduct additional genotoxicity tests.

Substances for which there is a harmonised classification in category 1A, 1B or 2 for germ cell mutagenicity and/or category 1A or 1B for carcinogenicity according to Annex VI to the CLP Regulation (EC) No 1272/2008 will usually not require additional testing in order to meet the requirements of Annex VIII for the *in vitro* cytogenicity study in mammalian cells. Provided that appropriate risk management measures are implemented, the carcinogenicity study to meet the requirements of Annex X (see Section R.7.7.2 of this Guidance) and the reproductive toxicity studies to meet the requirements of Annexes VIII to X (see Section R.7.7.6 of this

Guidance) may also be omitted for substances classified in category 1A or 1B for germ cell mutagenicity. In cases where a registrant is unsure of the formal position on the classification of a substance, or wishes to make a classification proposal himself, advice should be sought from an appropriate regulatory body before proceeding with any further testing.

In case additional testing is needed to meet the requirements of Annexes IX or X, the registrant must first submit a testing proposal to the European Chemicals Agency (ECHA) and obtain prior authorisation before any testing can be initiated.

It should also be noted that recommendations on a strategy for genotoxicity testing have also recently been published by other authoritative organisations (EFSA, 2011; EMA, 2012; UK COM, 2011). These strategies are based either on a step-wise approach or on a test-battery approach. Their principle is basically similar to the one detailed in this Guidance, *i.e.* the use of different pieces of information, including non-testing data and results from *in vitro* and *in vivo* testing, for a comprehensive assessment of the genotoxic potential a substance since no single test is capable of detecting all genotoxic mechanisms. However, as these strategies aim at serving different regulations and purposes, some differences can exist between them, in particular regarding the list of *in vitro* and *in vivo* tests recommended and the way to use them. For instance, while the UK COM and EFSA now both recommend the use of a core two-test battery (*i.e.* a bacterial reverse mutation test combined with an *in vitro* micronucleus test) for *in vitro* genotoxicity assessment, the REACH Regulation and this Guidance state the *in vitro* mammalian cell gene mutation test as a legal requirement in addition to the Ames test and the *in vitro* cytogenicity test if both are negative. Moreover, the *in vitro* chromosome aberration test is considered as a possible alternative option to the *in vitro* micronucleus test under REACH while it is now generally agreed that these tests are not equivalent since the *in vitro* chromosome aberration test is not optimal to measure numerical chromosome aberrations. Although this guidance aims at implementing the latest scientific developments in the field of genotoxicity testing, its main goal is to provide advice and support to the registrant in complying with the legal requirements under REACH and is thus in line with this Regulation.

R.7.7.6.2 Preliminary considerations

For a comprehensive coverage of the potential mutagenicity of a substance, information on gene mutations (base substitutions and deletions/additions), structural chromosome aberrations (breaks and rearrangements) and numerical chromosome aberrations (loss or gain of chromosomes, defined as aneuploidy) is required. This may be obtained from available data or tests on the substance itself or, sometimes, by prediction using appropriate *in silico* techniques (*e.g.* chemical grouping, read-across or (Q)SAR approaches).

It is important that whatever is known of the physico-chemical properties of the test substance is taken into account before devising an appropriate testing strategy. Such information may impact upon both the selection of test systems to be employed and/or modifications to the test protocols used. The chemical structure of a substance can provide information for an initial assessment of mutagenic potential. The need for special testing in relation to photomutagenicity may be indicated in some specific cases by the structure of a molecule, its light absorbing potential or its potential to be photoactivated. By using expert judgement, it may be possible to identify whether a substance, or a potential metabolite of a substance, shares or does not share structural characteristics with known mutagens. This can be used to justify a higher or lower level of priority for the characterisation of the mutagenic potential of a substance. Where the level of evidence for mutagenicity is particularly strong, it may be possible to make a conclusive hazard assessment in accordance with Annex I to REACH without additional testing on the basis of structure-activity relationships alone: in this case, the registrant still has to provide sufficient information to meet the requirements of Annexes VII to X but he may, if scientifically justified and duly documented in the registration dossier, invoke the general rules of Annex XI for adaptation of the standard testing regime by demonstrating, *inter alia*, that the results he wishes to use instead of testing in that context are adequate for the purpose of classification and labelling and/or risk assessment.

In vitro tests are particularly useful for gaining an understanding of the potential mutagenicity of a substance and they have a critical role in this testing strategy. They are not, however, without their limitations. Animal tests will, in general, be needed for the clarification of the relevance of positive findings and in case of specific metabolic pathways that cannot be simulated adequately *in vitro*.

The toxicokinetic and toxicodynamic properties of the test substance should be considered before undertaking, or appraising, animal tests. Understanding these properties will enable appropriate protocols for the standard tests to be developed, especially with respect to tissue(s) to be investigated, the route of substance administration and the highest dose tested. If little is understood about the systemic availability of a test substance at this stage, toxicokinetic investigations or modelling may be necessary.

Certain substances in addition to those already noted may need special consideration, such as highly electrophilic substances that give positive results *in vitro*, particularly in the absence of metabolic activation. Although these substances may react with proteins and water *in vivo* and thus be rendered inactive towards many tissues, they may be able to express their mutagenic potential at the initial site of contact with the body. Consequently, the use of test methods such as the comet assay or the gene mutation assays using transgenic animals that can be applied to the respiratory tract, upper gastrointestinal tract and skin may be appropriate. It is possible that specialised test methods will need to be applied in these circumstances, and that these may not have recognised, internationally valid, test guidelines. The validity and utility of such tests and the selection of protocols should be assessed by appropriate experts or authorities on a case-by-case basis.

Criteria for the evaluation and interpretation of results (e.g. how to define clear positive and clear negative results) are normally defined in the testing guidelines/methods. There is no requirement for verification of a clear positive or clear negative result. In cases where the response is neither clearly negative nor clearly positive and in order to assist in establishing the biological relevance of a result (e.g. a weak or borderline increase), the data should be evaluated by expert judgement and/or further investigations. A substance giving such a response should be reinvestigated immediately, normally using the same test method, but varying the conditions to obtain conclusive results. Only if, even after further investigations, the data set precludes coming to a conclusion of a positive or negative result, will the result be concluded as equivocal. Wherever possible, clear results should be obtained for one step in the strategic procedure before going on to the next. In cases where this does not prove to be possible and the study is inconclusive as a consequence of *e.g.* some limitation of the test or procedure, a further test should be conducted in accordance with the strategy.

Tests need not be performed if it is not technically possible to do so, or if they are not considered necessary in the light of current scientific knowledge. Scientific justifications for not performing tests required by the strategy should always be documented. It is preferred that tests as described in OECD Guidelines or Regulation (EC) No 440/2008 are used where possible. Alternatively, for other tests, up-to-date protocols defined by internationally recognised groups of experts, *e.g.* International Workshop on Genotoxicity Testing (IWGT, under the umbrella of the International Association of Environmental Mutagen Societies), may be used provided that the tests are scientifically justified. It is essential that all tests be conducted according to rigorous protocols in order to maximise the potential for detecting a mutagenic response, to ensure that negative results can be accepted with confidence and that results are comparable when tests are conducted in different laboratories. At the time of writing this guidance, a standard test guideline/method is still to be established for the *in vivo* comet assay described below. So if this test is to be conducted, and in waiting for the adoption of the comet OECD TG 489, consultation on the protocol with an appropriate expert or authority is advisable.

If a registrant wishes to undertake any tests for substances at the Annex IX or X tonnage levels that require the use of vertebrate animals, then there is a need to make a testing proposal to ECHA first. Testing may only be undertaken after ECHA has accepted the testing proposal in a formal decision.

R.7.7.6.3 Testing strategy for mutagenicity

Standard information requirement at Annex VII

A preliminary assessment of mutagenicity is required for substances at the REACH Annex VII tonnage level. All available information should be included but, as a minimum, there should normally be data from a gene mutation test in bacteria unless existing data for analogous substances indicates this would be inappropriate. For substances with significant toxicity to bacteria, not taken up by bacteria, or for which the gene mutation test in bacteria cannot be performed adequately, an *in vitro* mammalian cell gene mutation test may be used as an alternative test.

When the result of the bacterial test is positive, it is important to consider the possibility of the substance being genotoxic in mammalian cells. The need for further test data to clarify this possibility at the Annex VII tonnage level will depend on an evaluation of all the available information relating to the genotoxicity of the substance.

Standard information requirement at Annex VIII

For a comprehensive coverage of the potential mutagenicity of a substance, information on gene mutations, and structural and numerical chromosome aberrations is required for substances at the Annex VIII tonnage level of REACH.

In order to ensure the necessary minimum level of information is provided, at least one further test is required in addition to the gene mutation test in bacteria. This should be an *in vitro* mammalian cell test capable of detecting both structural and numerical chromosome aberrations.

There are essentially two different methods that can be viewed as alternative options according to REACH for this first mammalian cell test:

- An *in vitro* chromosome aberration test (OECD TG 473), *i.e.* a cytogenetic assay for structural chromosome aberrations using metaphase analysis. An increase in polyploidy may indicate that a substance has the potential to induce numerical chromosome aberrations, but this test is not optimal to measure numerical aberrations and is not routinely used for that purpose. Accordingly, this test guideline is not designed to measure numerical aberrations.

- An *in vitro* micronucleus test (OECD TG 487). This is a cytogenetic assay that has the advantage of detecting not only structural chromosomal aberrations but also aneuploidy. Use of a cytokinesis block, fluorescence *in situ* hybridisation with probes for centromeric DNA, or immunochemical labelling of kinetochore proteins can provide information on the mechanisms of chromosome damage and micronucleus formation. The labelling and hybridisation procedures can enable aneugens to be distinguished from clastogens. This may sometimes be useful for risk characterisation. If a substance is demonstrated to be an aneugen, it is assumed that its genotoxicity is thresholded, in contrast to non-thresholded genotoxicity. Both types of genotoxicity mechanisms trigger different ways to perform risk assessment.

Other *in vitro* tests may be acceptable as the first mammalian cell test, but care should be taken to evaluate their suitability for the substance being registered and their reliability as a screen for substances that cause structural and/or numerical chromosome aberrations. A supporting rationale should be presented for a registration with any of these other tests.

It is possible to present existing data from an *in vivo* cytogenetic test (*i.e.* a study or studies conducted previously) as an alternative to the first *in vitro* mammalian cell test. For instance, if an adequately performed *in vivo* micronucleus test is available already it may be presented as an alternative. There may however be specific cases where the *in vitro* mammalian cell test can still be justified even though *in vivo* cytogeneticity data exist. For example, in the *in vivo* micronucleus test, certain substances may not reach the bone marrow due to low bioavailability or specific tissue/organ distribution and would result negative. In addition, even if bioavailability of the parent compound in the bone marrow can be demonstrated, a clastogen requiring liver metabolism and for which the reactive metabolites formed are too short-lived to reach the bone marrow could give a negative result in the *in vivo* micronucleus test. In this case, *in vitro* testing could provide useful information on the mode of action of the substance, *e.g.* to understand whether the substance is clastogenic (or aneugenic) *in vitro*, and whether it requires a specific metabolism to be genotoxic. Justification of *in vitro* testing when *in vivo* data already exist should be considered on a case-by-case basis.

An *in vitro* gene mutation study in mammalian cells (OECD TG 476) is the second part of the standard information set required for registration at the Annex VIII tonnage level. For substances that have been tested already, this information should always be presented as part of the overall *Weight of Evidence* for mutagenicity with reference to induction of gene mutations in mammalian cells. For other substances, this second *in vitro* mammalian cell test will normally only be required when the results of the bacterial gene mutation test and the first study in mammalian cells (*i.e.* an *in vitro* chromosome aberration test or an *in vitro* micronucleus test) are negative. This is to detect *in vitro* mutagens that give negative results in the other two tests.

Under specific circumstances it may be possible to omit the second *in vitro* study in mammalian cells, *i.e.* if it can be demonstrated that this mammalian cell test will not provide any further useful information about the potential *in vivo* mutagenicity of a substance, then it does not need to be conducted. This should be evaluated on a case-by-case basis as there may be classes of compound for which conclusive data can be provided to show that the sensitivity of the first two *in vitro* tests cannot be improved by the conduct of the third test.

The *in vitro* mammalian cell gene mutation test will not usually be required if adequate information is available from a reliable *in vivo* study capable of detecting gene mutations. Such information may come from a TGR gene mutation assay. A comet assay or a liver UDS test may also be adequate. However, these two tests being indicator assays detecting putative DNA lesions, their use should be justified on a case-by-case basis, *e.g.* the UDS should be used only when it can be reasonably assumed that the liver is a target organ, since the UDS is restricted to the detection of primary DNA repair in liver cells.

Provided the *in vitro* tests have given negative results, normally, no *in vivo* tests will be required to fulfil the standard information requirements at Annex VIII. However, there may be rare occasions when it is appropriate to conduct testing *in vivo*, for example when it is not possible technically to perform satisfactory tests *in vitro*. Substances which, by virtue of, for example, their physico-chemical characteristics, chemical reactivity or toxicity cannot be tested in one or more of the *in vitro* tests should be considered on a case-by-case basis. In the same way, it may not always be possible with the S9 fraction used *in vitro* to mimic the *in vivo* metabolism of some substances, and the relevance of the *in vitro* negative results for those substances should be evaluated case by case. In addition, equivocal *in vitro* results or different results from different *in vitro* studies may require the consideration of further testing to reach a clear conclusion on mutagenicity. For those types of cases, expert judgement would be needed to determine whether *in vivo* testing is appropriate.

Requirement for testing beyond the standard levels specified for Annexes VII and VIII

Introductory comments

Concerns raised by positive results from *in vitro* tests usually require the consideration of further testing. The chemistry of the substance, data on analogous substances, toxicokinetic and toxicodynamic data, and other toxicity data will also influence the timing and pattern of further testing.

Unless there are appropriate results from an *in vivo* study already, testing beyond the standard set of *in vitro* tests is normally first directed towards investigating the potential for mutagenicity in somatic cells *in vivo*. Positive results in somatic cells *in vivo* constitute the trigger for consideration of investigation of potential expression of genotoxicity in germ cells. However, to avoid unnecessary testing of vertebrate animals and for cost reasons, as the TGR assays give the possibility to include sampling of somatic and male germ cells in a single study providing adapted sampling times (see OECD TG 488 for details), it is recommended to include such samples in the testing proposal for the TGR assays and to appropriately store the germ cell samples for later analysis in case there is a positive result in any of the somatic tissues tested.

Substances that are negative in the standard set of in vitro tests

In general, substances that are negative in the full set of *in vitro* tests specified in REACH Annexes VII and VIII are considered to be non-genotoxic. There are only a very limited number of substances that have been found to be genotoxic *in vivo*, but not in the standard *in vitro* tests. Most of these are pharmaceuticals designed to affect pathways of cellular regulation, including cell cycle regulation, and this evidence is judged insufficient to justify routine *in vivo* testing of industrial chemicals. However, occasionally, knowledge about the metabolic profile of a substance may indicate that the standard *in vitro* tests are not sufficiently reassuring and a further *in vitro* test, or an *in vivo* test, may be needed in order to ensure mutagenicity potential is adequately explored (*e.g.* use of an alternative to rat liver S9 mix, a reducing system, a metabolically active cell line, or genetically engineered cell lines might be judged appropriate).

Substances for which an in vitro test is positive

REACH Annex VII substances for which only a bacterial gene mutation test has been conducted and for which the result is positive should be studied further, according to the requirements of Annex VIII.

Regarding Annex VIII, when both the mammalian cell tests are negative but there was a positive result in the bacterial test, it will be necessary to decide whether any further testing is needed on a case-by-case basis. For example, suspicion that a unique positive response observed in the bacterial test was due to a specific bacterial metabolism of the test substance could be explored further by investigation *in vitro.* Alternatively, an *in vivo* test may be required (see below).

In REACH Annex VIII, following a positive result in an *in vitro* mammalian cell mutagenicity test, adequately conducted somatic cell *in vivo* testing is required to ascertain if this potential can be expressed *in vivo*. In cases where it can be sufficiently deduced that a positive *in vitro* finding is not relevant for *in vivo* situations (*e.g.* due to the effect of the test substances on pH or cell viability, *in vitro*-specific metabolism: see also Section R.7.7.4.1), or where a clear threshold mechanism coming into play only at high concentrations that will not be reached *in vivo* has been identified (*e.g.* damage to non-DNA targets at high concentrations), *in vivo* testing will not be necessary.

Annex VIII, Column 2 requires the registrant to consider appropriate mutagenicity *in vivo* studies already at the Annex VIII tonnage level, in cases where positive results in genotoxicity studies have been obtained. It should be noted that where this involves tests mentioned in Annexes IX or X, such as *in vivo* somatic cell genotoxicity studies, testing proposals must be submitted by the registrant and accepted by ECHA in a formal decision before testing can be initiated.

Standard information requirement according to Annexes IX and X

According to the requirements of Annexes IX and X, if there is a positive result in any of the *in vitro* studies from Annex VII or VIII and there are no appropriate results available from an *in vivo* study already, an appropriate *in vivo* somatic cell genotoxicity study should be proposed.

Before any decisions are made about the need for *in vivo* testing, a review of the *in vitro* test results and all available information on the toxicokinetic and toxicodynamic profile of the test substance is needed. A particular *in vivo* test should be conducted only when it can be reasonably expected from all the properties of the test substance and the proposed test protocol that the specific target tissue will be adequately exposed to the test substance and/or its metabolites. If necessary, a targeted investigation of toxicokinetics should be conducted before progressing to *in vivo* testing (*e.g*. a preliminary toxicity test to confirm that absorption occurs and that an appropriate dose route is used).

In the interest of ensuring that the number of animals used in genotoxicity tests is kept to a minimum, both males and females should not automatically be used. In accord with standard guidelines, testing in one sex only is possible when the substance has been investigated for general toxicity and no sex-specific differences in toxicity have been observed. If the test is performed in a laboratory with substantial experience and historical data, it should be considered whether a concurrent positive control and a concurrent negative control for all time points (*e.g*. for both the 24h and 48h time point in the micronucleus assay) will really be necessary (Hayashi *et al*., 2000).

For test substances with adequate systemic availability (*i.e*. evidence for adequate availability to the target cells) there are several options for the *in vivo* testing:

- A rodent bone marrow or mouse peripheral blood micronucleus test (OECD TG 474) or a rodent bone marrow chromosome aberration test (OECD TG 475). The micronucleus test has the advantage of detecting not only structural chromosomal aberrations (clastogenicity) but also numerical chromosomal aberrations (aneuploidy). Potential species-specific effects may also influence the choice of species and test method used.

- A transgenic rodent (TGR) mutation assay (OECD TG 488). TGR assays measure gene mutations and chromosomal rearrangements (the latter specifically in the plasmid and Spi- assay models) using reporter genes present in every tissue. In principle every tissue can be sampled, including target tissues and specific site of contact tissues.

- A comet (single cell gel electrophoresis) assay (OECD TG 489), which detects DNA strand breaks and alkali labile DNA lesions. In contrast to the above-mentioned *in vivo* micronucleus test and *in vivo* chromosome aberration test, this assay has the advantage of not being restricted to bone marrow cells. In principle every tissue from which single cell or nuclei suspensions can be prepared can be sampled, including specific site of contact tissues.

- Other DNA strand breakage assays may be presented as alternatives to the comet assay. All DNA strand break assays should be considered as surrogate tests, they do not necessarily detect permanent changes to DNA.

- A rat liver Unscheduled DNA synthesis (UDS) test (OECD TG 486). The UDS test is an indicator test measuring DNA repair of primary damage in liver cells but not a surrogate test for gene mutations *per se*. The UDS test can detect some substances that induce *in vivo* gene mutation because this assay is sensitive to some (but not all) DNA repair mechanisms. However not all gene mutagens are positive in the UDS test and it is thus useful only for some classes of substances. A positive result in the UDS assay can indicate exposure of the liver DNA and induction of DNA damage by the substance under investigation but it is not sufficient information to conclude on the induction of gene mutation by the substance. A negative result in a UDS assay alone is not a proof that a substance does not induce gene mutation.

Only the first two options for testing mentioned above can be used directly for providing evidence of *in vivo* chromosomal and gene mutagenicity, respectively. The other test methods require specific supporting information, for example results from *in vitro* mutagenicity studies, to be used for making definitive conclusions about *in vivo* mutagenicity and lack thereof.

In the framework of the 3Rs principles, the combination of *in vivo* genotoxicity studies or integration of *in vivo* genotoxicity studies into repeated dose toxicity studies, whenever possible and when scientifically justified, is strongly encouraged if this is to be performed to meet the requirements of the REACH Annex VIII tonnage level. All the above-mentioned *in vivo* tests for somatic cells are in principle amenable to such integration although sufficient experience is not yet available for all of the tests. It is possible for two or more endpoints to be combined into a single *in vivo* study, and thereby save on resources and numbers of animals used. The comet assay and the *in vivo* micronucleus test can be combined into a single acute study, although some modification of treatment and sampling times is needed (Hamada *et al*., 2001; Madrigal-Bujaidar *et al*., 2008; Pfuhler *et al.*, 2009; Bowen *et al*., 2011,). These same endpoints can be integrated into repeated dose (*e.g.* 28-day) toxicity studies (Pfuhler *et al*., 2009; Rothfuss *et al.*, 2011; EFSA, 2011).

Any one of these tests may be conducted, but this has to be decided using expert judgement on a case-by-case basis. The nature of the original *in vitro* response(s) (*i.e.* gene mutation, structural or numerical chromosome aberration) should be considered when selecting the *in vivo* study. For example, if the test substance showed evidence of *in vitro* clastogenicity, then it would be appropriate to follow this up with either a micronucleus test or chromosomal aberration test or a comet assay. However, if a positive result were obtained in the *in vitro* micronucleus test, the rodent micronucleus test would be appropriate to best address clastogenic and aneugenic potential.

For substances that appear preferentially to induce gene mutations, the TGR assays are the most appropriate and usually preferred tests to follow-up an *in vitro* gene mutation positive result and detect, *in vivo*, substances that induce gene mutation. With respect to the 3Rs principle and taking into account that a positive result in somatic cells triggers the need to consider the potential for germ cell testing, germ cells should always be collected, if possible, when a TGR study is performed. The rat liver UDS test has a long history of use and may in some specific cases be adequate to follow-up an *in vitro* gene mutation positive result, but not for tissues other than the liver. The sensitivity of the UDS test has been questioned (Kirkland and Speit, 2008) and the use of this test should be justified on a case-by-case basis, and take account of substance-specific considerations. The recommended use of the comet assay has been discussed at the OECD level and is indicated in the corresponding OECD TG (see http://www.oecd.org/env/testguidelines). The choice of any of these three assays can be justified only if it can be demonstrated that the tissue(s) studied in the assay is (are) sufficiently exposed to the test substance (or its metabolites). This information can be derived from toxicokinetic data or, in case no toxicokinetic data are available, from the observation of treatment-related effects in the organ of interest. Another type of data that can support evidence of organ exposure is knowledge on the target organ(s) of specific classes of substances (*e.g.* the liver for aromatic amines). In case the *in vivo* comet assay is used or proposed by the registrant, the test protocol followed or suggested should be described in detail and be in accordance with current scientific best practice, so as to ensure acceptability of the generated data. In waiting for the adoption of the comet OECD TG 489 the registrant should follow the EFSA guidance indicating the minimum criteria for acceptance of the comet assay (2012), as well as, for the combined comet-micronucleus test, the 3-day treatment schedule described by *e.g.* Bowen *et al.* (2011). The TGR and comet assays offer greater flexibility than the UDS test, most notably with regard to the possibility of selecting a range of tissues for study on the basis of what is known of the toxicokinetics and toxicodynamics of the substance. It should be realised that the UDS and comet tests are indicator assays: the comet assay detects DNA lesions whereas the UDS assay detects DNA repair patches (which depend on the DNA repair pathway involved and the proficiency of the cell type investigated), indirectly showing DNA lesions. In contrast, the TGR gene mutation assays measure mutations, *i.e.* permanent transmissible changes in the DNA.

Additionally, evidence for *in vivo* DNA adduct formation in somatic cells together with positive results from *in vitro* mutagenicity tests are sufficient to conclude that a substance is an *in vivo* somatic cell mutagen. In such cases, positive results from *in vitro* mutagenicity tests may not trigger further *in vivo* somatic tissue testing, and the substance would be classified at least as a category 2 mutagen. The possibility for effects in germ cells would need further investigation (see Section R.7.7.6.3, *Substances that give positive results in an in vivo test for genotoxic effects in somatic cells*).

Non-standard studies supported by published literature may sometimes be more appropriate and informative than established assays. Guidance from an appropriate expert or authority should be sought before undertaking novel studies. Furthermore, additional data that support or clarify the mechanism of action may justify a decision not to test further.

For substances inducing gene mutation or chromosomal aberration *in vitro*, and for which no indication of sufficient systemic availablity has been presented, or that are short-lived or reactive, an alternative strategy involving studies to focus on tissues at initial sites of contact with the body should be considered. Expert judgement should be used on a case-by-case basis to decide which tests are the most appropriate. The main options are the *in vivo* comet assay, TGR gene mutation assays, and DNA adduct studies. For any given substance, expert judgement, based on all the available toxicological information, will indicate which of these tests are the most appropriate. The route of exposure should be selected that best allows assessment of the hazard posed to humans. For insoluble substances, the possibility of release of active molecules in the gastrointestinal tract may indicate that a test involving the oral route of administration is particularly appropriate.

If the testing strategy described above has been followed and the first *in vivo* test is negative, the need for a further *in vivo* somatic cell test should be considered. The second *in vivo* test should only then be proposed if it is required to make a conclusion on the genotoxic potential of the substance under investigation; *i.e.* if the *in vitro* data show the substance to have potential to induce both gene and chromosome mutations and the first *in vivo* test has not addressed this comprehensively. In this regard, on a case-by-case basis, attention should be paid to the quality and relevance of all the available toxicological data, including the adequacy of target tissue exposure.

For a substance giving negative results in adequately conducted, appropriate *in vivo* test(s), as defined by this strategy, it will normally be possible to conclude that the substance is not an *in vivo* mutagen.

Substances that give positive results in an in vivo test for genotoxic effects in somatic cells

Substances that have given positive results in cytogenetic tests both *in vitro* and *in vivo* can be studied further to establish whether they specifically act as aneugens, and therefore whether thresholds for their genotoxic activity can be identified, if this has not been established adequately already. This should be done using *in vitro* methods and will be helpful in risk evaluation.

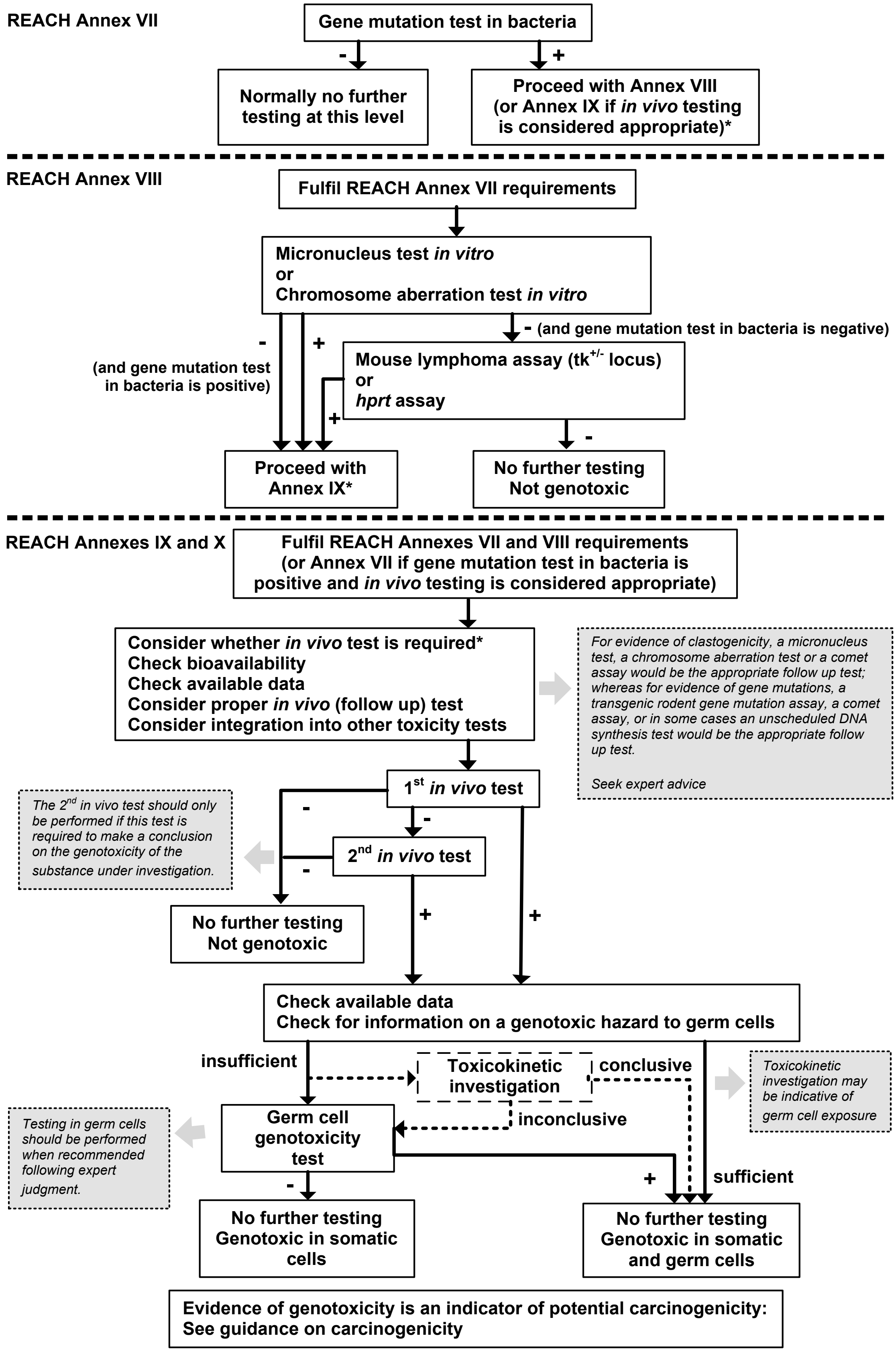
The potential for substances that give positive results in *in vivo* tests for genotoxic effects in somatic cells to affect germ cells should always be considered. The same is true for substances otherwise classified as category 2 mutagens under the CLP Regulation (EC) No 1272/2008 (for detailed information on the criteria for classification of substances for germ cell mutagenicity under the CLP Regulation (EC) No 1272/2008, see Section 3.5 of the *Guidance on the Application of the CLP criteria*). The first step is to make an appraisal of all the available toxicokinetic and toxicodynamic properties of the test substance. Expert judgement is needed at this stage to consider whether there is sufficient information to conclude that the substance poses a mutagenic hazard to germ cells. If this is the case, it can be concluded that the substance may cause heritable genetic damage and no further testing is justified. Consequently, the substance is classified as a category 1B mutagen. If the appraisal of mutagenic potential in germ cells is inconclusive, additional investigation will be necessary. In the event that additional information about the toxicokinetics of the substance would resolve the problem, toxicokinetic investigation (*i.e.* not a full toxicokinetic study) tailored to address this should be performed. Although the hazard class for mutagenicity primarily refers to germ cells, the induction of genotoxic effects at site of contact tissues by substances for which no indication of sufficient systemic availability or presence in germ cells has been presented are also relevant and considered for classification. For such substances, at least one positive *in vivo* genotoxicity test in somatic cells can lead to classification in Category 2 germ cell mutagens and to the labelling as 'suspected of causing genetic defects' if the positive effect *in vivo* is supported by positive results of *in vitro* mutagenicity tests. Classification as Category 2 germ cell mutagen may also have implications for potential carcinogenicity classification.

If specific germ cell testing is to be undertaken, expert judgement should be used to select the most appropriate test strategy. Internationally recognised guidelines are available for investigating clastogenicity in rodent spermatogonial cells and for the dominant lethal test. Dominant lethal mutations are believed to be primarily due to structural or numerical chromosome aberrations.

Alternatively, other methods can be used if deemed appropriate by expert judgement. These may include the TGR gene mutation assays (with modified sampling times as indicated in the OECD TG 488 to detect effects at the different stages of spermatogenesis), or DNA adduct analysis. In principle, it is the potential for effects that can be transmitted to the progeny that should be investigated, but tests used historically to investigate transmitted effects (the heritable translocation test and the specific locus test) use very large numbers of animals. They are rarely used and should normally not be proposed for substances registered under REACH.

In order to minimise animal use, it is recommended to include cell samples from both relevant somatic and germ cell tissues (*e.g*. testes) in *in vivo* mutagenicity studies: the somatic cell samples can be investigated first and, if they are positive, germ cell tissues can then also be analysed. Finally, the possibility to combine reproductive toxicity testing with *in vivo* mutagenicity testing could be considered.

Figure R.7.7–1 Flow chart of the mutagenicity testing strategy



* Registrants should note that a testing proposal must be submitted for a test mentioned in Annex IX or X, independently from the registered tonnage. Following examination of such testing proposal ECHA has to approve the test in its evaluation decision before it can be undertaken.

Table R.7.7–5 Examples of different testing data sets and follow-up procedures to conclude on genotoxicity/mutagenicity according to the mutagenicity testing strategy.

Depending on the *in vitro* and *in vivo* test results available and the REACH Annex(es) of interest, further testing may be required to meet the standard information requirements for mutagenicity and allow for a conclusion on genotoxicity/mutagenicity to be reached. Recommendations on what should be done or particurlarly looked at in those different cases are mentioned in the table, together with specific rules for adaptation when applicable (for detailed guidance see also main text).

	GM bact	Cyt vitro	GM vitro	Cyt vivo	GM vivo	Standard information required General follow up procedure	Conclusion	Specific rules for adaptation *[for detailed guidance, incl. timing of the tests, see main text]*	Comments
1	neg					Annex VII: no further tests are required. Annexes VIII, IX & X: conduct a CAbvitro or preferably a MNTvitro, and if this is negative, a GMvitro.	Annex VII: not genotoxic		Annexes VIII, IX & X: Select further tests in such a way that all the tests, together with other available information, enable thorough assessment for gene mutations and effects on chromosome structure and number.
2	neg	neg				Annex VII: no further tests are required. Annexes VIII, IX & X: conduct a GMvitro.	Annex VII: not genotoxic		Annexes VIII, IX & X: Select tests in such a way that all the tests, together with other available information, enable a thorough assessment for gene mutations and effects on chromosome structure and number.
3	neg		neg			Annex VII: no further tests are required. Annexes VIII, IX & X: conduct a CAbvitro or preferably a MNTvitro	Annex VII: not genotoxic		Annexes VIII, IX & X: Select tests in such a way that all the tests, together with other available information, enable a thorough assessment for gene mutations and effects on chromosome structure and number.

	GM bact	Cyt vitro	GM vitro	Cyt vivo	GM vivo	Standard information required General follow up procedure	Conclusion	Specific rules for adaptation *[for detailed guidance, incl. timing of the tests, see main text]*	Comments
4	neg	neg	neg			Annexes VII, VIII, IX & X: no further tests are required.	not genotoxic		The available metabolic evidence may, on rare occasions, indicate that *in vitro* testing is inadequate*; in vivo* testing is needed. Seek expert advice. Annexes VIII, IX & X: Select tests in such a way that all the tests, together with other available information, enable a thorough assessment for gene mutations and effects on chromosome structure and number.
5	pos					Annexes VII, VIII, IX & X: Complete *in vitro* testing with a CAbvitro or preferably a MNTvitro.			Consider need for further tests to understand the *in vivo* mutagenicity hazard, to make a risk assessment, and to determine whether C&L is justified.

	GM bact	Cyt vitro	GM vitro	Cyt vivo	GM vivo	Standard information required General follow up procedure	Conclusion	Specific rules for adaptation *[for detailed guidance, incl. timing of the tests, see main text]*	Comments
6	pos	neg				Annexes VII & VIII: *Complete in vitro* testing by conducting a GMvitro only under special conditions (see column 'Specific rules for adaption') Annexes IX & X: If systemic availability cannot be ascertained reliably, it should be investigated before progressing to *in vivo* tests. Select adequate somatic cell i*n vivo* test to investigate gene mutations *in vivo* (TGR, comet or if justified UDSvivo). If the TGR is to be conducted on somatic tissues, germ cell samples should be collected if possible, frozen and analysed for mutagenicity only in case of a positive result in somatic cells. If necessary seek expert advice.		Suspicion that a positive response observed in the GMbact was due to a specific bacterial metabolism of the test substance could be explored further by investigation *in vitro*.	Ensure that all tests together with other available information enable thorough assessment for gene mutations and effects on chromosome structure and number. Consider on a case-by-case basis need for further tests to understand the *in vivo* mutagenicity hazard, to make a risk assessment, and to determine whether C&L is justified.
7	neg	pos				Annexes VII, VIII, IX & X: If systemic availability cannot be ascertained reliably, it should be investigated before progressing to *in vivo* tests. Select adequate somatic cell *in vivo* test to investigate structural or numerical chromosome aberrations (MNTvivo or comet for *in vitro* clastogens and/or aneugens or CAbvivo for *in vitro*-clastogens) If necessary seek expert advice.			Ensure that all tests together with other available information enable thorough assessment for gene mutations and effects on chromosome structure and number. Consider need for further tests to understand the *in vivo* mutagenicity hazard, to make a risk assessment and to determine whether C&L is justified.

	GM bact	Cyt vitro	GM vitro	Cyt vivo	GM vivo	Standard information required General follow up procedure	Conclusion	Specific rules for adaptation *[for detailed guidance, incl. timing of the tests, see main text]*	Comments
8	pos	pos				Annexes VII, VIII, IX & X: If systemic availability cannot be ascertained with acceptable reliability, it should be investigated before progressing to *in vivo* tests. Select adequate somatic cell *in vivo* tests to investigate both structural or numerical chromosome aberrations and gene mutations. If necessary seek expert advice.		Generally, both genotoxic endpoints should be investigated. If the first *in vivo* test is positive, a second *in vivo* test to confirm the other genotoxic endpoint need not be conducted. If the first *in vivo* test is negative, a second *in vivo* test is required if the first test did not address the endpoints comprehensively.	Ensure that all tests together with other available information enable thorough assessment for gene mutations and effects on chromosome structure and number. Consider need for further tests to understand the *in vivo* mutagenicity hazard, to make a risk assessment, and to determine whether C&L is justified.
9	neg	neg	pos			Annexes VII, VIII, IX & X: If systemic availability cannot be ascertained reliably, it should be investigated before progressing to *in vivo* tests. Select adequate somatic cell i*n vivo* test to investigate gene mutations *in vivo* (TGR, comet or if justified UDSvivo). If the TGR is to be conducted on somatic tissues, germ cell samples should be collected if possible, frozen and analysed for mutagenicity only in case of a positive result in somatic cells. If necessary seek expert advice.			Ensure that all tests together with other available information enable thorough assessment for gene mutations and effects on chromosome structure and number. Consider on a case-by-case basis need for further tests to understand the *in vivo* mutagenicity hazard, to make a risk assessment, and to determine whether C&L is justified.
10	pos	neg			neg	Annexes VII, VIII, IX & X: no further tests are required.	not genotoxic		Further *in vivo* test may be necessary depending on the quality and relevance of available data.
	neg	pos		neg					

	GM bact	Cyt vitro	GM vitro	Cyt vivo	GM vivo	Standard information required General follow up procedure	Conclusion	Specific rules for adaptation *[for detailed guidance, incl. timing of the tests, see main text]*	Comments
11	pos	neg			pos	Annexes VII, VIII, IX & X: No further testing in somatic cells is needed. Germ cell mutagenicity tests should be considered. If necessary seek expert advice on implications of all available data on toxicokinetics and toxicodynamics and on the choice of the proper germ cell mutagenicity test.	genotoxic	Expert judgement is needed at this stage to consider whether there is sufficient information to conclude that the substance poses a mutagenic hazard to germ cells. If this is the case, it can be concluded that the substance may cause heritable genetic damage and no further testing is justified.	If the appraisal of mutagenic potential in germ cells is inconclusive, additional investigation may be necessary. Risk assessment and C&L can be completed.
	neg	pos		pos					
	neg	neg	pos		pos				
12	pos	pos	(pos)	pos		Annexes VII, VIII, IX & X: No further testing in somatic cells is needed. Germ cell mutagenicity tests should be considered. If necessary seek expert advice on implications of all available data on toxicokinetics and toxicodynamics and on the choice of the proper germ cell mutagenicity test.	genotoxic	Expert judgement is needed at this stage to consider whether there is sufficient information to conclude that the substance poses a mutagenic hazard to germ cells. If this is the case, it can be concluded that the substance may cause heritable genetic damage and no further testing is justified.	If the appraisal of mutagenic potential in germ cells is inconclusive, additional investigation may be necessary. Risk assessment and C&L can be completed.
	pos	pos	(pos)		pos				
13	pos	pos	(pos)	neg		Annexes VII, VIII, IX & X: Select adequate somatic cell *in vivo* tests to investigate both structural or numerical chromosome aberrations and gene mutations. If necessary seek expert advice.			
	pos	pos	(pos)		neg				

	GM bact	Cyt vitro	GM vitro	Cyt vivo	GM vivo	Standard information required General follow up procedure	Conclusion	Specific rules for adaptation *[for detailed guidance, incl. timing of the tests, see main text]*	Comments
14	pos	pos	(pos)	neg	neg	Annexes VII, VIII, IX & X: no further tests are required.	not genotoxic	Further *in vivo* test may be necessary pending on the quality and relevance of available data.	Risk assessment and C&L can be completed.
15	pos	pos	(pos)	neg	pos	Annexes VII, VIII, IX & X: No further testing in somatic cells is needed. Germ cell mutagenicity tests should be considered. If necessary seek expert advice on implications of all available data on toxicokinetics and toxicodynamics and on the choice of the proper germ cell mutagenicity test.	genotoxic	Expert judgement is needed at this stage to consider whether there is sufficient information to conclude that the substance poses a mutagenic hazard to germ cells. If this is the case, it can be concluded that the substance may cause heritable genetic damage and no further testing is justified.	If the appraisal of mutagenic potential in germ cells is inconclusive, additional investigation will be necessary. Risk assessment and C&L can be completed.
	pos	pos	(pos)	pos	neg				

Abbreviations: pos: positive; neg: negative; (pos): the follow up is independent from the result of this test; GM_{bact}: gene mutation test in bacteria (Ames test); Cyt_{vitro}: cytogenetic assay in mammalian cells; CAb_{vitro}: *in vitro* chromosome aberration test; MNT_{vitro}: *in vitro* micronucleus test; GM_{vitro}: gene mutation assay in mammalian cells; Cyt_{vivo}: cytogenetic assay in experimental animals; GM_{vivo}: gene mutation assay in experimental animals; CAb_{vivo}: *in vivo* chromosome aberration test (bone marrow); MNT_{vivo}: *in vivo* micronucleus test (erythrocytes); UDS_{vivo}: *in vivo* unscheduled DNA synthesis test; TGR: *in vivo* gene mutation test with transgenic rodent; comet: comet assay.

R.7.7.7 References on mutagenicity

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R.7.7.8 Carcinogenicity

R.7.7.8.1 Definition of carcinogenicity

Chemicals are defined as carcinogenic if they induce tumours, increase tumour incidence and/or malignancy or shorten the time to tumour occurrence. Benign tumours that are considered to have the potential to progress to malignant tumours are generally considered along with malignant tumours. Chemicals can induce cancer by any route of exposure (e.g. when inhaled, ingested, applied to the skin or injected), but carcinogenic potential and potency may depend on the conditions of exposure (e.g. route, level, pattern and duration of exposure). Carcinogens may be identified from epidemiological studies, from animal experiments and/or other appropriate means that may include (Quantitative) Structure-Activity Relationships ((Q)SAR) analyses and/or extrapolation from structurally similar substances (read-across). Each strategy for the identification of potential carcinogens is discussed in detail later in this report. The determination of the carcinogenic potential of a chemical is based on a *Weight of Evidence* approach. Classification criteria are given in the (EU Directive 67/548/EEC).[162]

The process of carcinogenesis involves the transition of normal cells into cancer cells *via* a sequence of stages that entail both genetic alterations (i.e*.* mutations[163]) and non-genetic events. Non-genetic events are defined as those alterations/processes that are mediated by mechanisms that do not affect the primary sequence of DNA and yet increase the incidence of tumours or decrease the latency time for the appearance of tumours. For example; altered growth and death rates, (de)differentiation of the altered or target cells and modulation of the expression of specific genes associated with the expression of neoplastic potential (e.g. tumour suppressor genes or angiogenesis factors) are recognised to play an important role in the process of carcinogenesis and can be modulated by a chemical agent in the absence of genetic change to increase the incidence of cancer.

Carcinogenic chemicals have conventionally been divided into two categories according to the presumed mode of action: genotoxic or non-genotoxic[163]. Genotoxic modes of action involve genetic alterations caused by the chemical interacting directly with DNA to result in a change in the primary sequence of DNA. A chemical can also cause genetic alterations indirectly following interaction with other cellular processes (e.g. secondary to the induction of oxidative stress). Non-genotoxic modes of action include epigenetic changes, i.e., effects that do not involve alterations in DNA but that may influence gene expression, altered cell-cell communication, or other factors involved in the carcinogenic process. For example, chronic cytotoxicity with subsequent regenerative cell proliferation is considered a mode of action by which tumour development can be enhanced: the induction of urinary bladder tumours in rats may, in certain cases, be due to persistent irritation/inflammation, tissue erosion and regenerative hyperplasia of the urothelium following the formation of bladder stones. Other modes of non-gentoxic action can involve specific receptors (e.g. PPARα, which is associated with liver tumours in rodents; or tumours induced by various hormonal mechanisms). As with other nongenotoxic modes of action, these can all be presumed to have a threshold.

R.7.7.8.2 Objective of the guidance on carcinogenicity

The objective of investigating the carcinogenicity of chemicals is to identify potential human carcinogens, their mode(s) of action, and their potency.

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162 Directive 67/548/EEC will be repealed and replaced with the EU Regulation on classification, labelling and packaging of substances and mixtures, implementing the Globally Harmonized System (GHS).

163 For a definition and for background information on the terms mutagnicity and genotoxicity see Section R.7.7.1.1.

With respect to carcinogenic potential and potency the most appropriate source of information is directly from human epidemiology studies (e.g. cohort, case control studies). In the absence of human data, animal carcinogenicity tests may be used to differentiate carcinogens from non-carcinogens. However, the results of these studies subsequently have to be extrapolated to humans, both in qualitative as well as quantitative terms. This introduces uncertainty, both with regard to potency for as well as relevance to humans, due to species specific factors such as differences in chemical metabolism and toxicokinetics and difficulties inherent in extrapolating from the high doses used in animal bioassays to those normally experienced by humans.

Once a chemical has been identified as a carcinogen, there is a need to elucidate the underlying mode of action, i.e. whether the chemical is directly genotoxic or not. In risk assessment a distinction is made between different types of carcinogens (see above).

For genotoxic carcinogens exhibiting direct interaction with DNA it is not generally possible to infer the position of the threshold from the *no-observed-effect level* on a dose-response curve, even though a biological threshold below which cancer is not induced may exist.

For non-genotoxic carcinogens, *no-effect-thresholds* are assumed to exist and to be discernable (e.g. if appropriately designed studies of the dose response for critical non-genotoxic effects are conducted). No effect thresholds may also be present for certain carcinogens that cause genetic alterations *via* indirect effects on DNA following interaction with other cellular processes (e.g. carcinogenic risk would manifest only after chemically induced alterations of cellular processes had exceeded the compensatory capacity of physiological or homeostatic controls). However, in the latter situation the scientific evidence needed to convincingly underpin this indirect mode of genotoxic action may be more difficult to achieve. Human studies are generally not available for making a distinction between the above mentioned modes of action; and a conclusion on this, in fact, depends on the outcome of mutagenicity/genotoxicity testing and other mechanistic studies. In addition to this, animal studies (e.g. the carcinogenicity study, repeated dose studies, and experimental studies with initiation-promotion protocols) may also inform on the underlying mode of carcinogenic action.

The cancer hazard and mode of action may also be highly dependent on exposure conditions such as the route of exposure. A pulmonary carcinogen, for example, can cause lung tumours in rats following chronic inhalation exposure, but there may be no cancer hazard associated with dermal exposure. Therefore, all relevant effect data and information on human exposure conditions are evaluated in a *Weight of Evidence* approach to provide the basis for regulatory decisions.

R.7.7.9 Information requirements on carcinogenicity

For the endpoint of carcinogenicity, standard information requirements are specifically described for substances produced or imported in quantities of ≥1000 t/y (Annex X). The precise information requirements will differ from substance to substance, according to the toxicity information already available and details of use and human exposure for the substance in question. The REACH Annexes VI to XI should be considered as a whole and in conjunction with the overall requirements of registration and evaluation.

Column 2 of Annex X lists specific rules according to which the required standard information may be omitted, replaced by other information, provided at a different stage or adapted in another way. If the conditions are met for adaptations under column 2 of this Annex, the fact and the reasons for each adaptation should be clearly indicated in the registration.

The standard information requirements for carcinogenicity and the specific rules for adaptation of these requirements are presented in Table R.7.7–6.

Table R.7.7–6 Standard information requirements for carcinogenicity and the specific rules for adaptation of these requirements

COLUMN 1 STANDARD INFORMATION REQUIRED	COLUMN 2 SPECIFIC RULES FOR ADAPTATION FROM COLUMN 1
Annexes VII-IX	
Annex X: 1. Carcinogenicity study.	1. A carcinogenicity study may be proposed by the registrant or may be required by the Agency in accordance with Articles 40 or 41 if: - the substance has a widespread dispersive use or there is evidence of frequent or long-term human exposure; and - the substance is classified as mutagen category 3 or there is evidence from the repeated dose study(ies) that the substance is able to induce hyperplasia and/or pre-neoplastic lesions. If the substance is classified as mutagen category 1 or 2, the default presumption would be that a genotoxic mechanism for carcinogenicity is likely. In these cases, a carcinogenicity test will normally not be required.

R.7.7.10 Information sources on carcinogenicity

There are many different sources of information that may permit inferences to be drawn regarding the potential of chemicals to be carcinogenic to humans. Clearly, these sources not only allow the identification of potential carcinogenic activity, but in case a substance is identified as a likely carcinogen they should also be informative with respect to the underlying mode of action as well as probable carcinogenic potency. The requirements of REACH call for proper classification and labelling, as well as for a quantitative assessment of risk that permits conclusions to be drawn regarding conditions under which safe use of the chemical may occur: i.e. the data should allow concluding on threshold or non-threshold mode of action, and on some dose descriptor (characterising the dose-response), preferably in quantitative terms.

It is noted (and indicated below), that the various sources inform differently on the aspects of hazard identification, mode of action, or carcinogenic potency.

R.7.7.10.1 Non-human data on carcinogenicity

Non-testing data on carcinogenicity

The capacity for performing the standard rodent cancer bioassay is limited by economic, technical and animal welfare considerations, such that an increased emphasis is being placed on the development of alternative, non-animal testing methods. However, carcinogenicity predictions through use of non-testing data currently represent an extreme challenge due to the multitude of possible mechanisms. Prediction of carcinogenicity in humans is especially problematic.

Although significant challenges remain, a broad spectrum of non-testing techniques exist for elucidating mechanistic, toxicokinetic or toxicodynamic factors important in understanding the carcinogenic process. These range from expert judgement, to the evaluation of structural similarities and analogues (i.e. read-across and grouping), to the use of (Q)SAR models for carcinogenicity. Such information may assist with priority setting, hazard identification, elucidation of the mode of action, potency estimation and/or with making decisions about testing strategies based on a *Weight of Evidence* evaluation.

Genotoxicity remains an important mechanism for chemical carcinogenesis and its definitive demonstration for a chemical is often decisive for the choice of risk assessment methodology. A commentary about non-testing options for genotoxicity is provided in Section R.7.7.1. It has long been known that certain chemical structures or fragments can be associated with carcinogenicity, often through DNA-reactive mechanisms. Useful guidance for structures and fragments that are associated with carcinogenicity *via* DNA reactive mechanisms has been provided by the US Food and Drug Administration's "Guideline for Threshold Assessment, Appendix I, Carcinogen Structure Guide" (US FDA, 1986); the Ashby-Tennant "super-mutagen model" (*e.g.* Ashby and Tennant, 1988); and subsequent builds on this model (*e.g.* Ashby and Paton, 1993; Munro *et al*., 1996). Additional information on structural categories can be found in the "IARC Monographs on the Evaluation of Carcinogenic Risk of Chemicals to Man" (IARC, 2006).

Models predicting test results for genotoxic endpoints for closely related structures are known as *local* or congeneric (Q)SARs. These congeneric models are less common for carcinogenicity than for mutagenicity. Franke *et al.* (2001) provide an example of such a model for a set of genotoxic carcinogens.

The situation is far more complex for non-genotoxic carcinogenicity due to the large number of different mechanisms that may be involved. However, progress is being made in predicting activity for classes of compounds that exert effect *via* binding to oestrogen receptors, induction of peroxisomal proliferation, and binding to tubulin proteins. Although many potentially useful models exist, their applicability will be highly dependant on the proposed mechanism and chemical class.

Several *global* (non-congeneric) models exist which attempt to predict (within their domain) the carcinogenic hazard of diverse (non-congeneric) groups of substances (*e.g.* Matthews and Contrera, 1998). These models may also assist in screening, priority-setting, deciding on testing strategies and/or the assessment of hazard or risk based on *Weight of Evidence*. Most are commercial and include expert systems such as Onco-Logic® (currently made available by US-EPA) and DEREK, artificial intelligence systems from MULTICASE, and the TOPKAT program. Historically, the performance of such models has been mixed and is highly dependent on the precise definition of carcinogenicity among those substances used to develop and test the model. These have been reviewed by ECETOC (2003) and Cronin *et al*. (2003).

Free sources of carcinogenicity predictions include the Danish EPA (Q)SAR database (accessible through the European Commission's Chemicals Bureau: ECB http://qsar.food.dtu.dk). Predictions in this database for 166,000 compounds include eight MULTICASE FDA cancer models, a number of genotoxicity predictions, rodent carcinogenic potency, hepatospecificity, oestrogenicity and aryl hydrocarbon (AH) receptor binding. Another source of carcinogenicity predictions is the Enhanced NCI database "*Browser*", which is sponsored by the US National Cancer Institute. This has 250,000 chemical predictions within it (http://cactus.nci.nih.gov), including general carcinogenicity, mutagenicity and additional endpoints, which may be of potential mechanistic interest in specific cases.

Further information on carcinogenicity models is available in the OECD Database on Chemical Risk Assessment Models where they are listed in an effort to identify tools for research and development of chemical substances. (http://www.olis.oecd.org/comnet/env/models.nsf/-MainMenu?OpenForm).

The guidance on the Grouping of Chemicals and on (Q)SARs (see Sections R.6.2 and R.6.1, respectively) explains basic concepts of grouping and (Q)SARs and gives generic guidance on validation, adequacy and documentation for regulatory purposes. The guidance also describes a stepwise approach for the use of read-across/grouping and (Q)SARs.

It is noted that all the above mentioned sources may potentially inform on possible carcinogenic hazard and on the underlying mode of action, as well as on carcinogenic potency.

Testing data on carcinogenicity

In vitro data

The following *in vitro* data, which provide direct or indirect information useful in assessing the carcinogenic potential of a substance and (potentially) on the underlying mode(s) of action, may be available. No single endpoint or effect in and of itself possesses unusual significance for assessing carcinogenic potential but must be evaluated within the context of the overall toxicological effects of a substance under evaluation as described in Section R.7.7.11.1. Except as noted, standardised protocols do not exist for most of the *in vitro* endpoints noted. Rather, studies are conducted in accordance with expert judgement using protocols tailored to the specific substance, target tissue and cell type or animal species under evaluation.

genotoxicity studies: the ability of substances to induce mutations or genotoxicity (as defined in Section R.7.7.1) can be indicative of carcinogenic potential. However, correlations between mutagenicity/genotoxicity and carcinogenesis are stronger when effects are observed in appropriately designed *in vivo* as opposed to *in vitro* studies.

***in vitro* cell transformation assay results:** such assays assess the ability of chemicals to induce changes in the morphological and growth properties of cultured mammalian cells that are presumed to be similar to phenotypic changes that accompany the development of neoplastic or pre-neoplastic lesions *in vivo* (OECD, 2006). The altered cells detected by such assays may possess, or can subsequently acquire, the ability to grow as tumours when injected into appropriate host animals. As *in vitro* assays, cell transformation assays are restricted to the detection of effects of chemicals at the cellular level and will not be sensitive to carcinogenic activity mediated by effects exerted at the level of intact tissues or organisms.

mechanistic studies, e.g. on:

- cell proliferation: sustained cell proliferation can facilitate the growth of neoplastic/pre-neoplastic cells and/or create conditions conducive to spontaneous changes that promote neoplastic development.

- altered intercellular gap junction communication: exchange of growth suppressive or other small regulatory molecules between normal and neoplastic/pre-neoplastic cells through gap junctions is suspected to suppress phenotypic expression of neoplastic potential. Disruption of gap junction function, as assessed by a diverse array of assays for fluorescent dye transfer or the exchange of small molecules between cells, may attenuate the suppression of neoplastic potential by normal cells.

- hormone- or other receptor binding; a number of agents may act through binding to hormone receptors or sites for regulatory substances that modulate the growth of cells and/or control the expression of genes that facilitate the growth of neoplastic cells. Interactions of this nature are diverse and generally very compound specific.

other targeted mechanisms of action:

- immunosuppressive activity: neoplastic cells frequently have antigenic properties that permit their detection and elimination by normal immune system function. Suppression of normal immune function can reduce the effectiveness of this *immune surveillance* function and permit the growth of neoplastic cells induced by exogenous factors or spontaneous changes.

- ability to inhibit or induce apoptosis: apoptosis, or programmed cell death, constitutes a sequence of molecular events that results in the death of cells, most often by the release of specific enzymes that result in the degradation of DNA in the cell nucleus. Apoptosis is integral to the control of cell growth and differentiation in many tissues. Induction of apoptosis can eliminate cells that might otherwise suppress the growth of neoplastic cells; inhibition of apoptosis can permit pre-neoplastic/neoplastic cells to escape regulatory controls that might otherwise result in their elimination.

- ability to stimulate angiogenesis or the secretion of angiogenesis factors: the growth of pre-neoplastic/neoplastic cells in solid tumours will be constrained in the absence of vascularisation to support the nutritional requirements of tumour growth. Secretion of angiogenesis factors stimulates the vascularisation of solid tumour tissue and enables continued tumour growth.

Animal data

A wide variety of study categories may be available, which may provide direct or indirect information useful in assessing the carcinogenic potential of a substance to humans. They include:

carcinogenicity studies (conventional long-term or life-time studies in experimental animals): Carcinogenicity testing is typically conducted using rats and mice, and less commonly in animals such as the Guinea pig, Syrian hamster and occasionally mini-pigs, dogs and primates. The standard rodent carcinogenicity bioassay would be conducted using rats or mice randomly assigned to treatment groups. Exposures to test substances may be *via* oral, inhalation or dermal exposure routes. The selection of exposure route is often dictated by *a priori* assumptions regarding the routes of exposure relevant to humans and/or other data sources (e.g. epidemiology studies or repeated dose toxicity studies in animals) that may indicate relevance of a given exposure route. Standardised protocols for such studies have been developed and are well validated (*e.g.* OECD TGs 451, 453 or US-EPA 870.4200).

short and medium term bioassay data (e.g. mouse skin tumour, rat liver foci model, neonatal mouse model): multiple assays have been developed that permit the detection and quantitation of putative pre-neoplastic changes in specific tissues. The induction of such *pre-neoplastic foci* may be indicative of carcinogenic potential. Such studies are generally regarded as adjuncts to conventional cancer bioassays, and while less validated and standardised, are applicable on a case-by-case basis for obtaining supplemental mechanistic and dose response information that may be useful for risk assessment (Enzmann *et al*., 1998).

genetically engineered (transgenic) rodent models (e.g. *Xpa*$^{-/-}$, *p53*$^{+/-}$, *ras*H2 or Tg.AC): animals can be genetically engineered such that one or more of the molecular changes required for the multi-step process of carcinogenesis has been accomplished (Tennant *et al.*, 1999). This can increase the sensitivity of the animals to carcinogens and/or decrease the latency with which spontaneous or induced tumours are observed. The genetic changes in a given strain of engineered animals can increase sensitivity to carcinogenesis in a broad range of tissues or can be specific to the changes requisite for neoplastic development in one or only a limited number of tissues (Jacobson-Kram, 2004; Pritchard *et al.*, 2003; ILSI/HESI, 2001). Data from these models may be used in a *Weight of Evidence* analysis of a chemical’s carcinogenicity.

genotoxicity studies *in vivo*: the ability of substances to induce mutations or genotoxicity (as defined in Section R.7.7.1.1) can be indicative of carcinogenic potential. There is, in general, a good correlation between positive genotoxicity findings *in vivo* and animal carcinogenicity bioassay results

repeated dose toxicity tests: can identify tissues that may be specific targets for toxicity and subsequent carcinogenic effects. Particular significance can be attached to the observation of pre-neoplastic changes (e.g. hyperplasia or metaplasia) suspected to be conducive to tumour development and may assist in the development of dose-effect relationships (Elcombe *et al.*, 2002).

studies on the induction of sustained cell proliferation: substances can induce sustained cell proliferation *via* compensatory processes that continuously regenerate tissues damaged by toxicity. Some substances can also be tissue-specific mitogens, stimulating cell proliferation in the absence of overt toxic effects. Mitogenic effects are often associated with the action of tumour promoters. Both regenerative cell proliferation and mitogenic effects can be necessary, but not sufficient, for tumour development but have sufficiently different mechanistic basis that care should be exercised in assessing which is occurring (Cohen and Ellwein, 1991; Cohen *et al.*, 1991).

studies on immunosuppressive activity: as noted earlier, suppression of normal immune surveillance functions can interfere with normal immune system functions that serve to identify and eliminate neoplastic cells.

studies on toxicokinetics: can identify tissues or treatment routes that might be the targets for toxicity and can deliver data on exposure and metabolism in specific organs. Linkages to subsequent carcinogenic impacts may or may not exist, but such data can serve to focus carcinogenesis studies upon specific tissue types or animal species.

other studies on mechanisms/modes of action, e.g. OMICs studies (toxicogenomics, proteomics, metabonomics and metabolomics): carcinogenesis is associated with multiple changes in gene expression, transcriptional regulation, protein synthesis and other metabolic changes. Specific changes diagnostic of carcinogenic potential have yet to be validated, but these rapidly advancing fields of study may one day permit assessment of a broad array of molecular changes that might be useful in the identification of potential carcinogens.

It is noted that the above tests differently inform on hazard identification, mode of action or carcinogenic potency. For example, conventional bioassays are used for hazard identification and potency estimation (i.e. derivation of a dose descriptor), whereas studies using genetically engineered animals are informative on potential hazard and possibly mode of action, but less on carcinogenic potency as they are considered to be highly sensitive to tumour induction.

R.7.7.10.2 Human data on carcinogenicity

Human data may provide direct information on the potential carcinogenicity of the substance. Relevant human data of sufficient quality, if available, are preferable to animal data as no extrapolations between species, or from high to low dose are necessary. Epidemiological data will not normally be available for new substances but may well be available for substances that have been in use for many decades. For substances in common use prior to the implementation of modern occupational hygiene measures, the intensity of human exposures to some carcinogens was sufficient to produce highly significant, dose-dependent increases in cancer incidence.

A number of basic epidemiological study designs exist and include cohort, case-control and registry based correlational (e.g. ecological) studies. The most definitive epidemiological studies on chemical carcinogenesis are generally cohort studies of occupationally exposed populations, and less frequently the general population. Cohort studies evaluate groups of initially healthy individuals with known exposure to a given substance and follow the development of cancer incidence or mortality over time. With adequate information regarding the intensity of exposure experienced by individuals, dose dependent relationships with cancer incidence or mortality in the overall cohort can be established. Case-control studies retrospectively investigate individuals who develop a certain type of cancer and compare their chemical exposure to that of individuals who did not develop disease. Case control studies are frequently nested within the conduct of cohort studies and can help increase the precision with which excess cancer can be associated with a given substance. Correlational or ecological studies evaluate cancer incidence/mortality in groups of individuals presumed to have exposure to a given substance but are generally less precise since measures of the exposure experienced by individuals are not available. Observations of cancer clusters and case reports of rare tumours may also provide useful supporting information in some instances but are more often the impetus for the conduct of more formal and rigorous cohort studies.

Besides the identification of carcinogens, epidemiological studies may also provide information on actual exposures in representative (or historical) workplaces and/or the environment and the associated dose-response for cancer induction. Such information can be of much value for risk characterisation.

Although instrumental in the identification of known human carcinogens, epidemiology studies are often limited in their sensitivity by a number of technical factors. The extent and/or quality of information that is available regarding exposure history (e.g. measurements of individual exposure) or other determinants of health status within a cohort is often limited. Given the long latency between exposure to a carcinogen and the onset of clinical disease, robust estimates of carcinogenic potency can be difficult to generate. Similarly, occupational and environmentally exposed cohorts often have co-exposures to carcinogenic substances that have not been documented (or are incompletely documented). This can be particularly problematic in the study of long established industry sectors (e.g. base metal production) now known to entail co-exposures to known carcinogens (e.g. arsenic) present as trace contaminants in the raw materials being processed.. Retrospective hygiene and exposure analyses for such sectors are often capable of estimating exposure to the principle materials being produced, but data documenting critical co-exposures to trace contaminants may not be available. Increased cancer risk may be observed in such settings, but the source of the increased risk can be difficult to determine. Finally, a variety of lifestyle confounders (smoking and drinking habits, dietary patterns and ethnicity) influence the incidence of cancer but are often inadequately documented for purposes of adequate confounder control. Thus, modest increases in cancer at tissue sites known to be impacted by confounders (e.g. lung and stomach) can be difficult to interpret.

Techniques for biomonitoring and molecular epidemiology are developing rapidly. These newly developed tools promise to provide information on biomarkers of individual susceptibility, critical target organ exposures and whether effects occur at low exposure levels. Such ancillary information may begin to assist in the interpretation of epidemiology study outcomes and the definition of dose response relationships. For example, monitoring the formation of chemical adducts in haemoglobin molecules (Birner *et al.*, 1990; Albertini *et al*., 2006), the urinary excretion of damaged DNA bases (Chen and Chiu, 2005), and the induction of genotoxicity biomarkers (micronuclei or chromosome aberrations; Boffetta *et al.*, 2007) are presently being evaluated and/or validated for use in conjunction with classical epidemiological study designs. Such data are usually restricted in their application to specific chemical substances but such techniques may ultimately become more widely used, particularly when combined with animal data that defines potential mechanisms of action and associated biomarkers that may be indicative of carcinogenic risk. Monitoring of the molecular events that underlie the carcinogenic process may also facilitate the refinement of dose response relationships and may ultimately serve as early indicators of potential cancer risk. However, as a generalisation, such biomonitoring tools have yet to demonstrate the sensitivity requisite for routine use.

R.7.7.10.3 Exposure considerations for carcinogenicity

Information on exposure, use and risk management measures should be collected in accordance with Article 9 and Annex VI of REACH.

It is indicated in REACH Annex X a carcinogenicity study may be required by the European Chemicals Agency (or proposed by the registrant) when the substance has a widespread dispersive use or there is evidence of frequent or long-term human exposure. Preliminary toxicokinetic studies may be required first to address specific questions regarding potential target tissues and relevant exposure routes relevant for the chemical of concern.

On the other hand, investigations on the carcinogenic properties of a chemical can be deferred, if it can be demonstrated to the satisfaction of the Agency that the chemical is used only in a closed system and that human exposures are negligible (i.e. risk reduction measures on the substance are already equivalent to those applied to high potency carcinogenic substances of category 1 and 2. Reasons for this could include the presence of other substances for which strict exposure regimes are implemented or enforced). The rationale for exemption from testing, of course, needs to be clearly documented upon registration.

Also, considerations on exposure may influence the search for information, e.g. applicable to the actual route of exposure. For example, if from exposure scenarios it is clear that only a single specific route is involved, toxicity data for this route is of higher relevance in data gathering and evaluation than for the other routes. Also, the involvement of inhalation exposure to particles will prioritise toxicity information needs in order to allow a proper hazard evaluation and risk assessment.

R.7.7.11 Evaluation of available information on carcinogenicity

This particular endpoint is complex and requires evaluation by expert judgement.

Note that the objective of this evaluation is to acquire information on the carcinogenic potential of the substance: i.e. is the substance carcinogenic or not, and, if so, what is the underlying mode of action (thresholded or not), and what is its carcinogenic potency (i.e. there is a need to define a dose descriptor).

An evaluation on the above mentioned properties requires a combining of various types of information, as indicated in Section R.7.7.10 (and below). Such an evaluation needs a *Weight of Evidence* approach for arriving at conclusions, i.e. a careful gathering, sorting and weighing of the various pieces of information available. This exercise is particularly complex and, therefore, requires expert judgement input.

R.7.7.11.1 Non-human data on carcinogenicity

Non-testing data for carcinogenicity

To date little experience is available for the evaluation of substances on non-testing data, since the use of non-testing data for regulatory decisions is rather new. Therefore, at every stage in the assessment for potential chemical toxicity, specialist judgement is essential. It is recognised though, that non-testing data may potentially inform on all carcinogenic properties, i.e. including mode of action and potency.

Documentation should include reference to a related chemical or groups of chemicals that give rise to concern or lack of concern. This can either be presented according to scientific logic (read-across) or as a mathematical relationship of chemical similarity.

In some cases, the carcinogenic potential posed by a substance can be assessed based upon analysis of the relative concentrations of constituents believed to present a risk in a complex mixture. For example, the classification of certain complex coal- and oil-derived substances as carcinogens can varies as a function of the content of marker carcinogens (benzene, 1,3-butadiene and benzene), whereas for others it depends on the level of polycyclic aromatic hydrocarbons measured following DMSO solvent extraction. (see Annex I of EU Directive 67/548/EEC). When properly validated, such chemical extraction and analysis techniques are highly predictive of the outcomes that would be obtained in animal carcinogenicity studies.

If well documented and applicable, (Q)SARs can be used to help reach the decision points described in the section below. The accuracy of such methods may be sufficient to help or allow either a testing or a specific regulatory decision to be made. Expert judgement is needed to make this determination.

Chemicals for which no test-data exist present a special case in which reliance on non-testing methods may be absolute. Many factors will dictate the acceptability of non-testing methods in reaching a conclusion based on no tests at all. A *Weight of Evidence* evaluation of carcinogenicity based on multiple genotoxicity and carcinogenicity estimates (*e.g.* from (Q)SAR models) may in some cases equal or exceed the decision basis which could be obtained by experimentally testing a chemical in one or two *in vitro* tests. This must be considered on a case-by-case basis by the registrant.

Further guidance on the use of Grouping of Chemicals and on (Q)SARs both for a qualitative (i.e. classification and labelling) as well as a quantitative assessment (i.e. identifying some dose descriptor value) is provided in Sections R.4.3.2 and R.6.2, respectively, and also includes basic concepts used, validation status, adequacy and documentation needs for regulatory purposes.

Testing data on carcinogenicity

In vitro data

In vitro data can only give preliminary information about the carcinogenic potential of a substance and possible underlying mode(s) of action. For example, *in vitro* genotoxicity studies may provide information about whether or not the substance is likely to be genotoxic *in vivo*, and thus a potential genotoxic carcinogen (see Section R.7.7.1), and herewith on the potential mode of action underlying carcinogenicity: with or without a threshold.

Besides genotoxicity data other *in vitro* data (described in Section R.7.7.10.1) such as *in vitro* cell transformation can help to decide, in a *Weight of Evidence* evaluation, whether a chemical possesses a carcinogenic potential. Cell transformation results in and of themselves do not inform as to the actual underlying mode(s) of action, since they are restricted to the detection of effects exerted at the level of the single cell and may be produced by mechanistically distinct processes.

Studies can also be conducted to evaluate the ability of substances to influence processes thought to facilitate carcinogenesis. Many of these endpoints are assessed by experimental systems that have yet to be formally validated and/or are the products of continually evolving basic research. Formalised and validated protocols are thus lacking for the conduct of these tests and their interpretation. Although it is difficult to give general guidance on each test due to the variety and evolving nature of tests available, it is important to consider them on a case-by-case basis and to carefully consider the context on how the test was conducted.

A number of the test endpoints evaluate mechanisms that may contribute to neoplastic development, but the relative importance of each endpoint will vary as a function of the overall toxicological profile of the substance being evaluated. It should further be noted that there are significant uncertainties associated with extrapolating *in vitro* data to an *in vivo* situation. Such *in vitro* data will, in many instances, provide insights into the nature of the *in vivo* studies that might be conducted to define carcinogenic potential and/or mechanisms.

Animal data

In vivo data can give direct information about the carcinogenic potential of a substance, possible underlying mode(s) of action, and its potency.

Testing for carcinogenicity is conventionally carried out in groups of rats or mice according to standard test protocols or guidelines (e.g. OECD TGs 451, 453 or US-EPA 870.4200) and a conclusion is based on a comparison of the incidence, nature and time of occurrence of neoplasms in treated animals and controls.

Knowledge of the historic tumour incidence for the strain of animal used is important (laboratory specific data are preferable). Also attention to the study design used is essential because of the requirement for statistical analyses. The quality, integrity and thoroughness of the reported data from carcinogenicity studies are essential to the subsequent analysis and evaluation of studies. A qualitative assessment of the acceptability of study reports is therefore an important part of the process of independent evaluation. Sources of guidance in this respect can be found in IEH (2002), CCCF (2004) and OECD (2002). If the available study report does not include all the information required by the standard test guideline, judgement is required to decide if the experimental procedure is or is not acceptable and if essential information is lacking.

The final design of a carcinogenicity bioassay may deviate from OECD guidelines if expert judgement and experience in the testing of analogous substances supports the modification of protocols. Such modifications to standard protocols can be considered as a function of the specific properties of the material under evaluation.

Carcinogenicity data may sometimes be available in species other than those specified in standard test guidelines (e.g. Guinea pig, Syrian hamster and occasionally mini-pigs, dogs and primates). Such studies may be in addition to, or instead of, studies in rats and mice and they should be considered in any evaluation.

Data from non-conventional carcinogenicity studies, such as short- and medium-term carcinogenicity assays with neonatal or genetically engineered (transgenic) animals, may also be available (CCCF, 2004; OECD, 2002). Genetically engineered animals possess mutations in genes that are believed to be altered in the multi-step process of carcinogenesis, thereby enhancing animal sensitivity to chemically induced tumours. A variety of transgenic animal models exist and new models are continually being development. The genetic alteration(s) in a specific animal model can be those suspected to facilitate neoplastic development in a wide range of tissue types or the alterations can be in genes suspected to be involved in tissue specific aspects of carcinogenesis. The latter must be applied with recognition of both their experimental nature and the specific mechanistic pathways they are designed to evaluate. For example, a transgenic animal model sensitive to mesothelioma induction would be of limited value in the study of a suspected liver carcinogen. While such animal model systems hold promise for the detection of carcinogens in a shorter period of time and using fewer animals, their sensitivity and specificity remains to be determined. Due to a relative lack of validation, such assays have not yet been accepted as alternatives to the conventional lifetime carcinogenicity studies, but may be useful for screening purposes or to determine the need for a rodent 2-year bioassay. Several evaluations of these types of study have been published (e.g. Jacobson-Kram, 2004; Pritchard *et al.*, 2003; ILSI/HESI (2001).

When data are available from more than one study of acceptable quality, consistency of the findings should be established. When consistent, it is usually straightforward to arrive at a conclusion, particularly if the studies were in more than one species or if there is a clear treatment-related incidence of malignant tumours in a single study. If a single study only is available and the test substance is not carcinogenic, scientific judgement is needed to decide on whether (a) this study is relevant or (b) additional information is required to provide confidence that it should not be considered to be carcinogenic.

Study findings also may not clearly demonstrate a carcinogenic potential, even when approved study guidelines have been followed. For example, there may only be an increase in the incidence of benign tumours or of tumours that have a high background incidence in control animals. Although less convincing than an increase in malignant and rare tumours, and recognising the potential over-sensitivity of this model (Haseman, 1983; Ames and Gold, 1990), a detailed and substantiated rationale should be given before such positive findings can be dismissed as not relevant.

Repeated dose toxicity studies may provide helpful additional information to the *Weight of Evidence* gathered to determine whether a substance has the potential to induce cancer, and for potential underlying modes of action (Elcombe *et al*., 2002). For example, the induction of hyperplasia (either through cytotoxicity and regenerative cell proliferation, mitogenicity or interference with cellular control mechanisms) and/or the induction of pre-neoplastic lesions may contribute to the *Weight of Evidence* for carcinogenic potential. Toxicity studies may also provide evidence for immunosuppressive activity, a condition favouring tumour development under conditions of chronic exposure.

Finally, toxicokinetic data may reveal the generation of metabolites with relevant structural alerts. It may also give important information as to the potency and relevance of carcinogenicity and related data collected in one species and its extrapolation to another, based upon differences in absorption, distribution, metabolism and or excretion of the substance. Species specific differences mediated by such factors may be demonstrated through experimental studies or by the application of toxicokinetic modelling.

Positive carcinogenic findings in animals require careful evaluation and this should be done with reference to other toxicological data (e.g. *in vitro* and/or *in vivo* genotoxicity studies, toxicokinetic data, mechanistic studies, (Q)SAR evaluations) and the exposure conditions (e.g. route). Such comparisons may provide evidence for (a) specific mechanism(s) of action, a significant factor to take into account whenever possible, that may then be evaluated with respect to relevance for humans.

A conceptual framework that provides a structured and transparent approach to the *Weight of Evidence* assessment of the mode of action of carcinogens has been developed (see Sonich-Mullin *et al.*, 2001; Boobis *et al*., 2006). This framework should be followed when the mechanism of action is key to the risk assessment being developed for a carcinogenic substance and can be particularly critical in a determination of whether a substance induces cancer *via* genotoxic or nongenotoxic mechanisms.

For example, a substance may exhibit limited genotoxicity *in vivo* but the relevance of this property to carcinogenicity is uncertain if genotoxicity is not observed in tissues that are the targets of carcinogenesis, or if genotoxicity is observed *via* routes not relevant to exposure conditions (e.g. intravenous injection) but not when the substance is administered *via* routes of administration known to induce cancer. In such instances, the apparent genotoxic properties of the substance may not be related to the mechanism(s) believed to underlie tumour induction. For example, oral administration of some inorganic metal compounds will induce renal tumours *via* a mechanism believed to involve organ specific toxicity and forced cell proliferation. Although genotoxic responses can be induced in non-target tissues for carcinogenesis *via* intravenous injection, there is only limited evidence to suggest that this renal carcinogenesis entails a genotoxic mechanism (IARC, 2006). The *burden of proof* in drawing such mechanistic inferences can be high but can have a significant impact upon underlying assumptions made in risk assessment.

In general, tumours induced by a genotoxic mechanism (known or presumed) are, in the absence of further information, considered to be of relevance to humans even when observed in tissues with no direct human equivalent. Tumours shown to be induced by a non-genotoxic mechanism are, in principle, also considered relevant to humans but there is a recognition that some non-genotoxic modes of action do not occur in humans (see OECD, 2002). This includes, for example, some specific types of rodent kidney, thyroid, urinary bladder, forestomach and glandular stomach tumours induced by rodent-specific modes of action, i.e., by mechanisms/modes of action not operating in humans or operative in humans under extreme and unrealistic conditions. Reviews are available for some of these tumour types providing a detailed characterisation that includes the key biochemical and histopathological events that are needed to establish these rodent-specific mechanisms that are not relevant for human health (see Technical Publication Series by IARC). Recently, the IPCS has developed a framework and provided some examples on how to evaluate the relevance to humans of a postulated mode of action in animals (ILSI RSI, 2003; Boobis *et al.*, 2006).

The information available for substances identified as carcinogenic based on testing and/or non-testing data should be further evaluated in an effort to identify underlying mode(s) of action and potency in order to subsequently allow a proper quantitative assessment of risk (see Section R.7.7.12.2). As already pointed out, the use of non-standard animal models (e.g. transgenic or neonatal animals) needs careful evaluation by expert judgement as to how to apply the results obtained for hazard and risk assessment purposes; it is not possible to provide guidance for such evaluations.

R.7.7.11.2 Human data on carcinogenicity

Epidemiological data may potentially be used for hazard identification, exposure estimation, dose response analysis, and risk assessment. The degree of reliability for each study on the carcinogenic potential of a substance should be evaluated using accepted causality criteria, such as that of Hill (1965). Particular attention should be given to exposure data in a study and to the choice of the control population. Often a significant level of uncertainty exists around identifying a substance unequivocally as being carcinogenic because of inadequate reporting of exposure data. Chance, bias and confounding factors can frequently not be ruled out. A clear identification of the substance, the presence or absence of concurrent exposures to other substances and the methods used for assessing the relevant dose levels should be explicitly documented. A series of studies revealing similar excesses of the same tumour type, even if not statistically significant, may suggest a positive association, and an appropriate joint evaluation (meta-analysis) may be used in order to increase the sensitivity, provided the studies are sufficiently similar for such an evaluation. When the results of different studies are inconsistent, possible explanations should be sought and the various studies judged on the basis of the methods employed.

Interpretation of epidemiology studies must be undertaken with care and include an assessment of the adequacy of exposure classification, the size of the study cohort relative to the expected frequency of tumours at tissue sites of special concern and whether basic elements of study design are appropriate (e.g. a mortality study will have limited sensitivity if the cancer induced has a high rate of successful treatment). A number of such factors can limit the sensitivity of a given study – unequivocal demonstration that a substance is not a human carcinogen is difficult and requires detailed and exact measurements of exposure, appropriate cohort size, adequate intensity and duration of exposure, sufficient follow-up time and sound procedures for detection and diagnosis of cancers of potential concern. Conversely, excess cancer risk in a given study can also be difficult to interpret if relevant co-exposures and confounders have not been adequately documented. Efforts are ongoing to improve the sensitivity and specificity of traditional epidemiological methods by combining cancer endpoints with data on established pre-neoplastic lesions or molecular indicators (biomarkers) of cancer risk.

Once identified as a carcinogenic substance on the basis of human data, well-performed epidemiology studies may be valuable for providing information on the relative sensitivity of humans as compared to animals, and/or may be useful in demonstrating an upper bound on the human cancer risk. Identification of the underlying mode(s) of action – needed for the subsequent risk assessment (see Section R.7.7.12.2) – quite often depends critically on available testing and/or non-testing information.

R.7.7.11.3 Exposure considerations for carcinogenicity

Exposure considerations may lead to adaptation of the evaluation of available information, and / or of the testing strategy.

As indicated before, waiving of carcinogenicity studies may apply, e.g. when it can be demonstrated that the substance is only produced and used in closed systems, which among other reasons may be due to the presence of other substances for which strict exposure regimes are implemented or enforced. On the other hand, a carcinogenicity study may be required (by the Agency or proposed by the registrant) when the substance has a widespread dispersive use or there is evidence of frequent or long-term human exposure, and information on its carcinogenic properties cannot be obtained by others means (from available effect information). Preliminary toxicokinetic studies may be required first to identify the potential target tissues and exposure routes that would guide the design of appropriate studies for the chemical of concern.

In the former case, i.e. when the substance is produced and used in closed systems only, conclusions on safe use and handling can be verified by use of read-across to risk assessments of structurally related carcinogens or to the so-called Threshold of Toxicological Concern (TTC) concept (see Appendix R.7-1): this concept identifies a *de minimis* exposure value for all chemicals, including genotoxic carcinogens, below which there is no appreciable risk to human health for any chemical. If it can be demonstrated that exposures are below these values, there is good reason for not performing the required tests. Clearly, good quality exposure information is essential in all these cases.

R.7.7.11.4 Remaining uncertainty on carcinogenicity

As indicated in the previous sections, adequate human data for evaluating the carcinogenic properties of a chemical are most often not available, and alternative approaches have to be used.

As also indicated in the previous sections and the Section R.7.7.1, test systems for identifying genotoxic carcinogens are reasonably well developed and adequately cover this property. There is also agreement that animal carcinogens which act by a genotoxic mode of action may reasonably be regarded as human carcinogens unless there is convincing evidence that the mechanisms by which mutagenicity and carcinogenicity are induced in animals are not relevant to humans. Unclear, however, and herewith introducing some uncertainty, is the relationship between carcinogenic potency in animals and in humans.

There is, on the other hand, a shortage of sensitive and selective test systems to identify non-genotoxic carcinogens, apart from the carcinogenicity bioassay. In the absence of non-testing information on the carcinogenicity of structurally related chemicals, indications for possible carcinogenic properties may come from existing repeated dose toxicity data, or from *in vitro* cell transformation assays. However, whereas the former source of data will have a low sensitivity (*e.g.* in case of a 28-day study), there is a possibility that the latter may lead to an over-prediction of carcinogenic potential. Insufficient data are available to provide further general guidance in this regard.

Non-genotoxic carcinogens may be difficult to identify in the absence of animal carcinogenicity test data. However, it could be argued that current conservative (cautious) risk assessment methodology should cover the risk for carcinogenic effects *via* this mode of action as well: i.e. current risk assessments for many non-genotoxic carcinogens are based on NOAELs for precursor effects or target organ toxicity with the application of conservative assessment factors to address uncertainty. For example, see the risk assessment for coumarin (EFSA, 2004; Felter *et al*., 2006). Such a risk assessment is not performed, though, in case this substance is not classified as dangerous for any other properties.

Once identified as a non-genotoxic carcinogen (from testing or non-testing data) there may be uncertainty as to the human relevance of this observation, i.e. to the human relevance of the underlying mode of action. In the absence of specific data on this, observations in the animal are taken as relevant to humans. However, additional uncertainty will exist for the relationship between carcinogenic potency in animals and in humans; this uncertainty, though, will be addressed in the procedure for deriving human standards (ILSI RSI, 2003).

Finally, conventional assays of carcinogenicity in animals have been found to be insensitive for some well-established human carcinogenic substances (e.g. asbestos and arsenic compounds). These substances can be shown to be carcinogenic when the test conditions are modified, thus illustrating that there will always be a possibility that a chemical could pose a carcinogenic hazard in humans but be missed in conventional animal studies. This is also true for other toxicological endpoints and should be taken into account by risk managers, especially when making decisions about the acceptability of scenarios showing particularly high exposures to workers and/or consumers.

R.7.7.12 Conclusions on carcinogenicity

R.7.7.12.1 Concluding on suitability for Classification and Labelling

In order to conclude on an appropriate classification and labelling position with regard to carcinogenicity, the available data should be considered using the criteria and guidance associated with the (EU Directive 67/548/EEC)[164].

R.7.7.12.2 Concluding on suitability for Chemical Safety Assessment

Besides the identification of a chemical as a carcinogenic agent from either animal data, epidemiological data or both, dose response assessment is an essential further step in order to characterise carcinogenic risks for certain exposure conditions or scenarios. A critical element in this assessment is the identification of the mode of action underlying the observed tumour-formation, as already explained in Section R.7.7.11.1: i.e. whether this induction of tumours is thought to be *via* a genotoxic mechanism or not.

In regulatory work, it is generally assumed that in the absence of data to the contrary an effect-threshold cannot be identified for genotoxic carcinogens exhibiting direct interaction with DNA, i.e., it is not possible to define a *no-effect level* for carcinogenicity induced by such agents. However, in certain cases even for these compounds a threshold for carcinogenicity may be identified in the low-dose region: e.g. it has in certain cases been clearly demonstrated that an increase in tumours did not occur at exposures below those associated with local chronic cytotoxicity and regenerative hyperplasia. It is also recognised that for certain genotoxic carcinogens causing genetic alterations, a practical threshold may exist for the underlying genotoxic effect. For example, this has been shown to be the case for aneugens

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[164] Directive 67/548/EEC will be repealed and replaced with the EU Regulation on classification, labelling and packaging of substances and mixtures, implementing the Globally Harmonized System (GHS).

(agents that induce aneuploidy – the gain or loss of entire chromosomes to result in changes in chromosome number), or for chemicals that cause indirect effects on DNA that are secondary to another effect (e.g. through oxidative stress that overwhelms natural antioxidant defence mechanisms).

Non-genotoxic carcinogens exert their effects through mechanisms that do not involve direct DNA-reactivity. It is generally assumed that these modes of actions are associated with threshold doses, and it may be possible to define no-effect levels for the underlying toxic effects of concern. There are many different modes of action thought to be involved in non-genotoxic carcinogenicity. Some appear to involve direct interaction with specific receptors (e.g. oestrogen receptors), whereas appear to be non-receptor mediated. Chronic cytotoxicity with subsequent regenerative cell proliferation is considered a mode of action by which tumour development can be induced: the induction of urinary bladder tumours in rats, for example, may, in certain cases, be due to persistent irritation/inflammation/erosion and regenerative hyperplasia of the urothelium following the formation of bladder stones which eventually results in tumour formation. Specific cellular effects, such as inhibition of intercellular communication, have also been proposed to facilitate the clonal growth of neoplastic/pre-neoplastic cells.

The identification of the mode of action of a carcinogen is based on a combination of results in genotoxicity tests (both *in vitro* and *in vivo*) and observations in animal experiments, e.g. site and type of tumour and parallel observations from pathological and microscopic analysis. Epidemiological data seldom contribute to this.

Once the mode of action of tumour-formation is identified as having a threshold or not, a dose descriptor has to be derived for the purpose of allowing to conclude on chemical safety assessment. For threshold mechanisms the No-Observed-Adverse-Effect-Level (NOAEL) or Lowest-Observed-Adverse-Effect-Level (LOAEL) (see general introduction for definition and derivation of these descriptors) for tumour-formation or for the underlying (toxic) effect should be established to allow the derivation of a so-called Derived-No-Effect-Level (DNEL) (Chapter R.8 of the *Guidance on IR&CSA*), that subsequently is used in the safety assessment to establish safe exposure levels.

If the mode of action of tumour formation is identified as non-thresholded, dose descriptors such as T25, BMD10 or BMDL10 (for definition and derivation of these descriptors, see Appendix R.8-6 to Chapter R.8 of the *Guidance on IR&CSA*) are to be established, that allow the derivation of a so-called Derived-Minimal-Effect-Level (DMEL) (for guidance, see Section R.8.5 in Chapter R.8 of the *Guidance on IR&CSA*), that subsequently is used in the safety assessment to establish exposure levels of minimal concern.

Though mainly derived from animal data, epidemiological data may also occasionally provide dose descriptors that allow derivation of a DNEL or DMEL, e.g. Relative Risk (RR) or Odds Ratio (OR).

Substance-specific data for carcinogenicity normally will be absent, especially for the lower tonnage level substances. As indicated in Section R.7.7.11.1, non-testing data (read-across, grouping and/or (Q)SAR) may occasionally be considered sufficient to conclude on this endpoint, i.e. for classification, but also for establishing the underlying mode of action and for estimating the carcinogenic potency. This may introduce some additional uncertainty, especially with respect to the dose descriptor value, the addressing of which requires expert judgement; it is noted that experience to date on this is extremely limited. Guidance on read-across and/or grouping, and the use of (Q)SAR is provided in Sections R.6.2 and R.6.1 of Chapter R.6 of the *Guidance on IR&*CSA, respectively.

R.7.7.12.3 Information not adequate

A *Weight of Evidence* approach comparing available adequate information with the tonnage-tiered information requirements by REACH may result in the conclusion that the information/data requirements are not fulfilled. In order to proceed in further information gathering, the following testing strategy can be adopted.

R.7.7.13 Integrated Testing Strategy (ITS) for carcinogenicity

R.7.7.13.1 Objective / General principles

The objective of this strategy is to describe where required how carcinogenicity should be assessed for all substances subject to registration under REACH: i.e. to identify substances with carcinogenic properties, their associated underlying mode of action, and their potency. Guidance is provided especially for those substances lacking pre-existing epidemiological or toxicological data on carcinogenicity.

The strategy provides the rationale for deciding whether or not a standard animal carcinogenicity study or any other further testing is required. It is recognised that standard carcinogenicity tests take considerable time to conduct and report, are expensive, and involve the use of a large number of animals. Consequently, it is preferable that decisions about the potential carcinogenicity of substances under REACH be taken as frequently as possible without the conduct of such tests.

The strategy recognises that the available information will differ from substance to substance. This may include various different types of toxicity information for the substance in question and/or for its analogues/structurally related chemicals. Details about the use and human exposure potential of the substance will also be available. All this will have an impact on the need for further data acquisition. Proposals for conducting a carcinogenicity test should be made with regard to the potential risk to human health and with consideration of the actual or intended production and/or use pattern.

REACH only specifies a carcinogenicity test for substances at the Annex X tonnage level (≥1000 t/y; see Section R.7.7.9). However, REACH also requires that carcinogenic substances at all tonnage levels be identified as substances of high concern, taking into account information from all available relevant sources (see Section R.7.7.10).

At the tonnage levels below 1000 t/y, the main concern is for those chemicals that are genotoxic. Chemicals may cause cancer secondary to other forms of toxicity, but protection of human health against the underlying toxicity (e.g. as identified from a repeat-dose toxicity study) will also protect against cancer that is secondary to that toxicity. It is noted, though, that some of these non-genotoxic carcinogens, when not classified for any other property and not identified as such in (limited) repeated dose toxicity studies will go unidentified; this also regards the risks associated with human exposures.

Finally, the strategy recognises that the carcinogenic process is a complex multi-step process. Chemically-induced cancer may be induced by any number of different pathways or modes of action and this allows for a variety of different approaches to carcinogenicity assessment. Substances that have the potential to act as genotoxic carcinogens can be identified by *in vitro* and *in vivo* mutagenicity tests, as described in Section R.7.7.1. Carcinogens that act by non-genotoxic modes of action are more difficult to identify because comparable, well-validated, short-term tests for the potentially numerous modes of actions involved are generally not available, and those tests that are available are not required as part of the standard information requirements of REACH.

A flow chart of the strategy is presented in Figure R.7.7–2.

R.7.7.13.2 Preliminary considerations

As a starting point, there will be the information collected with respect to mutagenicity. If they are available, test and non-test data from a literature search and, if possible, from members of an applicable chemical category or (Q)SAR analysis should be taken into account.

For substances for which there is no concern for mutagenic activity, and no other toxicological indicators of concern for carcinogenicity (i.e. for the substance itself or for structurally-related substances), there is no need for further consideration of its carcinogenic potential. This applies equally to those substances at the Annex X tonnage level as to those at lower tonnage levels.

If, however, for non-genotoxic substances toxicological indicators of concern are available (e.g. hyperplastic or pre-neoplastic lesions in repeated dose toxicity studies of the substance itself and/or of closely related substances), they should be investigated further on a case-by-case basis. Any decision on further testing is dependent upon the type and strength of the indications for carcinogenicity, the potential mechanism of action and their relevance to humans, and the type and level of human exposure (see Section R.7.7.10.2).

If no conclusion can be drawn regarding the potential genotoxicity of the substance then, in general, it will be determined on a case-by-case basis when and how the carcinogenic potential should be explored further. Again, this will then depend on the type and strength of the indications for carcinogenicity, the potential mechanism(s) of action, and the type and level of human exposure.

At least for substances at the higher tonnage levels, subchronic and/or chronic studies may provide additional important information on possible carcinogenic effects. There may, for example, be indications of peroxisomal proliferation or of hyperplastic or pre-neoplastic responses, including dose-response characteristics. These should be investigated further on the already indicated case-by-case basis, depending on the type and strength of the indications for carcinogenicity, the potential mechanism of action and relevance to humans, and the type and level of human exposure.

It may be appropriate on occasions to propose other tests to be undertaken, e.g. to test a read-across option with available non-testing data. These could include short-term tests, such as those for *in vitro* cell transformation or cell proliferation, or medium-term tests, like genetically engineered (transgenic) or neonatal models. It may well be that data generated in this way supports this read-across to available non-testing data, and herewith provides sufficient confidence in a read-across derived estimate of the carcinogenic potency for the substance and also for the magnitude of the risks associated with experienced exposure levels. The data generated may also weaken or even disprove the basis for read-across. It is noted that experience to date on this is very limited (as indicated in Section R.7.7.11.1). Guidance on read-across and/or grouping is provided in Section R.6.2 in Chapter R.6 of the *Guidance on IR&CSA*.

As validated testing procedures are not yet available and published in the OECD test guideline programme, it is essential that appropriate expert advice is sought regarding the application and suitability of any of these other tests.

Substances for which concern for carcinogenicity is solely based on positive genotoxicity data will, in a first step, be evaluated according to the approach outlined for identification of the genotoxicity hazard (see Section R.7.7.5).

Formally, for a substance classified as a category 1 or 2 mutagen, a carcinogenicity study will not normally be required (see Section R.7.7.9); *i.e.* it will be regarded as a genotoxic carcinogen. In order to allow an assessment of the magnitude of potential cancer risks associated with the prevailing human exposures, it may well be that available non-testing data (read-across, grouping, (Q)SAR) provide a sufficiently helpful estimate of the carcinogenic potency of the substance (i.e. by read-across) from which risks can be assessed. Guidance on read-across and/or grouping, and the use of (Q)SAR is provided in Sections R.6.2 and R.6.1, respectively.

In case such an approach is not possible, an estimate of acceptable exposure conditions may alternatively be obtained by use of the available data from animal toxicity studies: i.e. by identifying the minimal toxic dose in sub-chronic studies (if available, as some surrogate value for the dose descriptor) and by applying a large assessment factor; see for further guidance Gold *et al*. (2003). It is stressed that expert judgement is definitively needed here.

On very rare occasions, a case may be made to perform a carcinogenicity study in animals for substances that have been classified for mutagenicity in categories 1 or 2. Such a case would have to explain why the study was critically important; e.g. in the context of the clarification of carcinogenic risk associated with human exposures.

For substances classified as category 3 mutagens, and for which there is no carcinogenicity study, there should first be an evaluation of whether classification in category 2 for mutagenicity is possible. If such a classification is made, then the approach described above can be followed with regards to carcinogenicity. Occasionally, it may be established that classification as a category 2 mutagen is not appropriate. In such instances, it should not be assumed automatically that the substance has carcinogenic potential. However, unless there is clear evidence to indicate the contrary, it is expected that these substances will be regarded as genotoxic carcinogens.

As the previous paragraph implies, mutagenic potential *in vivo* is not always a reliable indicator of carcinogenic potential. If repeated dose toxicity studies indicate that pre-neoplastic changes (e.g. hyperplasia, precancerous lesions) occur, then the probability that carcinogenic activity will be expressed is increased. Non-testing data such as read-across and (Q)SAR may also contribute to this evaluation.

For substances at the REACH Annex X tonnage level, the need for or waiving of a standard animal test should be clearly explained, taking into account all the available toxicological and hygiene information on the substance and/or other relevant substances. For example, if it can be demonstrated that the substance is used only in a closed system and that human exposures are negligible, it is possible to propose no further testing for carcinogenicity.

It is recommended that when a carcinogenicity bioassay is required, study design and test protocol are well considered prior to delivering the test-proposal (e.g. OECD TG 453). Particular consideration, based on all the available data, should be given to the selection of the species and strain to be used in the carcinogenicity test, the route of exposure and dose level selection. It is also recommended that when a carcinogenicity test is to be conducted, an investigation of chronic toxicity should, whenever possible, form part of the study protocol. Finally, the limited value of a mouse assay as second species should be considered in this (Doe *et al.*, 2006).

The approaches outlined below may be used in the assessment of the potential carcinogenic risk of a substance to humans, and to help decide whether or not a carcinogenicity test will be required and, if so, when.

R.7.7.13.3 Testing strategy for carcinogenicity

As for other endpoints, the following three steps apply for the assessment of carcinogenicity (i.e. the hazard, underlying mode of action, and potency) for substances at each of the tonnage levels specified in Annexes VII to X of REACH.

i. Gather and assess all available test and non-test data from read-across/proper chemical category and suitable predictive models. Examine the Weight of Evidence that relates to carcinogenicity.

ii. Consider whether the standard information requirements are met.

iii. Ensure that the information requirements of Annexes VII and VIII are met; make proposals to conform with Annexes IX and X.

Further details about the procedures to follow at each of the different tonnage levels are described below.

Substances at Annexes VII, VIII and IX

A definitive assessment of carcinogenicity is usually not possible from the data available at the Annex VII, VIII and IX tonnage levels. However, for all substances, any relevant test data that are already available, together with information from predictive techniques such as read-across or chemical grouping, should be used to form a judgement about this important hazard endpoint.

The minimum information to be provided at the Annex VII, VIII and IX tonnage levels in relation to this endpoint is equivalent to that required for the mutagenicity endpoint (see Section R.7.7.2): positive results from *in vitro* mutagenicity studies provide an alert for possible carcinogenicity, and need confirmation *via* further testing *in vitro* and/or *in vivo* mutagenicity testing. As such, this will not lead to classification of a substance as a carcinogen, but this evidence should be taken into account in risk assessment: substances shown to be *in vivo* mutagens should be assumed to be potentially carcinogenic.

Furthermore, the results of repeated dose toxicity studies and /or reproductive/ developmental toxicity tests may be informative about a possible carcinogenic potential: hyperplasia or other pre-neoplastic effects may be observed in these studies. These observations may also be informative on potential mode(s) of action underlying the carcinogenic effect.

Although the criteria for carcinogenicity classification may not be met in the absence of substance-specific carcinogenicity data, the evidence from the available information alerting to possible carcinogenicity should be taken into account in the risk assessment for this endpoint: ways that allow an assessment of the magnitude of potential cancer risks associated with human exposures without performing the assay are indicated in indicated in Section R.7.7.13.2 (see Section for derivation of DMEL and DNEL values in Chapter R.8 of the *Guidance on IR&CSA*).

It is important to note that at the tonnage levels below 1000 t/y, the main concern is for those chemicals that are genotoxic. The repeated dose toxicity studies mentioned above may indicate cancers which are secondary to other forms of toxicity. For those the protection of human health against the underlying toxicity will also protect against cancer that is secondary to the toxicity. It is noted, though, that some of these non-genotoxic carcinogens, when not classified for any other property and not identified as such in (limited) repeated dose toxicity studies will go unidentified; this also regards the risks associated with human exposures.

Substances at Annex X

All substances at this tonnage should be evaluated for carcinogenicity.

All relevant data from all toxicity studies should be assessed to see whether a sufficiently reliable assessment about the carcinogenicity of the substance is possible, including alternative means, if needed: i.e. predictive techniques such as chemical grouping and read-across, and the use of (Q)SARs. On some occasions, it may be proposed to supplement these predictive approaches with *in vitro* or alternative shorter-term *in vivo* investigations in order to circumvent the need for a carcinogenicity study. This should usually be in the context of adding to the *Weight of Evidence* that a substance may be carcinogenic.

Formally, if the substance is classified as a category 1 or 2 mutagen (GHS category 1), a carcinogenicity study will not normally be required. For a substance classified as a category 3 mutagen (GHS category 2) it should first be established whether a case should be made for a higher level of classification.

For risk assessment, all the substances are then regarded as genotoxic carcinogens unless there is scientific evidence to the contrary. Ways that allow an assessment of the magnitude of potential cancer risks associated with human exposures without performing the assay are indicated in Section R.7.7.13.2. (see Chapter R.8 of the *Guidance on IR&CSA* for derivation of DMEL and DNEL values).

A carcinogenicity study may, on occasion, be justified. If there are clear suspicions that the substance may be carcinogenic, and available information (from both testing and non-testing data) are not conclusive in this, both in terms of hazard and potency, then the need for a carcinogenicity study should be explored. In particular, such a study may be required for substances with a widespread, dispersive use or for substances producing frequent or long-term human exposures. However, it should be considered only as a last resort.

It is noted, though, that some of non-genotoxic carcinogens, i.e. when not classified for any other property and not identified as potential carcinogens in (limited) repeated dose toxicity studies will go unidentified; this also regards the risks associated with human exposures.

If, in any case there is a need for further testing, the registrant must prepare and submit a well-considered test proposal (see Section R.7.7.6.2), and a time schedule for fulfilling the information requirements.

Figure R.7.7–2 Integrated Testing Strategy for carcinogenicity